

PROCEEDINGS OF THE INTERNATIONAL CONFERENCE ON MECHANICAL AND INDUSTRIAL TECHNOLOGIES, VOLUME-1

BOOK OF ABSTRACT

EDITORS

Dr. Ashish Kumar Srivastava

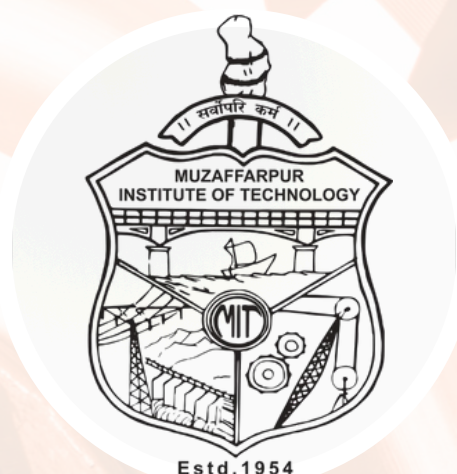
Dr. Pramod Kumar

Prof. (Dr.) Mithilesh Kumar Jha

Prof. (Dr.) Sandeep Tiwari

ICMIT
2025

MAY 24-25, 2025



ORGANIZED BY
DEPARTMENT OF MECHANICAL ENGINEERING
MUZAFFARPUR INSTITUTE OF TECHNOLOGY, MUZAFFARPUR
DEPARTMENT OF SCIENCE, TECHNOLOGY AND TECHNICAL EDUCATION,
GOVERNMENT OF BIHAR, PATNA

IN COLLABORATION WITH
INDIAN SOCIETY FOR TECHNICAL EDUCATION &
DARBHANGA COLLEGE OF ENGINEERING, DARBHANGA



About the Institute

The Muzaffarpur Institute of Technology (MIT), Muzaffarpur, is one of the oldest and prestigious technical institutions in Eastern India. The institute established in 1954, and operates under the Science, Technology and Technical Education Department, Government of Bihar. With a rich legacy and glory of more than 70 years the institute works with the motto of "Work is above everything else". The institute is affiliated with Bihar Engineering University, Patna, and offers various undergraduate programs (9) viz. Mechanical Engineering-60, Civil Engineering-60, Electrical Engineering-60, Electronics and Communication Engineering-40, and others. For encouraging research among the students of various engineering discipline, the institute offers different postgraduate programs (6) in Mechanical (Thermal Engineering-18 & Machine Design-18), Civil (Geotechnical Engineering-30 & Transportation Engineering-18), Electrical (Electrical Energy System-18), and Electronics (Advanced Electronics and Communication Engineering-18). Moreover, MIT Muzaffarpur also shares its huge and glorious history of its active Alumni which encourages the young minds for their future endeavors.

About the Department

The department of Mechanical Engineering was started from the session, 1960-61. Currently, the department is offering 4-years B.Tech Program in Mechanical Engineering with an intake capacity of 60. From the session 2017, the department has started the M.Tech program in the field of "Machine Design and Thermal Engineering", with an intake capacity of 18 each. The main intent of the department is to nurture the students with class tutorial and in hand practice with state-of-the-art laboratories & workshop. The department also encompasses the Central Research Facility lab (consist of some edge cutting equipment's viz. Advance CNC, 3D Printer, Stir Casting Machine, PIN on Disc Wear test Machine, CNC Wire-Cut EDM, Disc Polishing Machine, Optical Microscope, Vickers Micro Hardness Tester and others), Computational Lab (Vasishtha Narayan Singh Research Lab) for designing and performing numerical and analytical work in SolidWorks, Matlab, ANSYS 19.2 R1 and others.

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Volume I: May 2025

TITLE

Proceedings of the international conference on mechanical and industrial technologies

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MESSAGE FROM THE DESK OF CHAIRMAN



It gives me immense pleasure and pride that I welcome you all to the International Conference on Mechanical and Industrial Technologies (ICMIT-2025), organised by Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur in association with Darbhanga College of Engineering (DCE Darbhanga) and Indian Society for Technical Education, MITM Chapter

One of the most fundamental and evolving fields in engineering, mechanical engineering is still essential to advancing technological innovation, sustainability, and industry-wide progress. Leading researchers, academicians, business leaders, and scholars from all over the world are expected to attend this conference in order to exchange ideas, experiences and innovations in this rapidly evolving field. Under the theme, Mechanical and Industrial Technologies, this conference brings together a rich mix of voices through keynote address, technical paper presentations, and panel discussions. From advanced manufacturing and robotics to thermal systems, materials science, and automotive innovations, we've curated sessions that reflect the breadth and depth of our ever-evolving field.

What truly makes this gathering special isn't just the topics we explore, but the people who bring them to life. It's a chance for young innovators and seasoned experts to sit side by side, exchange ideas, learn from one another, and imagine what the future of mechanical engineering could look like—together, on a global scale.

I'm deeply grateful to everyone who made this event possible, our inspiring keynote speakers, dedicated authors and reviewers, the hardworking organizing committee. This conference is a reality today because of your passion, your time, and your unwavering support and for that, I'm genuinely thankful.

**Warm regards,
Prof. (Dr.) Mithilesh Kumar Jha
Principal, MIT Muzaffarpur**

MESSAGE FROM THE DESK OF CHAIRMAN



It is a pleasure to welcome you all to the International Conference on Mechanical and Industrial Technologies (ICMIT-2025). This gathering represents much more than a meeting of minds. It's a gathering that honors innovation, inquisitiveness, and the spirit of working together.

ICMIT brings together a diverse community of thinkers and practitioners—from students just beginning their journeys to seasoned experts who have shaped the field. Over the coming days, we will explore innovations spanning advanced manufacturing, robotics, materials science, thermal systems, automotive technologies, and beyond.

What excites me most about this conference is the unique opportunity it offers: to connect, share, and learn from one another in an environment that encourages both bold ideas and practical solutions. It is through these connections that we pave the way for breakthroughs that can truly transform industries and improve lives.

I would like to express my heartfelt gratitude to all the speakers, authors, reviewers, organizers, sponsors, and volunteers who have poured their passion and effort into making ICMIT a reality. Without you, this event simply wouldn't be possible.

I encourage everyone to take full advantage of the sessions, workshops, and networking opportunities, and to leave here inspired and empowered to contribute to the future of mechanical and industrial technologies.

**Warm regards,
Prof. (Dr.) Sandeep Tiwari
Principal, DCE Darbhanga**

MESSAGE FROM THE DESK OF CONVENER



I am pleased to convey a warm welcome to all of the International Conference on Mechanical and Industrial Technologies (ICMIT-2025) participants, speakers, delegates, and guests. Scholars, researchers, innovators, academicians, and business professionals from all over the world come together at this esteemed conference to share ideas, discuss new developments, and examine trends in mechanical and industrial engineering.

Today's fast-paced technological advancement is changing industries and how we think about productivity, sustainability, design, and manufacturing. Given this, the conference's theme, "Innovating for a Sustainable and Intelligent Future," is both pertinent and crucial. In fields like smart manufacturing, robotics and automation, materials engineering, thermal sciences, industrial management, and more, ICMIT 2025 seeks to address current issues and prospects.

The conference offers a singular chance for interdisciplinary cooperation, where theoretical understandings collide with practical implementations. We hope to inspire transformative solutions that can influence academia and industry through technical paper presentations, poster sessions, keynote addresses by distinguished speakers, and interactive panel discussions.

It takes a great deal of commitment and coordination to plan a conference this size and importance. I would like to take this opportunity to thank our organizing committee, reviewers, session chairs, volunteers, and supporting institutions for their tireless efforts and dedication, which have made this event possible. I also want to thank our sponsors and partners for their invaluable assistance.

I request each and every participant to actively participate, freely share ideas, and consider the possibility of future partnerships. I hope your time at ICMIT 2025 is memorable and enriching, regardless of whether you are here to share your research, get feedback, learn new things, or develop professional networks

Warm regards,
Dr. Ashish Kumar Srivastava
HOD, Mechanical Engineering

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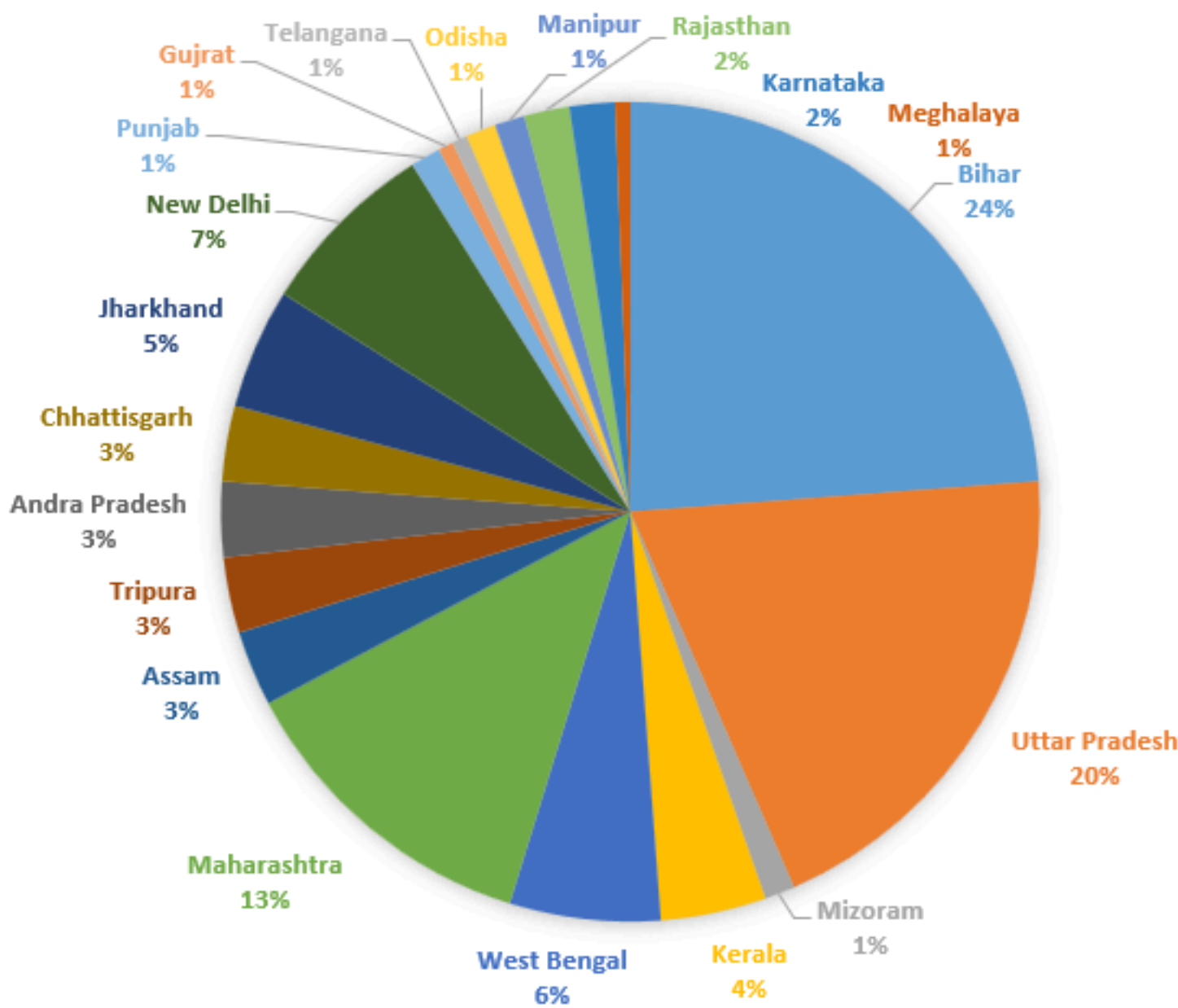
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**International Conference on Mechanical and Industrial Technologies
(ICMIT-2025)
Conference Summary**

Total Number of Abstracts received 275

Total Number of Abstracts accepted: 177

**State-Wise Paper Contribution
(National Diversity)**



**Country-Wise Paper Contribution
(International Diversity)**

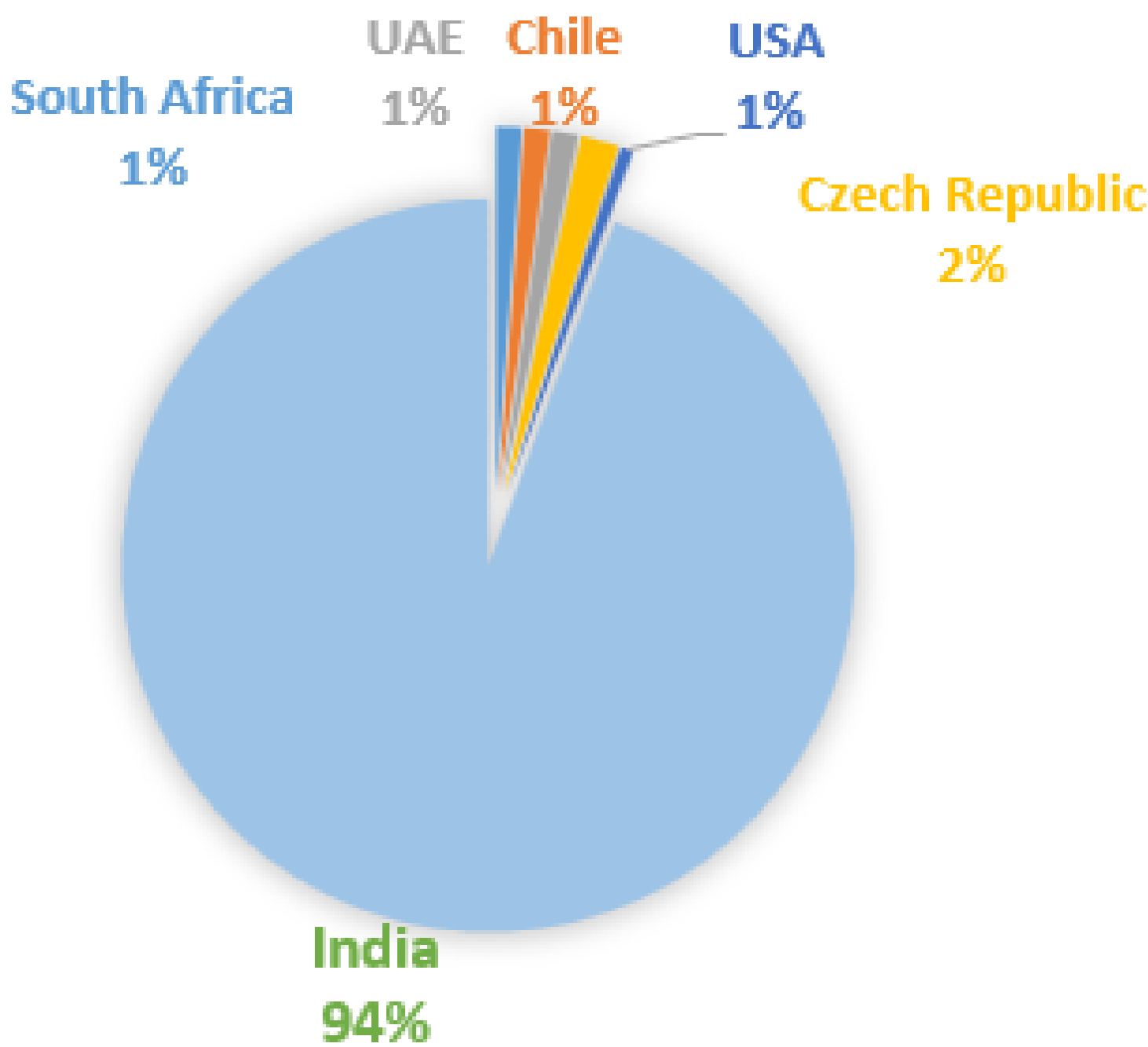


Table of Contents

<i>Message From the Desk of Chairman</i>	i-ii
<i>Message From the Desk of Convener</i>	iii
<i>Committee's</i>	iv
<i>International Advisory Board</i>	v
<i>National Advisory Board</i>	vi
<i>Editorial Board</i>	vii
<i>Technical Core Members</i>	vii
<i>Keynote Speakers & Session Chairs</i>	viii
<i>Conference Summary</i>	ix
<i>Table of Contents</i>	x-xx

Track 1: Materials Processing and Manufacturing Techniques

	Paper Id	Title	Authors Name	Page No.
1	ICMITMM02	Development of the composite component of Onyx Pro and carbon fibre through 3d printing.	Saurabh Kumar, Ashish Kumar Srivastava	2
2	ICMIT/MM03	Experimental Investigation and Mechanical Characterization of Al7075 with Nitinol Powder Metal Hybrid Composite by Stir Casting	Priya Bharti, Dr. Pramod Kumar	3
3	ICMIT/MM04	Optimization of Friction Stir Processing Parameter on Fabrication of Al6063 Reinforced Hybrid Composite Material Using Response Surface Measurement and Desirability Parameter.	Ashutosh Kumar, Dr. Pramod Kumar	4
4	ICMIT/MM07	Development of ECM Set-up and Parametric Optimization of Machining Parameters for High Strength Alloy.	Rakesh Kumar Singh, Anurag Pal, Dr. Ashish Srivastava, Pankaj Kushwaha, Anshu Anand, Sanjay Kumar	5
5	ICMIT/MM08	Microstructure and Mechanical characterization of Mg based surface composite produced by friction stir processing using titanium oxide and graphene as reinforcement	Rakesh Kumar Singh, Aakash Tiwari, Anurag Pal, Anshu Anand	6
6	ICMIT/MM10	Recent Challenges in machining of metal matrix composites- A review.	Bhaskar Chandra Kandpal, Nitin Johri, Subhash Mishra, Indrajeet Chavan, Pardeep Gahlot, Ajay Kumar	7
7	ICMIT/MM11	A Review of the Literature on the Mechanical Properties of Epoxy Composites Reinforced with Natural Fibers and Particulates.	Gaurav Bhati, Abdhesh Kumar, Pulkit Shrivastav	8
8	ICMIT/MM12	Fabrication and Characterization of AA 6082/ B ₄ C/ Pista Shell Powder Hybrid Composite Developed by Stir Casting Technique.	Shailendra Singh Chauhan, Manish Maurya, Neha Gupta, Aman Rajbhar, Danish Sheikh, Giriraj Kishor	9
9	ICMIT/MM14	Development and Performance Evaluation of Green Aluminium Metal Matrix Composites Reinforced with Graphene Nano-powder and Marble Dust.	Rahul Chaurasia, Saroj Kumar Sarangi, Abhishek Kumar, Rakesh Kumar Singh, Ambuj Saxena	10

10	ICMIT/MM15	A Review of Underwater Local Dry Welding: Technological Progress, Future Prospects.	<i>Ganta Venkateswara Rao, Saurav Suman,</i>	11
11	ICMIT/MM19	Photoluminescence properties of Eu ³⁺ doped BA ₂ Y _{0.66} V ₂ O ₇ Cl ₂ red phosphors for white-LEDs	<i>Sruthi S. Nair, Puja Kumari, Subrata Dasa</i>	12
12	ICMIT/MM20	Comparative Study of Mechanical and Functional Properties of Al ₆₀₆₁ -SiC-Cenosphere and Al ₆₀₆₁ -WC-Cenosphere Metal Matrix Composites with As-cast Al ₆₀₆₁ .	<i>Vyshnav P B, Renjith V B, C Sajith Babu, Jishnu K, Mohammed Roshik P A, Shebeer Ali K M</i>	13
13	ICMIT/MM21	Investigation of mechanical properties of 304L stainless steel welds using TIG, MIG and dual-sided TIG-MIG welding methods.	<i>Debnath Ghosh, Subhas Chandra Moi</i>	14
14	ICMIT/MM22	Comparative Study Of AlCrFeCoNiZn HEA Reinforced Composite With as cast Aluminium 6061.	<i>Renjith V B, Vyshnav P B, C Sajith Babu, Jishnu K, Mohammed Roshik P A, Shebeer Ali K M</i>	15
15	ICMIT/MM24	Optimizing Deposition Conditions to Improve Surface Roughness and Hardness of Electroless Ni-P Coatings.	<i>Sameer Lamichaney, Rupam Mandal, Subhasish Sarkar, Buddhadeb Oraon and Gautam Majumdar.</i>	16
16	ICMIT/MM25	Synthesis and Characterization of Electroless Ni-P Coating on Brass Substrate to Improve Hardness and Surface Roughness.	<i>Rupam Mandal, Nitesh Mondal, Ankit De, Subhasish Sarkar, Tapendu Mandal, Rajat Subhra Sen, Gautam Majumdar.</i>	17
17	ICMITMM26	Comparative Performance Analysis of Unprocessed Quartz and Commercial Ceramic Membranes Coated with Hydrophilic Nanoparticles for Oil-Water Separation and Industrial Wastewater Reuse.	<i>Ramanamane NJ, Pita M</i>	18
18	ICMIT/MM27	Microstructural and Mechanical Properties of 3D Printed Polylactic acid PLA Components.	<i>Sharda Kumari, Shalini Anand, Ashish Kumar Srivastava</i>	19
19	ICMIT/MM28	Fabrication and Characterization of Al ₇₀₇₅ B ₄ C/TiO ₂ Hybrid Metal Matrix Fabricated by Electromagnetic Stir Casting Process.	<i>Md Intakhab Alam, Sanjay Kumar</i>	20
20	ICMIT/MM29	Experimental investigation and vibration-based condition analysis for varying load of gear box with spur gear arrangement to increase reliability.	<i>Saicharan Suryavanshi, Patil Rupesh J., Jitendra Atmaram Hole, Nilesh S. Navagale</i>	21
21	ICMIT/MM30	Study on Hardness Behaviour & Microstructure of Al-0.88Mg-0.35Si alloys Micro Alloyed with 0.02 wt.% Sn.	<i>Hrishikesh Bhattacharyya, Monoj Baruah, Anil Borah</i>	22
22	ICMIT/MM31	Optimization of 3D printed parts using different infill patterns and densities by FDM method	<i>Utpal Sen, Mandeep Pathak, Dr. Borad M Barkachary</i>	23
23	ICMIT/MM33	Additive Manufacturing of Sustainable Polymers: Bridging Performance Gaps with Resource-Efficient Strategies and Circular Design.	<i>Ravi kumar, Birendra kumar</i>	24
24	ICMIT/MM34	Experimental Investigations on Wear Behavior of Novel Mg 5.5 Zn Bio Implant Alloy	<i>Praveen P, Bindu M D</i>	25
25	ICMIT/MM35	A Comprehensive Review of 3D Printing Technologies and Their Applications	<i>Sulaim Rizwi, Shahazad Ali, Anant Prakash Agrawal</i>	26
26	ICMIT/MM36	Fabrication of a 2-layer laminated steel composite by friction stir additive manufacturing	<i>Prashant Kumar, Dr. Pramod Kumar</i>	27

27	ICMIT/MM37	Mechanical and Tribological Evolution of AA7075 Aluminum Alloy Processed via ECAP.	<i>Krishna Gorai, Riya Roop, Rahul Bhandari</i>	28
28	ICMIT/MM38	The impact of tool rotational speed on the weld quality of FSW- welded Polystyrene sheet	<i>Rahul Kanti Nath, Vinayak Jha, John Deb Barma</i>	29
29	ICMIT/MM39	Investigation on the Physical, Mechanical and Tribological Behaviour of Palf Sisal based Polymer Hybrid Nanoparticle Composites: a review	<i>A. Kumar, S. S. Rana</i>	30
30	ICMIT/MM40	Analysing Flexural, Impact and Thermophysical properties of Biodegradable Natural Fiber Composites made of Pineapple leaf and Bamboo fiber.	<i>Chandan Kumar, Irfan Haider, Pramod Kumar</i>	31
31	ICMIT/MM41	An Investigation on Tensile and Compressive strength of Inhouse developed ABS material for 3D Printing	<i>Abhay Mishra, Rajeev Kumar, Anuj Kumar Sharma</i>	32
32	ICMIT/MM42	Synthesis And Characterization of Hydroxyapatite Extracted from Chicken Bone	<i>Jobin C J, Sajith Babu C</i>	33
33	ICMIT/MM43	Performance Evaluation of an "Ambient-Aware Adaptive Nano-MQL for Sustainable and Objective-Based Turning Operations"	<i>Mahip Singh, Prof. Amit Rai Dixit, Dr. Anuj Kumar Sharma</i>	34
34	ICMIT/MM44	Design and Additive Manufacturability of Voronoi based Bone scaffold	<i>Gourav Kumar, Hemant Kr. Choudhary</i>	35
35	ICMIT/MM45	Design and development of quality report from manufacturing component using image processing	<i>V. Rakesh, Ruchir Shrivastava, CH. Nithin Kumar, K. Manoj Reddy</i>	36
36	ICMIT/MM48	Effect of Notched Shape and Fiber Orientation on Combined Glass and Carbon Fiber-Reinforced Epoxy Matrix Composite Laminates Under Tensile Loading – FEA and Experimental Analysis	<i>Sandesh Anant Kale, Dr. Mangesh A.Ahire, Dr. P. R. Sonawane, and Dr. S. N. Khan</i>	37
37	ICMIT/MM51	Parametric Analysis and Optimization of Wear and Friction of Composites Produced by FDM Parameters	<i>Ravi Alandikar, Rachayya Arakerimath</i>	38
38	ICMIT/MM52	Comparative experimental study of various powder coating materials under cyclic corrosion aging test ISO 12944-6:1998	<i>Sandesh Anant Kale, Dr. Mangesh A.Ahire, Dr. P. R. Sonawane, and Dr. S. N. Khan</i>	39
39	ICMITMM53	Brief Overview on Alloy Fabrication: Opportunities and Challenges in Additive Manufacturing Process	<i>Suryank Dwivedi, Sangharatna M. Ramteke, Jorge Ramos Grez, and Max Marian</i>	40
40	ICMIT/MM54	Green Synthesis of Zinc-Based Nanocomposites from Agro-waste for Sustainable Nanomaterial Production	<i>Razan Turkmani, Sourjya Bhattacharjee, and Abdallah Shanableh</i>	41
41	ICMIT/MM58	Investigating the Effects of Warping Deformation and Adhesion in Fused Deposition Modeling Printing	<i>Resham Kumari, Shalini Anand, Ashish Kumar Srivastava</i>	42
42	ICMIT/MM59	Parametric Study on Print Orientation, Infill Configuration, and Pattern Design for FFF-Manufactured ABS Parts	<i>Sulaim Rizwi, Shahazad Ali, Anant Prakash Agrawal</i>	43
43	ICMIT/MM60	Development of Epoxy-Based Sustainable Material Using Date Palm Fiber- A Review	<i>Mohd Affan Ashraf, Dr. Shailendra Kumar Verma, Shiv Narayan Prajapati,</i>	44
44	ICMIT/MM61	Development of Epoxy-Based Sustainable Material Using Date Palm Fiber Mat and Graphene: Mechanical and Microstructural Characterization	<i>Mohd Affan Ashraf, Shailendra Kumar Verma, Shiv Narayan Prajapati, Anant Prakash Agrawal</i>	45

45	ICMIT/MM63	Materials and Mechanics in Safety Helmet Design: A Focus on Composites and Hybrids	<i>Gayathri Namboothiri, Dr S.N. Khan, Pavankumar Sonawane</i>	46
46	ICMIT/MM64	21st-century revolutionary growth of biodegradable 3d printed composite polymers.	<i>Shalini Anand, Yogesh Kumar, Ashish Kumar Srivastava</i>	47
47	ICMIT/MM65	Enhancing EDM Efficiency with Hastelloy C-276 and Bio-Based Dielectrics	<i>Narendra Kumara, Dr. P.Sudhakar Rao</i>	48
48	ICMIT/MM66	Microstructural and Mechanical Characterisation of Wire-Fed Stainless Steel in Direct Energy Deposition Additive Manufacturing	<i>Sachindra Shankar, Akash Nag, Jana Petru</i>	49
49	ICMIT/MM67	Mechanical performance of 3D-printed AlSi10Mg honeycomb-structured protective gears	<i>Madhulika Srivastava, R Sai Ruthwique, Preeti Gautam, Akash Nag</i>	50
50	ICMIT/MM68	Microstructural and mechanical properties of the Al-Zn-Al sheet lamination through friction stir additive manufacturing	<i>Preeti Gautam, Akash Nag, Jana Petru</i>	51
51	ICMIT/MM69	Multi-Modal Characterization of Crack Initiation Zones in Brittle Materials Under Grinding Stresses	<i>Tanmay Tiwari, Chandra Sekhar Rakurty</i>	52
52	ICMIT/MM71	The Role of WAAM Process Parameters in Engineering Titanium- Based Functionally Graded Materials	<i>Amit Kumar Mandape, Srinivasu Gangi Setti</i>	53
53	ICMIT/MM72	Thermal Analysis of Dissimilar Friction Stir Welding of Al 6061-T6 and Cu 1100 Using Finite Element Method	<i>Kumar Shambhav, Vishal, Rakesh Kumar, Somnath Chattopadhyaya, Raj Das</i>	54
54	ICMIT/MM73	Developing Scheduling Policies to reduce the Total Weighted Completion Time in additive manufacturing	<i>Advitiya Singh, Krishan Walia, Ujjwal Wadhwa, Prof/Dr. A. K. Madan</i>	55
55	ICMIT/MM74	Impact of Process Variables on Roughness of Turned Surfaces using Biodegradable Cutting Fluids	<i>Deepak Kumar, Rajeev Ranjan, Subhas Chandra Moi, Manoj Kundu</i>	56
56	ICMIT/MM77	Advancements in 3D Printing of Composite Materials: Challenges and Prospects	<i>Narinder Kumar, Hardeep Singh, Manjit Singh</i>	57
57	ICMIT/MM78	Development of TiB2-Al7Si0.3Mg in-situ Functionally Graded material through Centrifugal Casting Process	<i>Amrit Mallick, Raj Kumar Sahu, Srinivasu Gangiseti</i>	58
58	ICMIT/MM79	Effects of Process Variables and Tool Geometry on Friction Stir Welding of Similar Aluminium Alloy Joints AA5083 and AA6061	<i>Chandrakant Swain, Akash Shukla, Kedar Badheka</i>	59
59	ICMIT/MM80	Optimization of EDM Process Parameters for Al-MMC by Composite Desirability Approach	<i>Vaddepalli Pradeep, Anil Kumar Bodukuri, Pankaj Kumar</i>	60
60	ICMIT/MM81	Analysis of Calotropis Gigantea Plant Stem Fiber Reinforced Epoxy Composite Material	<i>Rajesh Kumar, Brajesh Kumar Bharti, Dharamendra Kumar, Gopal Kumar, Manish Kumar</i>	61
61	ICMIT/MM86	Influence of Process Parameters on Surface Modification Using Electric Discharge Machining: A systematic review	<i>Roli Verma, Rabesh Kumar Singh, Anuj Kumar Sharma, Ashish Kumar Srivastava, Saurabh Chauhan</i>	62
62	ICMIT/MM87	Numerical Simulation and Experimental Analysis on Cold Forging of Commercially Pure Aluminum.	<i>Khemraj, Manvandra Kumar Singh, Hemant Kumar Choudhary, and Raj Bahadur Singh</i>	63

63	ICMIT/MM90	A Review Paper on Parametric Optimization of TIG Welding	<i>Shashi Bhushan Kumar, Dr. S.S Dhami</i>	64
64	ICMIT/MM91	A Critical Review on Reinforcement Effects in SAC-Based Lead-Free Solders: Base Metal Degradation and IMC Formation	<i>Mitesh Pratap, Ambrish Maurya</i>	65
65	ICMIT/MM92	Chemically functionalized MoS ₂ Nano additive-enriched bio-lubricants for superior tribological performance	<i>Gaurav Raj, Hemant Kr. Choudhury</i>	66
66	ICMIT/MM93	Comparative Study of Mechanical And Functional Properties of Al6061-SiC-Cenosphere And Al6061-WC-Cenosphere Metal Matrix Composites With Ascast Al6061	<i>Vyshnav P B, Renjith V B, C Sajith Babu, Jishnu K, Mohammed Roshik P A, Shebeer Ali K M</i>	67
67	ICMIT/MM99	Colloidal Formulation and Robocasting of Silicon Nitride Scaffolds with Tailored TPMS Structures	<i>Amit Kumar, Siddharth Yadav, Anuj Kumar Sharma, Santosh Kumar, Govind Kumar Verma</i>	68
68	ICMIT/MM100	Designing a Boron Nitride Polyethylene Composite for Shielding Neutrons	<i>Raviraj Kumar</i>	69
69	ICMIT/MM101	Nano Finishing of Additively Manufactured Parts	<i>Atul Singh Rajput</i>	70
70	ICMIT/MM102	Performance Enhancement of μ -ECDM for Exotic Non-Ferrous Materials Using On-Process Sonication of Electrolyte Solution	<i>Nitesh Kumar</i>	71
71	ICMIT/MM103	Effect of Alkali Treatment and Fiber Orientation on the Tensile Properties of Jute Fiber Reinforced Composites	<i>Avinash Petta, Sabindra Kachhap</i>	72
Track 2: Fluid Flow and Thermal Engineering				
72	ICMIT/TT04	Experimental Investigations of Eigen Frequencies on Bellow Pipes used in thermal plants with free-free conditions	<i>Md Asjad Mokhtar, Ankit Kumar</i>	74
73	ICMIT/TT05	Combustion Modeling of CH ₄ -Air Mixture in an Axisymmetric Channel with an Obstacle	<i>Vinayak Jha, Kumar Amit, Prof. (Dr) Sandeep Tiwari</i>	75
74	ICMIT/TT06	Design, Analysis & Fabrication an IoT-based refrigerator monitoring system	<i>N Sreekanth Reddy, L Mohan Sai Charan, D Srikanth Goud, P harshavardhan Goud, Santosh Madeva Naik</i>	76
75	ICMIT/TT07	Performance Assessment of Eco-Friendly Refrigerant R513A Versus R134a: Heat Transfer Coefficient and Pressure Drop Analysis	<i>Neeraj Kumar Vidhyarthi, Subhrajit Debbarma, Ishika Kumari, Vedant Vaidya, Diya Sarkar, Manjusha Jalli, Sandipan Deb, Sagnik Pal, Ajoy Kumar Das</i>	77
76	ICMIT/TT08	A Comparative Study and Analysis of Shell and Tube Heat Exchanger Using Nanofluid	<i>Deka K. K., Baruah M.</i>	78
77	ICMIT/TT09	A Sustainable Comparative Investigation on Diesel Pumps and Solar Irrigation Pumps	<i>Akash Kumar, Abha Kumari, Ramjee Prasad Gupta</i>	79
78	ICMIT/TT12	Redesigning an Aircraft Engine Bracket for an Enhanced Performance	<i>Atul, Prabhanjan P Bharadwaj, R Sahana, Roopa R, S Pavish Kumar</i>	80

79	ICMIT/TT13	Assessing the Socio-Economic Impact of Cow Dung Briquette Production: A Sustainable Approach to Biowaste Conversion and Energy Utilisation	<i>Narayan Kumar, Vivek Kumar, Suyash Vikram, Dharmendra Kumar, Kumar Abhinav</i>	81
80	ICMIT/TT14	Smart Grid Simulation with AI, Renewable Energy, and Battery Integration	<i>Nikhil Chaurasia, Rubyansh Yadav, Sachin, Sahil Chouhan, Ajay Solanki</i>	82
81	ICMIT/TT15	Sustainable Briquetting of Cow Dung: A Low-Cost Biofuel Production Technique for Rural & Industrial Energy Needs	<i>Narayan Kumar, Vivek Kumar, Jigesh Yadav, Pawan Kumar</i>	83
82	ICMIT/TT18	Optima Design of Wavy Channel for Heat Transfer with Nano-Fluid Flow	<i>Md. Fazan Akhtar, Gulshan Kumar</i>	84
83	ICMIT/TT20	Design and Fabrication of a Peltier-Based Air-Cooling Device	<i>Vivek Narain, Gaurav Pal, Suhail Saifi, Faheem Ahasan</i>	85
84	ICMIT/TT22	Advances in Bubble Dynamics and Heat Transfer in Confined Two-Phase Flow Systems: A Comprehensive Review	<i>Jyoti Kumar, Abhik Majumder, Vishwajeet Kumar</i>	86
85	ICMIT/TT23	A comparative review of shape of splitter plate for enhancement of heat transfer	<i>Jyoti Kumar, Abhik Majumder, Vishwajeet Kumar, Alok Ranjan</i>	87
86	ICMIT/TT25	Implication of Ethanol in Diesel-Moringa blend: An experimental and numerical investigation	<i>Jyoti Kumar, Abhishek MS, Love Kumar, Rishav Kumar Gupta, Alok Ranjan</i>	88
87	ICMIT/TT26	Experimental analysis of V-shaped with dimple and protrusion rib on absorber plate in solar air heater	<i>Himanshu Kumar, Pappu Kumar, and Sudhir Kumar</i>	89
88	ICMIT/TT27	Performance Evaluation of Microchannels with Integrated Microinserts for Enhanced Heat Transfer	<i>Shailesh Ranjan Kumar, Satyendra Singh</i>	90
89	ICMIT/TT30	Heat Distribution in Proton Exchange Membrane (PEM) Fuel Cells using Monte Carlo Simulation.	<i>Ajay Solanki, Shreyas Sah, Sudhanshu Kumar, Bhakti Jain</i>	91
90	ICMIT/TT31	Experimental Investigation and Performance Analysis of Solar Powered Electric Vehicle System	<i>Som Nath Saha, Dhaneshwar Mahto, Sanjay Kumar Singh and Ranjeet Ranjan</i>	92
91	ICMIT/TT32	Experimental cooling performance analysis of thermoelectric module	<i>Dhaneshwar Mahto, Ranjeet Ranjan, Som Nath Saha, Chandan Kumar</i>	93
92	ICMIT/TT33	Prismatic Solar Still Using Forced Convection and Steel Balls: An Experimental Approach	<i>Dhaneshwar Mahto, Ranjeet Ranjan, Som Nath saha, Chandan Kumar</i>	94
93	ICMIT/TT36	Experimental investigation of diesel pool fires inside a naturally ventilated compartment	<i>Avinash Chaudhary, Mahesh Kumar Tiwari, Ravinder Kumar</i>	95
94	ICMIT/TT38	Design, Development, Testing and Performance Evaluation of an Earth-Subsurface Air-Cooling System Using Buried Copper Pipes	<i>Shailesh Ranjan Kumar, Amarjeet Kumar, Ravi Shankar Kumar, Vikash Kumar, Amit Kumar, Aayush Aanchal, Shubhanshu Raj, Abhishek Kumar Ranjan, Aditya Raushan</i>	96
95	ICMIT/TT39	Some Studies on the flow channel design optimisation in fuel cells	<i>Ajay Solanki, Assistant Professor, Nikhil Yadav, Piyush Vishwakarma, Rishabh Yadav,</i>	97

96	ICMIT/TT41	Performance Evaluation of a 1.14 kWp Solar Panel for Lucknow Region Using “PVsyst”	Anuj Singh, Rabesh Kumar Singh, Anuj Kumar Sharma, Brajendra Singh Sengar	98
97	ICMIT/TT43	A Numerical Study on Heat Transfer Enhancement in Solar Air Heater with Circular Perforated Finned Absorber	Dr. Rajesh Kumar, Vikash Kumar	99
98	ICMIT/TT45	A Comprehensive Review of Ocean Energy Sources and Their Challenges in India	Ravi Ranjan Manjul, Pradip Deb Roy	100
99	ICMIT/TT46	Adaptive Design, Fabrication, and Performance Analysis of a Thermosyphon Solar Water Heater Using a Flat Plate Collector	Uday kant Kumar	101
100	ICMIT/TT47	Computational Analysis of Soaking of Steel Bloom in Reheating Furnace	Arijit Banerjee, Ganesh Ghosh	102
101	ICMIT/TT48	Analysis of thermo-viscoelastic waves inside a Polymer micro-rod caused by instantaneous heat under Atangana-Baleanu fractional derivatives framework	Shobha	103
102	ICMIT/TT49	Combustion Characteristics of Fuel Briquettes Produce from Coal Fines and Mangifera Indica Sawdust with Jaggery as a Binder	Kripa Sindhu, Vijay Singh, Anuj Kumar Sharma	104
103	ICMIT/TT50	Multi-objective optimisation of shell and tube heat exchanger using nsga-ii model	Aaryan mendiratta, Aayush raghav, Aryan jain, K. manjunath	105
104	ICMIT/TT51	Experimental Investigation of Nanoparticle Integration in Biodiesel Blends for Enhanced Performance and Reduced Emissions	Dhirendra Kumar, Sabindra Kachhap	106
105	ICMIT/TT52	Flow Past an Oscillating Cylinder: Influence of Reynolds Number and Oscillation Frequency on Wake Development	Aditya Kumar singh, Aditya Jain, Archisha Patnaik, Raj Kumar Singh	107
106	ICMIT/TT53	Emerging Innovations in Thermal Insulation Materials to Promote Energy Conservation	Smita Sharma	108
Track 3: Design and Simulations				
107	ICMIT/DS03	Simulation of passive air-cooling system for an arc welding machine	Jay Prakash, Chandan Chatteraj	110
108	ICMIT/DS05	Theoretical Simulations and Experimental Studies on the Crack Growth of Cr-Mo alloy steel using Finite Element Method	Md Asjad Mokhtar, Ankit Kumar, Dr. Sandeep Tiwari	111
109	ICMIT/DS06	Role of artificial intelligence and machine learning in metal casting- A Review	B.C. Kandpal, Nitin Johri, Natraj Mishra	112
110	ICMIT/DS08	Fault Diagnosis and Condition Monitoring in Robotic arm joint using DCNN	Naman Kedia, Nikhil Yadav, Ajay Solanki	113
111	ICMIT/DS10	Effect of molding sand constituents on casting quality: an RSM-based study	Partha Paul, Subhas Chandra Moi, Manoj Kundu	114
112	ICMIT/DS11	Design and Performance analysis of sustainable automotive bumpers using glass reinforced plastic and recycled rubber composites	Snehal Jadhav, Pavankumar Sonawane, Subim Khan, Dnyaneshwar Mate, Mangesh Ahire, Nilesh Navgale	115

113	ICMIT/DS12	Optimizing Front Axle Design for Commercial Vehicles: Enhancing Strength-to-Weight Ratio through FEA and Experimental Validation	Saurabh Vaidya, Sagar Dhage, Pavankumar Sonawane	116
114	ICMIT/DS17	Design and Simulation Analysis of an In-Plane Comb Drive MEMS Accelerometer	Mohd Omar, Naman Gupta, Nikhilesh Ratna, Sanjay Kumar	117
115	ICMIT/DS18	Experimental Study and Vibration-Based Condition Monitoring of Spur Gearbox Under Variable Load Conditions for Enhanced Reliability and Performance Optimization	Saicharan Suryavanshi, Patil Rupesh J., Jitendra Atmaram Hole, Nilesh S. Navagale	118
116	ICMIT/DS19	Design and Implementation of an IoT-Based Smart Water Quality Monitoring System Using Turbidity Sensors and Node-RED	Sujit Kumar, Alka Raj, Md. Zafar, Hrithik Raushan, Adarsh Kumar	119
117	ICMIT/DS20	Parametric Analysis and optimization of Sound absorption coefficient in anechoic chamber	Shivaraj Naik, Rachayya Arakerimath, Santosh P Bhosle	120
118	ICMIT/DS21	Finite Element Analysis and Structural Improvement in Industrial Helmet Using Composite Material (hemp/Kevlar)	Gayathri Namboothiri, Subim Khan, Pavankumar Sonawane	121
119	ICMIT/DS22	Modelling and Analysis of Self-Balancing and Anti-Collision System in Two-Wheeler	Neha Gupta, Shailendra Singh Chauhan, Syed Faizan Rasool, Veer Pratap Singh, Rimjhim	122
120	ICMIT/DS23	Evaluation of Johnson-cook material model parameters of aa5083	Navneet Kumar, Md Adnan Ali Khan, Nilesh Bihana1, Sanjay Kumar	123
121	ICMIT/DS24	A Review on Analysis of Automobile lower control arm	Shrikant Pandurang Katakdhond, Dr.N.G. Jawarkar, Dr. P. R. Sonawane, Dr. S. N. Khan	124
122	ICMIT/DS25	Stress Optimization and Shape Optimization of Automobile lower control arm Using FEA and UTM Testing	Shrikant Pandurang Katakdhond, Dr.N.G. Jawarkar	125
123	ICMIT/DS26	A Review on Optimization of Engine Mounting Bracket Using Finite Element Analysis	Rahul Jagdish Zunj, Dr. Shailesh S. Pimpale, Dr. Pruthviraj D. Patil	126
124	ICMIT/DS29	Classification of eye refractive error using machine learning Techniques	Dharmendra Kumar, Narayan Kumar, Suyash Vikram, Vivek Kumar, Kumar Abhinav3	127
125	ICMIT/DS30	Design and Finite Element Analysis of Banana Coir Fiber-Reinforced Composites for Automobile Door Intrusion Beams	Abhijit Vasant Chabukswar, Dr. J A Hole, Prof. Abhinay Dube, Dr. P R Sonawane, Dr. S N Khan	128
126	ICMIT/DS31	Design and Shape Optimization of Engine Mounting Bracket of a Truck Using Hybrid Material	Rahul Jagdish Zunj, Dr. Shailesh S. Pimpale and Dr. Pruthviraj D. Patil	129
127	ICMIT/DS34	Real-Time Simulation-Based Performance Assessment of Photovoltaic Systems	Ramakant Sharma, Km Brajesh, Kirti Pal, Omveer Singh	130
128	ICMIT/DS36	Urban Airflow Modelling for Pollution Dispersion in Cities	Amit Kumar, Jigesh Yadav	131
129	ICMIT/DS37	Residual Stress Evaluation in LPBF-Printed Stainless Steel 316L Lattice Structures: Experimental and Simulation-Based Approach	Deepak Kumar, Pratik Kumar Shaw, Amit Rai Dixit, Ashish Kumar Srivastava	132
130	ICMIT/DS39	Design, Finite Element Analysis, and Experimental Validation of Banana Coir Fiber-Reinforced Composites for Automotive Door Intrusion Beams	Abhijit Vasant Chabukswar, Dr. J A Hole, Prof. Abhinay Dube, Dr. P R Sonawane, Dr. S N Khan	133

131	ICMIT/DS41	Design and Fabrication of a Multi-Directional Air-Cooling System for Enhanced Thermal Comfort in Living Spaces	<i>Vishal V. kakade, Shriram B. Joshi, Pallavi D. Parsode, Shreya A. Gotmukle, Dr. Pruthviraj D. Patil, Dr. Pimpale S. S.</i>	134
132	ICMIT/DS42	Design and Fabrication of Portable Pulse Jet Bag Filter House	<i>K. Jaswanth, K Sai Rahul, L. Naveen, A. Syamala</i>	135
133	ICMIT/DS44	Design and Fabrication of Nozzle for Improved Layer Adhesion in Concrete 3d printing -Using solid works.	<i>Mr. P. Praveen, K. Chaitanya Pranith, B. Satish ,S. Sri Harsha</i>	136
134	ICMIT/DS45	Investigation of Stresses and Fatigue Life of Crankshafts with Oilways using Finite Element Fatigue Analysis	<i>Prashant Rawat, Anant Prakash Agarwal, Shiv Narayan Prajapati</i>	137
135	ICMIT/DS46	Design Optimization and Fatigue Behavior of Crankshafts with Hollow Crankpins Subjected to Torsional and Bending Loads using FEM	<i>Prashant Rawat, Anant Prakash Agarwal, Shiv Narayan Prajapati</i>	138
136	ICMIT/DS47	Selection of Flywheel Materials Using Mechanical Properties and Weighted Decision-Making Techniques	<i>Dinesh Sarkar, Rajeev Ranjan</i>	139
137	ICMIT/DS48	The Science Behind Prince Rupert's Drop: A Review of Fracture Mechanics	<i>Karan B. Khare, Subim N. Khan, Pavankumar R. Sonawane, Pravin R. Kale</i>	140
138	ICMIT/DS49	Optimized Design Approaches for Enhanced Performance of Brake Pedals: A Review	<i>Aditya Sonavani, Ashish Kumar, Pavankumar Sonawane</i>	141
139	ICMIT/DS52	Design and Prototyping of Advanced Driver Assistance System	<i>Lakhan Chavan , Prathamesh Vasekar, Gayatri Mane, Prajwal Pawar , Dr. S. S. Pimpale, Dr. Pruthviraj D. Patil</i>	142
140	ICMIT/DS53	Finite Element-Based Comparative Analysis of Core Loss in PMSMs Using Various Magnetic Materials	<i>Ajay Anand, Raja Gandhi, Alok Ranjan, Sadhan Gope, and Rakesh Roy</i>	143
141	ICMIT/DS54	Biomedical Scaffolds from 3D-Printed PLA/MgO Composites: A Novel Approach	<i>Ravinder Singh, Rachin Goyal, Manjit Singh, Gurkirat Singh, Sanjeev Sharma</i>	144
142	ICMIT/DS55	Assessing the Crashworthiness of 4th Generation Pressure Vessels in Hydrogen Cars Using Finite Element Analysis	<i>Arnav Gupta, Aditya Kumar Singh, Md Zunaid</i>	145
143	ICMIT/DS56	Effect of Insoles on Kinetics Responses during Walking	<i>Rizwan Ahmad, Shailendra Singh Chauhan, Madhusudan Pal, Neha Gupta</i>	146
144	ICMIT/DS57	Deep Learning-Based Weld Defect Detection in Underwater Welding on AH36 Using Convolutional Neural Networks	<i>Ganta Venkateswara Rao, Saurav Suman</i>	147
Track 4: Industrial Technologies & Miscellaneous				
145	ICMIT/IT01	Performance investigation of highly sensitive Si3N4 material based MRR sensor for biosensor applications	<i>Pankaj Kumar Ray Suman Ranjan, Vikash Kumar, Alok Kumar</i>	149
146	ICMIT/IT02	Piezoelectric Energy Harvesting: A Review	<i>Atul</i>	150

147	ICMIT/IT04	Robotic Bidirectional Path Planning in Static and Dynamic Environment	<i>Narayan Kumar, Amit Kumar, Ankit Kumar</i>	151
148	ICMIT/IT05	Impact of Membrane Housing Height on Oil Retention and Filtration Efficiency in Quartz-Based Separation Systems	<i>Ramanamane N, Pita M</i>	152
149	ICMIT/IT06	Effect Of Graphene Nanoparticles on Sn-Zn-Bi Solder Alloy	<i>Ajall Hussain A.H , Sreekumar C.R</i>	153
150	ICMIT/IT08	Multi Point Motion Planning of Service Robot for Cleaning and Sanitization in Dynamic Obstacles Condition Using A-Star Algorithm	<i>Anup Kumar, Narayan Kumar, Tarun Dev</i>	154
151	ICMIT/IT10	Credit Card Fraud Detection through Machine Learning Techniques: A Comparative study	<i>Raju Ranjan Kumar, Dr. Priyanka Srivastava</i>	155
152	ICMIT/IT11	Optimization of process parameters of fused deposition modeling to improve quality of PC-ABS part using genetic algorithm	<i>Naveen Kumar Suniya1, Arvind Kumar Verma1</i>	156
153	ICMIT/IT12	Towards Sustainable Development: Prioritizing Lean, Green, Circular, and Sustainable (LGCS) Practices to Mitigate Fruit and Vegetable Wastage Using AHP	<i>Irfan Haider, Dr Vikas Kumar Choubey, Dr. Mukesh Kumar</i>	157
154	ICMIT/IT19	Development of Electrochemically Deposited and Annealed Surfaces for the Study of Wettability through Evaluation of Contact Angle, Contact Angle Hysteresis, and Roll-Off Angle	<i>Vishwajeet Kumar, Abhik Majumder, Jyoti Kumar, Alok Ranjan</i>	158
155	ICMIT/IT21	Electromechanical characterization of VHB-like dielectric elastomer-based out-of-plane actuator for soft robotics	<i>Deepak Gupta, Raj Kumar Sahu</i>	159
156	ICMIT/IT22	Mechanical and Dielectric Properties of Porous Elastomer for Flexible Sensors	<i>Tejasvi Anant1, Raj Kumar Sahu1,</i>	160
157	ICMIT/IT23	A Review of Photostrictive Actuators	<i>Ankit Sharma, Manish Kumar Sharma, Rajeev Kumar, Himanshu Pathak, Kumar Sambhav</i>	161
158	ICMIT/IT25	Infrared Suppression Strategies for Warships: An Analytical Efficacy Evaluation	<i>Khagesh Kumar Choudhary, Dr. P K Sharma, Capt. (Dr.) AVSN Murty</i>	162
159	ICMIT/IT26	Revolutionizing Agri-Food Supply Chain Efficiency With Blockchain Technology	<i>Shraddha Agrawal, Sanjiv Kumar Tiwari, Ritesh Kumar Singh, Anshu Anand</i>	163
160	ICMIT/IT27	Haptic Feedback based Machine Learning models for improving accuracy in surgical Robots	<i>Mayank Mishra, Rishabh Bharadwaj, Netavya Kapoor, Krovvidi Srinivas</i>	164
161	ICMIT/IT28	Design and Development of a Low-Cost Bioelectrical Impedance Analysis System Using Arduino	<i>Km Brajesh, Abha Kumari, Kirti Pal, Munna Khan</i>	165
162	ICMIT/IT29	Simulation of a LiDAR-Based Obstacle Avoidance Robot in a Hospital Environment using Webots: Enhancing Efficiency and Safety	<i>Emmanuel James Singh, Anuj Kumar Sharma</i>	166
163	ICMIT/IT30	Performance Evaluation of Swarm Robotics for 3-D Mapping	<i>Rajan Singh, Anuj Kumar Sharma</i>	167
164	ICMIT/IT31	A Case Study On Performance Improvement Using Dmaic Methodology Of Six Sigma	<i>Suvendhu Ghosh, Dr. Rupali</i>	168

165	ICMIT/IT32	Artificial Intelligence in Composite Materials: A Comprehensive Review	<i>Snehal Jadhav, Pavankumar Sonawane, Mayur Jakhete, Subim Khan</i>	169
166	ICMIT/IT33	Prediction of Surface Roughness for AlSi10Mg Fabricated via Laser Powder Bed Fusion using Machine Learning Models	<i>Kunal Dewangan, T. Ram Prabhu, G. Srinivasu</i>	170
167	ICMIT/IT34	Design of VR system to study the internal object in 360° direction	<i>Nitesh kumar, Rashi Tyagi, Surendra Kumar Yadav</i>	171
168	ICMIT/IT36	Experimental Investigation on Delamination of Waste PCBs Using a DMF-Based Solvent	<i>Shayam Anwer, Akansha Raj, Arunabh Meshram, Anuj Kumar Sharma, Rabesh Kumar Singh</i>	172
169	ICMIT/IT38	Deep Convolutional Neural Network Models for Adaptive Anomaly Detection in the Face of Zero-Day Network Attacks	<i>Pragya Bajpayi, Sandeep Mathur, Madhu Sharma Gaur</i>	173
170	ICMIT/IT39	Green-Waste Composting: Sustainable Approach	<i>Amar Nath Dey, Chandan Chatteraj, Subrata Samanta, Pabitra Mandal</i>	174
171	ICMIT/IT40	Design Analysis & Fabrication of Autonomous mobile robot for material Handling in assembly line	<i>I.Rekha, M.Siriza and T.Srilatha</i>	175
172	ICMIT/IT42	Context-Aware Object Detection In Hospital Environments Using YOLOv11	<i>Piyush Awasthi, Anuj Kumar Sharma</i>	176
173	ICMIT/IT43	AI-Driven Object Detection for Sustainable Surveillance Using YOLOv12	<i>Shivam Singh, Anuj Kumar Sharma</i>	177
174	ICMIT/IT/44	Transition from localization to a deep localization strategy: A pathway towards sustainability	<i>Ankit Agrawal, S. K. Pradhan, B.K. Rout Nitesh Kumar</i>	178
175	ICMIT/IT46	Depression Detection from Twitter Data Using Machine Learning	<i>Shweta Singh, Siddharth Yadav, Usha Sharma, Rabesh Kumar Singh</i>	179
176	ICMIT/IT50	Optimal PMU Placement Using Binary PSO Based Optimization Algorithm for Multi-Objective Constrained Functions	<i>Santosh Kumari Meena, Dr. Akhil Ranjan Garg</i>	180
177	ICMIT/IT51	A Deep learning enabled intrusion detection system for Internet of things “IOT” environment	<i>Siddhant Banger, Shashwat Dash, Sashwat Sinha, Jitendra N Rai</i>	181

Track 1

Materials Processing & Manufacturing Techniques

ICMITMM02

Development of composite component of Onyx Pro and carbon fibre through 3d printing

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ABSTRACT

This research paper focuses on the development of a novel composite material combining onyx pro and carbon fibre through advanced 3D printing techniques. The study investigates the mechanical properties, durability, and applications of this innovative composite, aiming to enhance the performance characteristics of traditional materials. By leveraging the strengths of both onyx pro, known for its toughness and aesthetic appeal, and carbon fibre, recognized for its high strength-to-weight ratio, the research outlines the fabrication process, optimization parameters, and post-processing techniques. Experimental results demonstrate significant improvements in tensile strength, flexibility, and thermal resistance, positioning the onyx pro-carbon fibre composite as a viable candidate for various industries, including aerospace, automotive, and consumer products. This work contributes to the growing field of additive manufacturing and material science, providing insights into the potential of hybrid materials in meeting the demands of modern engineering applications.

Keywords: 3D Printing, Onyx Pro, Additive Manufacturing, FDM, Fibre composite

ICMIT/MM03

**Experimental Investigation and Mechanical Characterization of Al7075 with Nitinol Powder
Metal Hybrid Composite by Stir Casting**

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ABSTRACT

In the manufacturing field, there is a need for advanced materials to benefit from newly developed advanced technologies. Aluminium matrix composites (AMCs) developed with micro/nano reinforcements emerge as an attractive candidate for innumerable applications, including automotive, aerospace, electronics, biomedical, and many more, owing to their high strength-to-weight ratio and outstanding mechanical characteristics. This experimental analysis involved the fabrication of lightweight metal matrix composites made up of Aluminium 7075 and Nitinol powders as reinforcements via stir casting. Aluminium 7075 with Nitinol leads to materials that are better suited for demanding manufacturing applications, enabling innovative designs and enhanced performance. Among the various nonferrous materials, Aluminium based materials are unique because of their corrosion resistance, specific weight, and strength. This material will become more promising if it is hybridized using a variety of reinforcements. The fabricated hybrid composites subsequently mechanically characterized. This mechanical characterization involved evaluating the tensile strength, yield strength, compression strength, and hardness of samples prepared with different weight percentages of reinforcements. The experiment results exhibit that the hardness, ultimate compression/tensile strength, and yield strength of the AMCs Compared to the unreinforced aluminium, the increase occurred while elongation and the coefficient of thermal expansion (CTE) decreased. The fabricated Aluminium 7075 Nitinol nanocomposite exhibited significantly higher damping capacity and elastic modulus compared to pure Aluminium. This can also be due to the superior matrix reinforcement interfacial bonding.

Keywords: Aluminium 7075, Nitinol powders, Stir casting, Metal matrix composites, Mechanical characterization.

ICMIT/MM04

Optimization of Friction Stir Processing Parameters on the Fabrication of Al6063 Reinforced Hybrid Composite Material Using Response Surface Measurement and Desirability Parameter

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ABSTRACT

Friction stir processing (FSP) is an innovative solid-state technique in which the material remains unmelted and un-recast, with process parameters such as tool rotational speed, tool feed, and axial force significantly influencing the mechanical properties. Recent studies have included metal oxides or carbides in the FSP process, yielding surface composites of aluminum alloys. In addition to the process parameters, it is posited that the composition of additives may influence the mechanical properties. Traditionally, statistical analyses focused on modeling process parameters to enhance the response behavior of composites. In this investigation, however, both process parameters (tool rotational speed and tool feed) and composition parameters (SiC wt.% and Graphene wt. %) were incorporated to achieve optimal mechanical properties of the composites. The research involves the synthesis of AA6103 aluminum composites by the modulation of tool rotational speed and feed, while concurrently adjusting the concentration of reinforcement additives (SiC wt.% and Graphene wt.%). The ultimate tensile strength, flexural strength, and hardness of the produced composites will be evaluated using a universal testing machine and a Vickers hardness tester. The central composite design technique and mathematical model will be developing using response surface methodology.

Keywords: Response Surface Methodology, Al6063, Mechanical properties, Process Parameter.

ICMIT/MM07

Development of ECM Set-up and Parametric Optimization of Machining Parameters for High Strength Alloy

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ABSTRACT

In the present work, an ECM (Electrochemical Machining) setup was developed, and the machining parameters were optimized for high-strength alloys. ECM is an advanced, non-traditional machining process specifically designed for materials that are difficult to machine using conventional methods. It offers several advantages, including a high material removal rate, minimal tool wear, and superior surface finish. This process is particularly beneficial for machining hard and exotic metals such as titanium alloys, Inconel, and high-nickel, cobalt, and rhenium alloys. Additionally, ECM is effective in creating intricate geometries, including internal cavities and complex contours, which are challenging to achieve through traditional machining techniques. The ECM process operates by advancing a positively charged workpiece (anode) towards a negatively charged tool (cathode) while a controlled flow of pressurized electrolyte is introduced into the machining zone. As electrons transfer between the tool and the workpiece, the material dissolves in a controlled manner, shaping the workpiece accordingly. The byproducts, primarily metal hydroxides, are removed through the electrolyte flow. Often referred to as "reverse electroplating," ECM shares similarities with Electrical Discharge Machining (EDM) in its use of an electrical process to remove material. However, unlike EDM, ECM does not cause tool wear, making it an efficient and cost-effective alternative for precision machining. In this work, Ti-6Al-4V alloy was machined using the developed ECM setup, and the machining parameters were optimized using the Taguchi L9 array. The highest material removal rate (MRR) achieved was 0.128 g/min under the optimal conditions of 140 V, 30% electrolyte concentration, and the addition of 1% alumina powder to the electrolyte solution.

Keywords: Electrochemical machining (ECM), Ti6Al4V, MRR,

ICMIT/MM08

Microstructure and Mechanical characterization of Mg-based surface composite produced by friction stir processing using titanium oxide and graphene as reinforcement

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ABSTRACT

Friction stir processing (FSP) is an eco-friendly process capable of inducing a wide range of mechanical and microstructural changes in a specific area of interest. FSP uses the stirring motion of the rotating tool on the workpiece to improve its properties. The goal of this research is to develop a lightweight material with enhanced and improved properties that are required for various engineering applications. Friction stir processing (FSP) was applied to fabricate Magnesium (Mg) based alloy with TiO₂ and graphene reinforcement to predict the mechanical properties at 95% confidence level. The experiments were analysed based on response surface methodology to influence the process parameters volume percentage reinforcement, tool rotation speed, tilt angle on the MMCs. Microstructures were investigated the presence of the reinforcement particles on the MMCs through Optical microscopy, Field emissions scanning electron microscopy and Energy Dispersive X-ray mapping. The after experimentation and result analysis the ultimate tensile strength is 291 MPa and microhardness 124.22 obtained using process parameters tool speed, tilt angle and reinforcement % by weight are 2100 rpm, 10 and 6%, respectively.

Keywords: FSP, Reinforcement, TiO₂, Graphene

ICMIT/MM10

Recent Challenges in machining of metal matrix composites- A review

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ABSTRACT

Composites play a crucial role across industries like automotive, defence, and biomedical due to their superior properties. This review focuses on the machining of different composites examining their characteristics, parameters, materials, and advanced machining methods. The paper explores the advancements in machining of composites with reinforcements like sugarcane ash, eggshell, orange peel, and rice husk ash for eco-friendly applications in construction, automotive, and aerospace. Various machining processes such as turning, milling, drilling, grinding are commonly used for machining. Apart from traditional machining processes non conventional machining methods such as electrical discharge machining, laser machining, abrasive jet machining, etc. are discussed, along with recent advancements like hybrid machining such as electro discharge diamond grinding. Insights into composites machining process parameters and optimization processes with predictive modelling contribute to a comprehensive understanding of their potential applications and material engineering advancements.

Keywords: Composites, metal matrix composites, polymer matrix composites, machining

ICMIT/MM11

A Review of the Literature on the Mechanical Properties of Epoxy Composites Reinforced with Natural Fibers

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ABSTRACT

Natural fibers are abundant, biodegradable, renewable, and cost-effective resources derived from various plant and animal sources. Their eco-friendly nature and mechanical properties make them ideal reinforcement materials in polymer composites. When combined with epoxy resin, these fibers enhance tensile strength, flexural strength, impact resistance, and thermal stability, offering lightweight and sustainable alternatives to synthetic materials. This study compares the mechanical performance of epoxy resin reinforced with different natural fibers and agricultural by-product particles, providing insights into their suitability for industrial applications. By offering a comprehensive overview, this review aims to foster further research and innovation in natural fiber-reinforced composites, promoting the development of sustainable and high-performance polymer materials.

Keywords: Natural fiber, Agricultural residue, Particulates, Epoxy, Mechanical properties

ICMIT/MM12

**Fabrication and Characterization of AA 6082/B₄C Pista Shell Powder Hybrid Composite
Developed by Stir Casting Technique.**

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ABSTRACT

Metal matrix composites (MMCs) have gained attention due to the rising demand for materials that are lightweight, strong, and thermally stable. This study explores the fabrication and characterization of hybrid composite using AA 6082/ B₄C/ pista shell composite by stir casting method. Comprehensive mechanical and thermal analyses—such as tensile strength, hardness, wear resistance, and thermal conductivity tests—are conducted to evaluate the mechanical performance of the fabricated composite. The results indicate significant enhancement in both mechanical and thermal properties, supporting potential applications in the automotive and aerospace sectors. This work highlights the viability of incorporating industrial/agricultural waste as reinforcement in metal matrix composites, contributing to sustainable material development.

Keywords: Hybrid composite, Aluminum 6082, Boron carbide, Pista shell powder, Stir casting, Thermal analysis.

ICMIT/MM14

Development and Performance Evaluation of Green Aluminium Metal Matrix Composites Reinforced with Graphene Nanopowder and Marble Dust

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ABSTRACT

Sustainable and high-performance composites have gained sufficient attention in recent years. In this regard, an attempt is made to develop aluminium metal matrix composites reinforced with graphene nanopowder and marble dust by the stir casting technique. Marble dust powder was kept constant at 0.5% in all composite samples while graphene varied between 0.5% and 2.5%. This unique synergic combination of reinforcement shows improvement in the mechanical and tribological properties of composites. The optimum properties of composites are recorded at 0.5% marble dust and 2% graphene nanopowder-reinforced samples. A maximum drop of 41% in wear rate was observed at 60 N loading conditions at 0.63 m/s speed within a range of 20-100 N loading and 0.63 and 1.89 m/s rotational speed conditions. Specific wear rate also shows the same trend. Coefficient of friction values also decreased in composite samples when compared to as-cast aluminium alloy. Toughness values decreased sharply in composite samples; however, a little increase in value is found as reinforcement increases, reaching 9 J in the 0.5% marble dust and 2% graphene samples, which is 25% less than the as-cast alloy sample. Microhardness is improved by 56.9% and tensile strength by 32.5% in the 0.5% marble dust and 2% graphene-reinforced sample, and elongation decreased by 33.7% in the 2.5 % graphene-reinforced sample. SEM characterization of fractured tensile specimens and toughness samples reveals the nature of the failure of composites. SEM of wear surface shows wear behaviour of composites. Overall findings demonstrate that the marble dust and graphene powder hybrid reinforcement offer an optimal balance between mechanical and tribological properties along with sustainability, paving the way for next-generation green metal matrix composites.

Keywords: Sustainable composites, graphene nanopowder, marble dust, mechanical and tribological properties, Aluminium metal matrix composites.

ICMIT/MM15

A Review of Underwater Local Dry Welding: Technological Progress, Future Prospects

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ABSTRACT

Underwater local dry welding (ULDW) a vital alternative for underwater joining and repairs, effects a balance between the operational flexibility needed in wet conditions and the weld quality of dry hyperbaric techniques. The technological development of ULDW is reviewed in this paper, with particular attention paid to process foundations, enclosure systems, gas distribution, and advanced ULDW welding techniques like ultrasonic assisted, laser beam and Flux cored TIG welding by lowering frequent underwater weld flaws such porosity and hydrogen-induced cracking, localized dry zones enhance arc stability and shielding. Finally, a roadmap that highlights prospects in robotic integration, AI-enhanced welding, and sustainability is place ahead with inclusion of challenges and limitations. Positioning ULDW as a game-changing method for offshore maintenance and underwater building in the future is the objective of this review.

Keywords: Underwater welding, local dry welding, mechanical properties, future trends.

ICMIT/MM19

Photoluminescence properties of Eu^{3+} doped $\text{Ba}_2\text{Y}_{0.66}\text{V}_2\text{O}_7\text{Cl}_2$ red phosphors for white-LEDs**Sruthi S. Nair¹, Puja Kumari^{2*}, Subrata Dasa³**¹*Materials Science and Technology Division, CSIR-National Institute for Interdisciplinary Science and Technology, Thiruvananthapuram, Kerala-695019, India*²*Department of Physics, New Government Polytechnic Patna, Bihar 800013, India*³*Academy of Scientific and Innovative Research (AcSIR), Ghaziabad-201002, India***Corresponding Author Email ID: pp.phy.ism@gmail.com***ABSTRACT**

Here, the photoluminescence excitation and emission mechanism of undoped and Eu^{3+} doped $\text{Ba}_2\text{Y}_{2/3}\text{V}_2\text{O}_7\text{Cl}_2$ phosphors have been investigated in detail along with the influence of Eu^{3+} concentration on the luminescence behaviour. The undoped and a series of Eu^{3+} doped $\text{Ba}_2\text{Y}_{2/3}\text{V}_2\text{O}_7\text{Cl}_2$ samples have been produced via the solid-state synthesis route at high temperatures. The SEM image shows that the particles formed are non-uniform and agglomerated. The reflectance spectra of undoped and Eu^{3+} doped $\text{Ba}_2\text{Y}_{2/3}\text{V}_2\text{O}_7\text{Cl}_2$ samples exhibited a broad absorption band around 300 nm owing to the charge transfer of $\text{O}_2^- \rightarrow \text{V}_5^+$ within VO_4^{3-} tetrahedron. The excitation spectrum of the undoped sample monitoring at 463 nm consists of a shoulder at 300 nm with a maximum excitation peak at around 320 nm. While, the emission spectra recorded under the excitation maxima at around 320 nm shows a broad emission band ranging from 380 to 580 nm and peaked at around 463 nm. However, Eu^{3+} doped samples showed strong red-orange emission lines of Eu^{3+} ions under the UV and blue excitations ranging from 320 nm to 464 nm. The present results also demonstrate elevated Eu^{3+} emission intensity because of the strong energy transfer phenomenon from the host to Eu^{3+} ions. The photoluminescence intensity was found to increase along with the increase in Eu^{3+} concentration. The emission intensity was observed to be maximum for $\text{Ba}_2\text{Y}_{0.44}\text{Eu}_{0.2}\text{V}_2\text{O}_7\text{Cl}_2$ compositions. The investigated results indicated that the $\text{Ba}_2\text{Y}_{0.44}\text{Eu}_{0.2}\text{V}_2\text{O}_7\text{Cl}_2$ phosphor is a suitable red-emitting phosphor for making white light emitting diodes under near-UV and blue excitations.

Keywords: Phosphors, orthovanadates, Eu^{3+} activation, red-orange emission, white-LEDs

ICMIT/MM20

Comparative Study Of Mechanical And Functional Properties Of Al6061-SiC-Cenosphere And Al6061-WC-Cenosphere Metal Matrix Composites With As-Cast Al6061

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ABSTRACT

Al6061 is a strong and versatile aluminum alloy extensively used in various industrial sectors. This research focuses on the fabrication and evaluation of Al6061-based metal matrix composites reinforced with Tungsten Carbide and Cenosphere, as well as Silicon Carbide and Cenosphere, using the stir casting method. The primary goal is to explore their suitability for automotive engineering applications. Mechanical tests revealed notable enhancements in hardness, tensile strength, and compressive strength compared to as-cast Al6061. Additionally, wear resistance improved and density reduced with the incorporation of reinforcements. These improvements demonstrate the composite's potential for high-performance automotive uses, where strength, durability, and lightweight properties are crucial.

Keywords: Cenosphere; Metal matrix composites; Silicon Carbide; Tungsten Carbide

ICMIT/MM21

Investigation of mechanical properties of 304L stainless steel welds using TIG, MIG and dual-sided TIG-MIG welding methods

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ABSTRACT

Welding of stainless steel, particularly 304L grade is widely used in industries demanding high corrosion resistance and mechanical strength. But the quality of welded joint depends not only on the input process parameters but also on the different welding methods. The present work represents the study of effect of different welding methods on the joint quality / mechanical-metallurgical characteristics of AISI 304L stainless steel specimens. Three different welding techniques namely Tungsten Inert Gas (TIG), Metal Inert Gas (MIG) and a combined dual-sided TIG-MIG welding approach have been applied during the preparation of welded samples. Butt joint configuration has been prepared using this three welding methods and joint quality is estimated by ultimate tensile strength and correlated with hardness and microstructure. Due to the development of very fine skeletal δ -ferrite grains in the weld zone and mild precipitation, the results show that the dual-sided TIG-MIG welding approach demonstrates the maximum joint efficiency when compared to individual TIG and MIG welding processes. In all situations, the weld metal's hardness value is found to be significantly higher than that of the base metal and the HAZ zone, and the MIG welding process shows the highest hardness value when compared to other techniques.

Keywords: TIG Welding, MIG Welding, Joint Quality, Hardness, Stainless Steel

ICMIT/MM22

Comparative Study Of AlCrFeCoNiZn HEA Reinforced Composite With Ascast Aluminium 6061

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ABSTRACT

Aluminum 6061 is well-known for its lightweight characteristics, robust strength, and excellent corrosion resistance, making it a favored material in structural and transportation sectors. In contrast, high-entropy alloys (HEAs), made up of several primary elements, are recognized for their outstanding mechanical and thermal properties. This study involved the synthesis of a nanocrystalline AlCrFeCoNiZn HEA through high-energy ball milling. X-ray diffraction (XRD) analysis confirmed the formation of a face-centered cubic (FCC) structure and a uniform particle distribution. Mechanical testing demonstrated considerable improvements in hardness, tensile and compressive strength in the HEA-reinforced Al6061 matrix composite produced via stir casting, when compared to the as-cast Al6061. Furthermore, the addition of reinforcements led to enhanced wear resistance. These advancements underscore the composite's potential for high-performance applications in both aerospace and automotive industries, where a balance of strength, lightweight, and durability is crucial.

Keywords: Aluminium 6061; High-entropy alloy; Metal matrix composite; Stir casting

ICMIT/MM24

Optimizing Deposition Conditions to Improve Surface Roughness and Hardness of Electroless Ni-P Coatings**Sameer Lamichaney^{1*}, Rupam Mandal¹, Subhasish Sarkar², Buddhadeb Oraon¹ and Gautam Majumda^{1,3}***Department of Mechanical Engineering, Jadavpur University, India, 700032**Assistant Engineer, Public Health Engineering, West Bengal, India**Techno India University, West Bengal, India, 700091***Corresponding Author Email ID: sameer.ccct@gmail.com***ABSTRACT**

This study aims to enhance the surface quality and mechanical properties of copper substrates by applying an electroless Ni-P binary alloy coating. A Taguchi L27 design of experiments was employed to systematically investigate the effects of process parameters. The concentrations of Sodium Hypophosphite, Nickel Sulphate, and the operating temperature were varied according to the Taguchi methodology to develop predictive models for surface roughness and microhardness of the coated samples. Analysis of Variance (ANOVA) revealed that the developed models are statistically significant, with R^2 values exceeding 0.8 and p values below 0.005, indicating strong correlations between input parameters and responses. The results demonstrate that the proposed models effectively capture the influence of processing variables. Contour plot analysis further suggests that lower concentrations of both Nickel Sulphate and Sodium Hypophosphite contribute to achieving smoother surfaces and enhanced hardness in the coatings. The coating deposited under optimal conditions, as confirmed by Energy Dispersive X-ray (EDX) analysis, revealed the successful deposition of Nickel and Phosphorus on the copper substrate. Scanning Electron Microscopy (SEM) of the optimally coated surface showed a characteristic cauliflower-like morphology of the Ni-P electroless coating, which contributes to the observed enhancement in hardness.

Keywords: Electroless Ni-P coating, Taguchi, Surface roughness, Micro-hardness, ANOVA, Signal-to-noise ratio plots, contour plot

ICMIT/MM25

Synthesis and Characterization of Electroless Ni-P Coating on Brass Substrate to Improve Hardness and Surface Roughness

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ABSTRACT

Hardness and surface roughness stand out as crucial material properties and are continually evolving within the domain of surface engineering. This article is based on the improvement of surface properties. An experimental arrangement is considered to study, synthesize, and characterize the effect of electroless accumulation of Ni-P over the brass substrate. The result is produced after crossing several experimental steps, a brass alloy (Cu- 70%, Zn- 30%) cleaned by 30% HCL solution is considered a substrate and activated with Palladium Chloride (PdCl₂) solution at 55° C. Nickel Sulfate (NiSO₄.6H₂O), Sodium Hypophosphite (NaH₂PO₂), Sodium Citrate (Na₃C₆H₅O₇.2H₂O), and Sodium Acetate (CH₃COONa) these four chemical solutions are being utilized. The surface roughness is measured using a Talysurf profilometer, while the hardness is assessed with a Vickers hardness tester. The hardness and surface roughness value of substrate and Ni-P coating are measured individually for different data sets and identify the importance of coating. Finally, SEM and EDAX analyses were also done on the substrate and Ni-P coating to see the chemical changes.

Keywords: Electroless Coating; Micro-Hardness; Surface Roughness; SEM; EDAX

ICMIT/MM26

Comparative Performance Analysis of Unprocessed Quartz and Commercial Ceramic Membranes Coated with Hydrophilic Nanoparticles for Oil-Water Separation and Industrial Wastewater Reuse

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ABSTRACT

The treatment of industrial wastewater containing emulsified oils is essential for promoting water reuse and achieving environmental sustainability. Although commercial ceramic membranes are widely used for oil–water separation, their high production and maintenance costs pose significant limitations to broader adoption. This study explores the feasibility of quartz-based membranes as a cost-effective alternative. Three membrane types—raw quartz, chemically washed quartz, and hydrophilic nanoparticle-coated quartz—were developed and systematically compared to a commercial ceramic membrane. Surface morphology was characterized using Scanning Electron Microscopy (SEM), while wettability and oil affinity were assessed through oil and grease analysis. Filtration performance was evaluated using synthetic oily wastewater with oil concentrations ranging from 200 to 500 mg/L. Key performance indicators included permeate flux, oil rejection efficiency, and residual oil concentration in the filtrate. The nanoparticle-coated quartz membrane demonstrated the highest performance, achieving 96% oil rejection and a permeate flux of 64 L/m²·h, closely rivalling the 94% rejection rate of the commercial ceramic membrane. In contrast, the raw and washed quartz membranes exhibited lower rejection rates of 67% and 79%, respectively. Furthermore, the coated membrane produced the lowest residual oil concentration (11 mg/L), indicating superior separation capability. These findings affirm the potential of nanoparticle-coated quartz membranes as scalable, affordable solutions for industrial oily wastewater treatment and sustainable water recovery.

Keywords: Quartz Membranes, Oil–Water Separation, Surface Wettability, Permeate Flux, Oil Rejection Efficiency, Water Reuse

ICMIT/MM27

Microstructural and mechanical properties analysis of 3D printed polylactic acid (PLA) component

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ABSTRACT

A developing technology called 3D printing uses a 3-dimensional CAD model and a layer-by-layer additive process to create complicated geometry. The primary benefits of 3D printing based on Additive Manufacturing (AM) are mass customization, design freedom, and waste reduction. Nowadays, 3D printing is utilized in a wide range of industries across the globe, including electronics, automotive, aerospace, agriculture, and medicine, as well as other household goods. However, the limited materials and qualities of the generated parts limit the use of the AM process in big industrial sectors. This review discusses the many 3D printing techniques, materials used to make the testing specimen, microstructural and mechanical properties analysis of testing specimen (tensile and 3point bending specimen). Additionally, covered is the impact of process variables on 3D printing, including nozzle temperature, platform temperature, and layer thickness. This will assist improve the quality of 3D printed products and their applications across a range of industries.

Keywords: Additive manufacturing (AM), Computational Aided Design (CAD), 3D printing, PLA,

ICMIT/MM28

Fabrication and Characterization of Al7075 B₄C/TiO₂ Hybrid Metal Matrix Fabricated by Electromagnetic Stir Casting Process**Md Intakhab Alam, Sanjay Kumar****Noida Institute of Technology, Greater Noida, Uttar Pradesh***Corresponding Author Email ID: Sanjay.kumar@niet.co.in***ABSTRACT**

The current study investigates the use of electromagnetic stir casting to fabricate hybrid metal matrix composites based on Al7075 reinforced with titanium dioxide (TiO₂) and boron carbide (B₄C). The study methodically examines how different casting temperatures (900°C, 1000°C, and 1100°C), stirring durations (10, 15, and 20 seconds), and stirring speeds (600, 700, and 800 rpm) affect the composite's quality and homogeneity. The matrix was supplemented with reinforcement compositions that ranged from 2 weight percent to 6 weight percent of a mixed B₄C and TiO₂ mixture. It was assessed how processing parameters affected the microstructural properties, reinforcement dispersion, and overall composite integrity. The results show that the homogeneity of particle distribution and interfacial bonding are greatly improved by the ideal control of stirring temperature, speed, and duration. These findings support the creation of high-performance hybrid composites based on aluminum for use in automotive and aerospace applications by offering insightful information about the interaction between process and microstructure. Through the development of lightweight, high-strength structural materials appropriate for the automotive, aerospace, and military industries, the results contribute to the larger area of materials engineering by emphasizing the role of hybrid reinforcements and optimal casting procedures. The performance of composites may be further improved by future research into nano-sized reinforcements and post-processing techniques including heat treatment and hot isostatic pressing.

Keywords: Al7075 alloy, Hybrid metal matrix composites, Electromagnetic stir casting, Boron carbide (B₄C), Titanium dioxide (TiO₂), Stirring parameters, Microstructure, Composite fabrication

ICMIT/MM29

Experimental investigation and vibration-based condition analysis for varying load of gear box with spur gear arrangement to increase reliability

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ABSTRACT

For many industrial applications, the dependability of gearbox systems is essential, especially when spur gear combinations are involved. In order to increase the reliability of gearboxes, this study focuses on their experimental examination and vibration-based condition analysis under various load settings. Advanced signal processing techniques are used to collect and evaluate vibration data from a gearbox with spur gear arrangements by applying various load scenarios in a methodical manner. The goal of the study is to pinpoint distinctive vibration signatures connected to various operational loads and fault states. The findings show a significant relationship between load fluctuations and particular vibration patterns, which offers important information for the early identification of possible problems. These discoveries can be used to improve maintenance plans and stop unplanned malfunctions, which will extend gearbox lifespan and operating reliability. This research contributes to the field of predictive maintenance and condition monitoring, offering practical solutions for improving the reliability of mechanical systems in industrial settings.

Keywords: Gearbox reliability, Spur gear arrangement, Vibration analysis, Condition monitoring, Load variation, Predictive maintenance, Signal processing, Fault detection, Mechanical systems, Industrial applications

ICMIT/MM30

Study on Hardness Behaviour & Microstructure of Al-0.88Mg-0.35Si alloys Micro Alloyed with 0.02 wt.% Sn

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ABSTRACT

The hardness and microstructure of Al-0.88Mg-0.35Si alloy on micro-alloying with 0 wt.% and 0.02 wt.% Sn was investigated. The alloys attained maximum hardness at peak aged condition. The addition of 0.02 wt.% Sn has accelerated the peak age hardness obtaining period from 12 hrs to 6 hrs and increased the peak age hardness from 83 HV to 105 HV of the Al-Mg-Si alloy. Addition of 0.02 wt.% Sn resulted in increasing hardness of the alloy by 26.8% and 26.5% in as-cast and peak-aged conditions respectively. The microstructural study of the alloy revealed coarse eutectic plate-like Mg₂Si and coarse plate-like Al (Fe, Mn) Si particles respectively. Also, 0.02 wt.% Sn addition decreased plate-like and needle-like Fe-rich Al (Fe, Mn) Si phase in comparison to 0 wt.% Sn-added alloy. Micro-alloying Sn favoured spheroid-like Mg₂Si morphology, indicating precipitation of the Mg₂Si particles in the alloy and presence of Sn in dark refined plate-like and finer spheroid-like phase.

Keywords: Microstructure, FESEM, Micro-alloying Sn

ICMIT/MM31

Optimization of 3D printed parts using different infill patterns and densities by FDM method

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ABSTRACT

Additive Manufacturing (AM) has emerged as a transformative alternative to conventional production methods, yet the mechanical performance of 3D-printed components remains a critical area of investigation. Polylactic Acid (PLA), a biodegradable thermoplastic widely used in Fused Deposition Modelling (FDM), is favored for its printability and environmental benefits. This study evaluates the tensile properties of PLA specimens fabricated according to ASTM D638 Type I standards, focusing on the influence of infill pattern and density. A total of nine specimens featuring honeycomb and grid infill structures at 20%, 40%, 60%, and 80% densities alongside one fully solid sample were designed in SolidWorks and printed using an Olivetti 3D Desk printer. Tensile testing was performed using an MTS ACCUMEN Universal Testing Machine (UTM), and corresponding simulations were conducted in ANSYS to analyze stress distribution and deformation behavior. Results demonstrated a strong correlation between experimental and simulated data. Notably, the 80% honeycomb infill exhibited the most favorable balance between mechanical strength and weight efficiency. These findings underscore the significance of infill optimization in enhancing the structural reliability of FDM-printed PLA components, particularly in the context of structural optimization through infill manipulation.

Keywords: Polylactic Acid, FDM, 3d printed, ASTM D638 Type I, infill patterns, Additive Manufacturing, Finite Element Analysis

ICMIT/MM33

Additive Manufacturing of Sustainable Polymers: Bridging Performance Gaps with Resource-Efficient Strategies and Circular Design**Ravi Kumar*, Birendra Kumar***Department of Mechanical Engineering, Motihari College of Engineering***Corresponding Author E-mail ID: ravik.phd19.me@nitp.ac.in***ABSTRACT**

Additive manufacturing (AM), or 3D printing, is transforming how we process sustainable materials like bioplastics, natural fibers, and recycled plastics—materials that often struggle with weak mechanical performance, low heat resistance, and environmental drawbacks. This review explores how AM technologies are solving these challenges, making it possible to turn eco friendly materials into durable, functional products. By analyzing recent research, we highlight innovative approaches such as mixing bioplastics with natural additives (e.g., plant-based fibers), optimizing printing processes to save energy, and creating recycling systems that reuse waste materials. Key results show that AM can cut material waste by up to 40% and reduce carbon emissions by 30–50% compared to traditional methods, especially for small-scale production. AM also allows designers to create complex, lightweight shapes that were previously impossible, such as custom medical implants or car parts made from biodegradable materials. Real-world examples include 3D-printed building blocks using mushroom-based composites and recycled plastic filaments that perform nearly as well as new materials. However, hurdles remain, like improving the heat resistance of bioplastics during printing and scaling up recycling processes. This review emphasizes how AM supports greener manufacturing by combining renewable materials, smarter design, and less waste. It also calls for more research and collaboration to help industries adopt these technologies and meet global sustainability targets.

Keywords: Additive manufacturing, Sustainable polymers, Circular economy, Lifecycle assessment, Resource efficiency

ICMIT/MM34

Experimental Investigations on Wear Behavior of Novel Mg 5.5 Zn Bio Implant Alloy

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ABSTRACT

Magnesium-zinc (Mg-Zn) alloys are widely recognized as promising bioimplant materials due to their excellent biocompatibility, biodegradability, and mechanical properties that closely resemble those of natural bone. They are commonly employed in orthopaedic implants and temporary fixation devices. However, their application in load-bearing environments is limited by poor wear resistance. Zinc addition, up to 6 wt%, has been shown to improve wear resistance by forming hard Mg Zn precipitates. Furthermore, age hardening enhances the alloy's hardness and wear performance. In this study, a novel Mg-5.5Zn alloy was fabricated via the stir casting method and subjected to age hardening treatment. The wear behaviour of the developed alloy was evaluated under varying conditions: loads of 20 N and 40 N, sliding distances of 500 m and 1000 m, and sliding velocities of 0.5 m/s and 1 m/s using pin on disc apparatus. Across all test conditions, the age-hardened alloy demonstrated superior wear resistance. The results indicate that sliding distance and applied load are the most significant factors influencing wear performance. Interestingly, at higher loads, the as-cast alloy showed better wear resistance, while at lower loads, the age-hardened alloy outperformed the cast counterpart.

Keywords: Age hardening; Biomedical implants; Biocompatibility; Magnesium Zn alloys: Wear Resistance

ICMIT/MM35

A Comprehensive Review of 3D Printing Technologies and Their Applications**Sulaim Rizvi, Shahazad Ali*, Anant Prakash Agrawal***Department of Mechanical Engineering, Noida Institute of Engineering & Technology Greater Noida (INDIA)***Corresponding Author Email ID: shahzadali0786@gmail.com***ABSTRACT**

The fundamental change in product design and manufacturing methods has occurred due to 3D printing technology. The disruptive production method creates structures with complex geometries by successively adding layers which traditional methods cannot achieve. 3D printing innovations within medical applications have demonstrated substantial advancement of biological constructs as well as prosthetics and patient-specific implants which provide better anatomical compatibility. Due to dental applications the production speed of crowns and aligners improved resulting in shorter periods of patient wait time. The integration of aerospace technology allows orbital platforms to produce replacement components in space and reduces their need for terrestrial product delivery. The International Space Station keeps active and operational AM systems for generating products when needed. Scientific studies are active for determining how extraterrestrial regolith can serve as material to construct habitable infrastructure on lunar and Martian surfaces. The application space of 3D printing remains almost unexplored since it lacks agricultural solutions for resourceful food creation and architectural solutions for new construction systems and 4D printing technologies which enable objects to adjust due to environmental signals.

Keywords: 3D Printing, Additive Manufacturing, CAD, Dentistry, Biomedical Applications, Aerospace, Space Technology, Food Technology, Automotive Industry, Education, Technological Advancements

ICMIT/MM36

Fabrication of a 2-layer laminated steel composite by friction stir additive manufacturing

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ABSTRACT

Friction Stir Additive Manufacturing (FSAM) has emerged as a promising solid-state technique for fabricating multi-layer laminated composites with enhanced mechanical properties this is solid state method used in the fabrication of laminated composites is the friction stir additive manufacturing (FSAM) method which is a sheet lamination process in which two-layer bonding is established by applying severe plastic deformation on one of sheets. In this method, the layers are bonded to each other using the friction stir welding (FSW) process according to the principles outlined in the lap joining. In the FSAM method, a tool with pin and shoulder is plunged onto the top surface of the overlap sheets and, which moves forward resulting in the sheets bonding due to frictional heat and severe plastic deformation. In order to control interfacial reaction, pin length and tool geometrical parameters that control frictional heat and plastic deformation during the process play a prominent role. The process involves sequential friction stir welding of steel layers, ensuring metallurgical bonding without bulk melting. Microstructural characterization reveals refined grain structures due to dynamic recrystallization, while mechanical testing demonstrates improved strength and toughness compared to conventional steel.

Keywords: Friction Stir Additive Manufacturing, Laminated Steel Composite, Solid-State Processing, Microstructure, Mechanical Properties

ICMIT/MM37

Mechanical and Tribological Evolution of AA7075 Aluminum Alloy Processed via ECAP

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ABSTRACT

The mechanical and tribological properties of cast ingot AA7075 aluminum alloy were significantly enhanced through Equal Channel Angular Pressing (ECAP) conducted at room temperature using Route A and Route BC. The study aimed to evaluate the alloy's performance post-processing with a focus on tensile strength, wear behavior, and hardness. ECAP specimens (12 mm diameter and 90 mm length) were machined from as-cast billets and preheated at a temperature of 250°C for 1 hour. Simultaneously, the ECAP die was preheated at 300°C. Experimental results demonstrated remarkable improvements in tensile strength, with increases of 46.03%, 55.03%, and 58.5% for 1-pass, 2-pass Route A, and 2-pass Route BC, respectively. However, a gradual decline in ductility was observed after the first pass. Wear loss exhibited a progressive increase under applied loads of 15 N, 25 N, and 35 N, while the coefficient of friction showed localized rises in regions corresponding to minimal wear loss, indicating intricate interactions between surface hardening and contact mechanics. Vickers hardness measurements revealed substantial enhancements, with increments of 10.3%, 16.3%, and 18.8% for 1-pass, 2-pass Route BC, and 2-pass Route A, respectively. These findings confirm the efficacy of ECAP in achieving grain refinement, improving mechanical strength, wear resistance, and hardness of AA7075 aluminum alloy.

Keywords: ECAP, Grain Refinement, Microstructure, AA7075 Al

ICMIT/MM38

The impact of tool rotational speed on the weld quality of FSW- welded Polystyrene sheet

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ABSTRACT

The solid-state joining technique known as friction stir welding (FSW) is widely used for both metallic and thermoplastic materials due to its minimal energy input and ability to produce high-quality welds. In the present study, FSW was used to connect sheets of 3 mm-thick Polystyrene with the primary goal of investigating the effects of tool rotation speed on the quality of the weld. To examine their effects on the mechanical and microstructure characteristics of the weld joints, four tool rotating speeds were used. The welding procedure was carried out in a controlled environment while the tool's rotating speed was varied and other parameters remained constant. Mechanical tests, including tensile strength and hardness, as well as scanning electron microscopy, were used to assess the final joints' microstructure. The results showed a significant relationship between weld quality and tool rotational speed. An ideal range of rotating speed resulted in improved mechanical qualities, a defect-free connection, and improved material flow. Inadequate mixing or heat breakdown of the polymer caused poor weld quality when rotary speeds were set above or below the ideal range. The study demonstrates that the FSW joint quality in thermoplastic materials like polystyrene is significantly influenced by the tool's rotating speed.

Keywords: *Friction Stir Welding (FSW); Polystyrene (PS); Tool Rotational Speed; Weld Quality; Polymer Joining.*

ICMIT/MM39

Investigation on the Physical, Mechanical and Tribological Behaviour of Palf Sisal based Polymer Hybrid Nanoparticle Composites: a review

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ABSTRACT

This review presents a synthesis of recent research findings and developments in the field while offering a thorough analysis of the physical, mechanical, and tribological behaviour displayed by PALF sisal polymer hybrid nanoparticle composites. Sisal fibres, which come from the Agave sisalana plant, are becoming more and more popular as an environmentally benign reinforcement for polymer composites. The study covers a wide range of topics, such as characterisation approaches, preparation methods, and important discoveries on the manufacture, testing, and mechanical performance of PALF sisal polymer hybrid nanoparticle composites. The interfacial interactions between PALF sisal fibres reinforced with silica nanoparticle composites, including shape and tribological behaviour, are given particular attention. The impact of factors such as fibre content and surface treatments on the overall characteristics of sisal fibre composites is shown through a critical examination of the body of existing literature. In order to direct future developments in the optimisation of PALF sisal polymer hybrid nanoparticle composites for a variety of applications, the study ends by describing current issues and suggesting future lines of research.

Keywords: Sisal fiber, PALF fibre, nanoparticle, hybrid composites, physical behavior, mechanical behavior, tribological behavior.

ICMIT/MM40

Analysing Flexural, Impact, and Thermophysical properties of Biodegradable Natural Fiber Composites made of Pineapple leaf and Bamboo fiber**Chandan Kumar, Irfan Haider*, Pramod Kumar***Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur.**Corresponding Author Email ID: irfan@mitmuzaffarpur.org***ABSTRACT**

The goal of this study is to analyse the flexural, impact and thermophysical properties of biodegradable natural fiber composites (NFCs). Here NFCs is made from bamboo, pineapple leaf fiber (PALF), rattan and a composite that consisted of both rattan and bamboo (RBFC). In this study, the binder (or fiber) of varying weight fraction and alkaline treatment (NaOH) of the fibers were common factors that led to an increase in flexural strength. In fact, PALF that was also treated in NaOH at a varying weight fraction showed an impressive increase in flexural strength. The other composite, RBFC, had low water absorption for the flexibility and density of the composite as it expressed great thermal conductivity. However, bamboo fiber composites (BFC) had the highest water absorption and swelling from immersion in water. In addition, the Fourier Transform Infrared (FTIR) analysis of the samples from the RBFC also confirmed hydroxyl groups and lignocellulosic components. The study has shown the possibility of NFCs being a more sustainable alternative to synthetic composites, including their respective mechanical, thermal, and environmental advantages and benefits, such as low energy in production and biodegradability. This study can promote the NFC application in automotive manufacturers, lightweight structures as well as 3D printing applications in general. This study is aligned with the increasing global demand for eco friendly composite materials.

Keywords: Natural fiber composites, flexural strength, thermal conductivity, alkaline treatment, Eco friendly composite.

ICMIT/MM41

An Investigation on Tensile and Compressive Strength of In-house Developed ABS Material for 3D Printing

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ABSTRACT

Acrylonitrile Butadiene Styrene (ABS) filament is produced using a filament extruder and subsequently compared to commercially available ABS filament. Improvements in filament extruder technology directly contribute to enhanced material properties. The ABS filament created with this extruder is evaluated alongside its commercial counterpart. Optical microscopy analysis reveals that the extruded filament contains fewer impurities than the commercially available filament. The tensile and compression strengths of the extruded ABS filament demonstrate exceptional performance, with a compression yield strength of approximately 26 MPa and a compressive strength of 57 MPa at a strain of 50%. Tensile test demonstrates that the maximum tensile strength is 32 MPa when the layer thickness is 0.2 mm. The results indicate that the created filament extruder improves the quality of ABS material. Further research may be directed towards the growth of filaments using other materials and composites.

Keywords: Filament Extruder, Acrylonitrile Butadiene Styrene, Optical microscope, Performance, Mechanical properties

ICMIT/MM42

Synthesis And Characterization of Hydroxyapatite Extracted from Chicken Bone**Jobin C J, Nishanth S and Sajith Babu C***Department of Mechanical Engineering, Government Engineering College, Thrissur, Kerala**Email: nishanth2000@gmail.com**Email: babusajith@gmail.com***Corresponding Author Email ID: jobinchungath@gmail.com***ABSTRACT**

The global consumption of soft drinks and beverages has increased tremendously in the last two decades. These drinks, having high calories, sugar, and salts but very low essential nutrients, lead to health issues like tooth and bone decay. Hydroxyapatite is a naturally occurring mineral form of calcium apatite, which is a biocompatible material for repairing or replacing bone and also helps as dental implants. One of the most convenient methods for synthesizing HA is using chicken bones. The method includes cleaning of the bones, drying of the cleaned bones at high temperature, calcination, and characterization of the structure and purity of HA using FTIR and XRD techniques. The chicken bones are boiled for two hours in distilled water following the removal of bone marrow and excess meat. Later the chicken bones are dipped in acetone for the next two hours, and after cleaning with distilled water several times, they are heated to about 80°C by using a hot air oven. The calcination process resulting in the crystallization of HA takes place inside a muffle furnace for over 2 hours with a temperature range of about 950-1000°C. The resulting white-coloured powder is the bio- derived HA, which is then characterized using FTIR and XRD techniques.

Keywords: Hydroxyapatite, chicken bone, bio material, calcination

ICMIT/MM43

Performance Evaluation of an "Ambient-Aware Adaptive Nano-MQL for Sustainable and Objective-Based Turning Operations"

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ABSTRACT

This study evaluates the performance of an intelligent and adaptive Nano-MQL system AutoTune Nano-MQL designed for sustainable turning operations under ambient-aware control. Unlike conventional systems that rely on fixed nanoparticle concentrations and static coolant flow rates, this system dynamically adjusts both parameters in real-time, responding to machining conditions and ambient environment. The AutoTune system integrates a dual-reservoir smart mixing unit and an embedded control algorithm that interprets live data from multiple sensors, including tool-work interface temperature, cutting forces, fluid flow rate, surface roughness, and environmental conditions such as temperature and humidity. A microprocessor processes these inputs to determine the optimal nanoparticle concentration and coolant flow rate, aligning them with the selected machining objective. Operators can choose from several pre-defined modes such as optimum surface finish, minimum coolant use, economical machining, or sustainable operation. Each mode triggers a distinct decision logic, ensuring tailored nanofluid delivery for the desired performance outcome. Experimental validation during turning operations of AISI 304, demonstrated significant improvements in machining quality, tool wear, thermal stability, and coolant efficiency compared to conventional MQL and dry cutting. The system showcases a major step toward smart, sustainable, and data- driven manufacturing environments.

Keywords: Minimum Quantity Lubrication (MQL), Nanofluid-based Cooling, Adaptive Lubrication System, Real-time Process Control, Dual-Reservoir Mixing, Dynamic Nanoparticle Concentration, Cutting Fluid Optimization, Precision Machining, Ultrasonic Agitation, Peristaltic Pumping System, Mist Atomization

ICMIT/MM44

Design and Additive Manufacturability of Voronoi based Bone scaffold**Gourav Kumar*, Hemant Kr. Choudhary***Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur, Bihar, India.***Corresponding Author Email ID: imgouravkr@gmail.com***ABSTRACT**

The porous scaffold can mimic the porous structure of natural bone for bone-tissue regeneration. The Voronoi tessellation method is used to generate bone scaffold design with controlled porosity, pore size and interconnectivity. This gives both the mechanical strength and biological performance. Parametric design variations were analyzed to balance the structural strength and permeability which is critical for bone-tissue growth and nutrition transport. The bone scaffold ($10 \times 10 \times 10$ mm) with controlled porosity ($P \% = 60\%$) and mean pores size ($D_p = 0.600$ mm) was designed and manufactured with a biocompatible material (Onyx Micro carbon fiber filled nylon) using Continuous Fiber Fabrication (CFF) method. Manufacturing challenges, including Resolution limitation, Surface finish, Support structure requirements were analyzed to assess the manufacturability of more complex Voronoi geometries. Mechanical testing and computational simulations evaluated the scaffolds' compressive strength and stiffness. In vitro study assessed biocompatibility and osteogenic potential. Results shows that Voronoi-based scaffolds achieve a favourable combination of porosity (60-80%) and mechanical performance (comparable to trabecular bone), with additive manufacturing (AM) enabling precise replication of intricate designs. This work highlights the potential of Voronoi-based scaffolds for bone tissue engineering (BTE) and provides insights into optimizing design parameters and manufacturing processes for clinical applications.

Keywords: Bone-tissue Regeneration, Voronoi tessellation, Continuous Fiber Fabrication (CFM), Trabecular bone, Additive manufacturing (AM).

ICMIT/MM45

Design and development of quality report from manufacturing component using image processing**V. Rakesh, Ruchir Shrivastava*, CH. Nithin kumar, K. Manoj Reddy***Department of Mechanical Engg, Hyderabad institute of technology and Management, Hyderabad, India***Corresponding Author Email ID: ruchirshrivastava.mech@hitam.org***ABSTRACT**

Accurate dimensional measurement of manufacturing components under mechanical load is vital for quality assurance. Conventional methods—CMMs, laser scanners, and strain gauges—though precise, are often costly, contact-based, and environment-sensitive. This study presents a real-time, non-contact measurement system using computer vision and deep learning. A monocular camera setup records video of the test article, analyzed using YOLO object detection and OpenCV for edge detection, contour analysis, and feature extraction. Principal parameters such as length, width, and deformation are computed precisely, with camera calibration ensuring real-world scaling through pixel-to-millimeter conversion. Unlike laser-based systems, the proposed method tolerates alignment variation and operates in typical shop-floor conditions. It is computationally efficient, scalable, and significantly cheaper than 3D scanning or photogrammetry. The continuous video stream enables time-series analysis, aiding structural integrity assessments. YOLO integration adds automation and adaptability across diverse shapes and sizes, enhancing inspection processes in automotive, aerospace, and heavy machinery sectors. This hybrid approach—combining computer vision, AI, and video analytics—redefines industrial metrology by delivering high precision, flexibility, and automation in a cost-effective package.

Keywords: Computer Vision Metrology, Non-Contact Measurement, YOLO Object Detection, Real-Time Dimensional Analysis

ICMIT/MM48

Effect of Notched Shape and Fiber Orientation on Combined Glass and Carbon Fiber-Reinforced Epoxy Matrix Composite Laminates Under Tensile Loading – FEA and Experimental Analysis

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ABSTRACT

Recycling and reusing carbon fibers have become increasingly practical and sustainable solutions to address the challenge of carbon fiber waste. Composite materials are now widely used in the automotive and aerospace industries due to their high strength-to-weight ratio—an essential parameter for ensuring efficiency, reliability, and sustainable performance. This paper analytically and experimentally investigates the effect of notch shape and size on damage evolution and tensile failure strength in combined glass and carbon fiber-reinforced epoxy matrix (GCFRP) laminates. Using Hashin's damage criteria and Finite Element Method (FEM) simulations, the study examines the specific effects of U- and V-shaped notches and various fiber orientations (0-30-60, 0-60-0, 0-90-0) on principal stresses and elongation behavior. The results indicate that V-notched specimens with 0-60-0 and 0-90-0 fiber orientations significantly enhance the load-carrying capacity of GCFRP laminates. Furthermore, tensile strength increases when fiber orientation aligns with the direction of the externally applied load. The paper provides a detailed analysis of how notch geometry and fiber orientation influence mechanical performance.

Keywords: Sustainability; Glass-Carbon Fiber-Reinforced Polymers (GCFRPs); Notch Effect; Finite Element Method (FEM); Damage Propagation; Fiber Orientation.

ICMIT/MM51

Parametric Analysis and Optimization Wear and Friction of Composites Produced By FDM

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ABSTRACT

This study investigates the effect of key Fused Deposition Modelling (FDM) process parameters on the wear and frictional behaviour of two distinct materials—epoxy resin (M1) and aramid Kevlar (M2)—with the objective of optimizing their tribological performance. As additive manufacturing becomes increasingly important in producing functional and load-bearing components, understanding how parameters such as layer thickness, infill density, print speed, and nozzle temperature influence individual material behaviour is essential. Both materials were fabricated independently using FDM and tested under dry sliding conditions using a pin-on-disc tribometer in accordance with ASTM G99 standards. A Design of Experiments (DOE) approach utilizing Taguchi's L9 orthogonal array was employed to analyse the influence of process parameters on wear rate and coefficient of friction (COF) for each material. The results revealed that epoxy resin (M1) demonstrated higher wear resistance, particularly influenced by nozzle temperature and infill density, whereas aramid Kevlar (M2) exhibited a lower coefficient of friction, with frictional performance more sensitive to layer thickness and print speed. Optimization was carried out using Signal-to-Noise (S/N) ratio analysis for both materials separately, identifying the ideal parameter settings for minimizing wear and friction. Predictive regression models were developed for each material, showing strong correlation with experimental results. This research provides critical insights into the tailored optimization of FDM parameters based on the specific material being used, supporting material-specific applications in tribology and functional component design.

Keywords: FDM, Tribology, Wear, Friction, Epoxy Resin, Aramid Kevlar, Taguchi Method, Pin-on-Disc, Parametric Optimization

ICMIT/MM52

Comparative experimental study of various powder coating materials under cyclic corrosion aging test ISO 12944-6:1998**Sandesh Anant Kale*, Dr. Mangesh A. Ahire, Dr. P. R. Sonawane, Dr. S. N. Khan**¹Department of Mechanical Engineering, Rajarshi Shahu College of Engineering, Tathawade, Pune, India*Corresponding Author Email ID: sandesh.pune@gmail.com**ABSTRACT**

Rusting is a crucial issue in most industries, and a significant amount of money is invested to ensure the operational safety, longevity, operational efficiency, and performance of all equipment. Most structural parts in the chemical and textile industries are exposed to extremely acidic and humid environments. There are numerous materials, such as SS316 and alloy steel, that can easily defend against corrosion; nevertheless, this solution is costly. Powder coating is an affordable alternative that can provide comparable performance to alloy steel. This research studies and compares the performance of several powders, including polyurethane (PU), fusion-bonded epoxy powder, CoR, and a dual coat method (zinc primer + pure epoxy powder). All powders are tested for accelerated corrosion in accordance with ISO 12944-6:1998. adhesion test ISO 204, pull of adhesion test ISO 4624, cyclic aging test ISO 16474, salt spray test ISO 9227, and the findings of every powder is outlined in this paper. The results indicate that dual coat (zinc primer and epoxy powder) system gives better results compared with other powders. However, cost of coating is much higher than other single coat powder which will be economical. To keep the balance between corrosion protection life and cost of production, Fusion Bonded Epoxy powder would be best suitable in chemical and textile industry application.

Keywords: Corrosion, accelerated corrosion test, adhesion, pull off adhesion, Fusion Bonded Epoxy, Powder coating, Zinc rich primer

ICMITMM53

Brief Overview on Alloy Fabrication: Opportunities and Challenges in Additive Manufacturing Process

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ABSTRACT

The additive manufacturing (AM) technique has revolutionized material design and fabrication by enabling unprecedented control over alloy composition and part geometry. Through powder bed fusion (PBF) based techniques like laser powder bed fusion (LPBF), metal binder jetting (MBJ), electron beam melting (EBM), and direct energy deposition (DED), alloys can be engineered with tailored properties to meet specific requirements for various applications. These methods allow for in-situ alloying, enabling the development of novel materials with enhanced material's bulk and surface properties such as surface roughness, chemical purity, improved mechanical strength, tribological characteristics, and bio-functionality. Among PBF techniques, LPBF has emerged as a pivotal technology in industries such as aerospace, biomedical, and energy. Recent advancements in LPBF have expanded its versatility, making it a promising approach for alloy fabrication across various applications. Therefore, this review work focuses on the current challenges and future opportunities in alloy fabrication through the LPBF technique.

Keywords: Additive Manufacturing, Powder Bed Fusion, Laser Powder Bed Fusion, Alloy Fabrication, Bulk and Surface Properties.

ICMIT/MM54

Green Synthesis of Zinc-Based Nanocomposites from Agro-Waste for Sustainable Nanomaterial Production

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ABSTRACT

Zinc-based nanomaterials are widely employed in catalysis, antimicrobial coatings, electronics, and water treatment due to their favourable reactivity and biocompatibility. However, conventional synthesis methods are often chemically intensive, relying on toxic reagents and generating hazardous waste, which limits scalability and poses environmental challenges. In this study, we report a green synthesis route for zinc-based nanocomposites using phytochemical-rich extract from waste date-palm seeds acting as both a reducing and stabilizing agent. The method eliminates the need for harmful chemicals, aligning with principles of green chemistry and sustainable manufacturing. The synthesized Zn-based nanocomposites, characterized using Field Emission Scanning Electron Microscopy (FESEM), exhibited spherical nanostructures in the range of 100-250 nm which appeared aggregated. X-ray Photoelectron Spectroscopy (XPS) and Energy Dispersive Spectroscopy (EDS) confirmed the incorporation of Zn^{2+} within an organic matrix rich in phosphate and sulphate moieties. The nanocomposite particles were evaluated for the removal of diclofenac—a widely used pharmaceutical and a micropollutant commonly found in effluents from pharmaceutical manufacturing. Batch adsorption experiments demonstrated up to 90% removal efficiency across a range of water chemistry conditions, underscoring their potential in decentralized treatment of industrial wastewater. This work presents a green approach to nanomaterial fabrication using agro-waste, with the resulting Zn-based nanomaterial offering potential for pollutant removal at the source and supporting sustainable manufacturing practices.

Keywords: Green synthesis, sustainable manufacturing, Zinc-based nanoparticles, Date seeds, Pharmaceuticals Contaminants

ICMIT/MM58

Investigating the Effects of Warping Deformation and Adhesion in Fused Deposition Modeling Printing

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ABSTRACT

In the continuously changing scenario of manufacturing industries, the demand for additive manufacturing (AM) techniques is increasing day by day. Fused filament fabrication (FFF) is one of the most affordable and widespread AM technologies used for rapid prototyping and manufacturing, but it faces challenges such as warping and poor bed adhesion. The present work discusses the development of onyx (micro carbon fiber filled nylon) based composite component reinforced by glass fiber, varying from 2-4% by density, through the FFF Mark forged Mark Two 3D printer. Further, it investigates the microstructural changes, warping deformation behavior, and dimensional accuracy in terms of the different specimens based on the various proportions of glass fiber. The process parameters layer thickness 100 microns, printing speed 280mm/s, and infill density were kept constant, but not temperature. By varying printing parameters, fiber orientation, and reinforcement pattern, the research evaluates how these factors influence thermal-induced deformation and bonding strength between layers. Warping is quantified through dimensional analysis and deformation tracking, while using ASTM standards tensile testing for fiber. Microscopic analysis is conducted to examine the quality of interlayer fusion and fiber matrix interfaces. The result shows that the uniform silk type structure with uniform dispersion of glass fiber and the warping deformation decreases with an increase in the layer thickness. However, as the printing speed increases, the warping deformation will also increase.

Keywords: Warping Deformation, Layer Thickness, Printing Speed, Glass Fiber, Dimensional Accuracy, Markforged Mark Two Fused Filament Fabrication (FFF)

ICMIT/MM59

**Parametric Study on Print Orientation, Infill Configuration, and Pattern Design for FFF-
Manufactured ABS Parts**

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ABSTRACT

The industrial sector gains high value through additive manufacturing technology which allows fast prototyping and functional production of printed parts. This research determines the joint performance between print orientation levels (0°, 45°, 90°), infill density settings (60%, 75%, 90%), and filling patterns (grid, honeycomb, gyroid) for Fused Filament Fabrication (FFF) manufactured Acrylonitrile Butadiene Styrene (ABS) parts. Tensile, flexural, and impact tests conducted by following ASTM standards revealed the complete mechanical performance assessment. Experimental data shows that the strength characteristics of tensile, flexural and impact depend immensely on the combination of print orientation with filling pattern selection. The most successful outcome was achieved by using a 45-degree orientation alongside 75% infill density with a gyroid filling pattern. The testing conditions produced tensile and flexural and impact results that reached 132% and 105% above 0° parameters and 118% and 95% beyond 90° parameters while the impact scores obtained 147% and 73% enhancement. Engineers and designers working on ABS component optimization in additive manufacturing applications can benefit from this research due to its important discoveries.

Keywords: Additive Manufacturing Fused Filament Fabrication (FFF) 3D Printing ABS, Infill Density, Filling Pattern

ICMIT/MM60

Development of Epoxy-Based Sustainable Material Using Date Palm Fiber- A Review**Mohd Affan Ashraf, Dr. Shailendra Kumar Verma*, Shiv Narayan Prajapati,***Department of Mechanical Engineering, Noida Institute of Engineering and Technology, Greater Noida, India***Corresponding Author Email ID: shailendra.niet@gmail.com***ABSTRACT**

This review article emphasizes the promise of date palm fiber (DPF), sourced from Phoenix dactylifera, as a sustainable and environmentally friendly option for various industrial uses. Plentiful, eco-friendly, and affordable, DPF presents a favorable strength-to-weight ratio, lightweight properties, and compatibility with both thermoset and thermoplastic resins, positioning it as a compelling substitute for synthetic, non-biodegradable substances. Its uses extend across construction, automotive, aerospace, and packaging sectors, aiding in minimizing environmental impact and promoting sustainable manufacturing methods. The combination of DPF with bio- epoxy resins, recognized for their biodegradability, thermal stability, and mechanical strength, further improves the properties of composites. These composites show enhanced tensile strength, durability, and heat resistance, rendering them ideal for high-performance applications. This paper examines progress in DPF composites, emphasizing chemical treatments, integration with other fibers, and improved fiber-matrix adhesion to enhance mechanical and thermal characteristics. The text covers applications like eco-friendly building materials, heat insulation, and innovative packaging options. By utilizing agricultural waste and promoting the circular economy, DPF provides a sustainable advancement in material science. This review offers an in-depth perspective on DPF's potential, tackling existing challenges and highlighting its significance in promoting green technologies and sustainable engineering approaches.

Keywords: Date Palm Fiber; Epoxy Composites; Sustainable Materials; Energy Absorption; Mechanical Properties.

ICMIT/MM61

Development of Epoxy-Based Sustainable Material Using Date Palm Fiber Mat and Graphene: Mechanical and Microstructural Characterization

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ABSTRACT

The development of sustainable composite materials has gained momentum due to growing environmental concerns and the need for biodegradable alternatives to synthetic fibers. This study focuses on the fabrication and characterization of an epoxy-based hybrid composite reinforced with date palm fiber (DPF) mat and graphene. DPF, an abundant agricultural waste, serves as a natural fiber reinforcement, while graphene is introduced to enhance mechanical performance and microstructural integrity. The composites were prepared using the hand lay-up technique, followed by post-curing. Mechanical tests, including tensile, hardness and impact strength, were conducted to evaluate performance. Microstructural characterization using Scanning Electron Microscopy (SEM) revealed effective dispersion of graphene and improved interfacial bonding between fiber and matrix. The addition of graphene significantly improved the composite's strength and stiffness while maintaining lightweight properties. The results highlight the synergistic effect of DPF and graphene in enhancing composite behavior, suggesting its potential use in automotive, construction, and low-load structural applications. This study underscores the viability of combining natural fibers with nanomaterials for creating eco-friendly, high-performance composites.

Keywords: Composite Material, Date Palm fiber, Graphene, SEM

ICMIT/MM63

Materials and Mechanics in Safety Helmet Design: A Focus on Composites and Hybrids

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ABSTRACT

Composite and hybrid materials are playing an increasingly important role in the development of safety helmets, offering improved protection, reduced weight, and enhanced durability. This paper explores the latest developments in materials and design strategies for safety helmets, with a focus on meeting key performance requirements such as impact resistance, strength-to-weight ratio, and long-term reliability. Developments in carbon fiber-reinforced polymers, epoxy-based laminates, and hybrid material systems are explored for their potential to enhance mechanical performance and align with safety standards. The paper also discusses key mechanical properties and design metrics relevant to helmet performance, including energy absorption and structural behavior under dynamic loads. It also considers real-world industrial uses, like construction, mining, and manufacturing, to highlight requirements and performance. Current challenges include balancing material performance with sustainability goals and adapting to changing industry needs of industries. Future directions focus on leveraging simulation tools and data-driven approaches to support material innovation and improve helmet design across a range of environments.

Keywords: Industrial Safety Helmets, Composite materials, Hybrid materials, Mechanical properties

ICMIT/MM64

21st-century revolutionary growth of biodegradable 3d printed composite polymers**Shalini Anand^{1,2*}, Yogesh Kumar¹, Ashish Kumar Srivastava¹**¹National Institute of Technology, Patna, India²Muzaffarpur Institute of Technology, Muzaffarpur-842003, India*Corresponding Author Email ID: shalini@mitmuzaffarpur.org**ABSTRACT**

Today's fourth industrial revolution, the demand for 3d biodegradable material in 3d printing has gained attention due to its potential in addressing environmental concerns in the manufacturing industry. Many of the biodegradable materials, such as polycaprolactone (PCL), polybutylene adipate-co-terephthalate (PBAT), PLA, etc., have their unique properties in the fields of social, industrial, medical, defence and automobiles. By integrating natural fibres and bio-based additives, these materials enhance mechanical strength, thermal properties, and microstructural, tribological, and structural properties that are suitable for diverse applications. The present work deals with the detailed discussion and futuristic approach in 3d printing and explains outcomes of the micro-level studies based on the printing parameters, material degradation profile and environmental impact, addressing challenges such as hydrophilicity and processing limitations by using different types of polymer-based 3d printing processes. The discussion explores that using biodegradable materials in 3d printing is its potential to reduce the carbon footprint of the production process. Process parameters such as nozzle diameter, infill density, layer height, working temperature, and infill pattern have their effect on the properties of the product produced. Biodegradable composite polymer has a wide range of applications across various fields, including medical, industrial, aerospace engineering, home decor, and social applications etc. At last, the present work also suggested the futuristic development and forecasting of growth in different fields acquired by 3d printing and other additive manufacturing techniques

Keywords: Biodegradable materials, Polymer composite, 3d printing, Additive Manufacturing, Infill density, layer thickness, printing speed

ICMIT/MM65

Enhancing EDM Efficiency with Hastelloy C-276 and Bio-Based Dielectrics

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ABSTRACT

This paper examines some of the materials and their combinations as used in electric discharge machining (EDM) for performance enhancement and sustainability. The study indicates that the objective parameters such as material removal rate (MRR) and surface roughness (SR) were reasonably improved through using Hastelloy C276 combined with graphite electrodes and eco-friendly water dielectric to achieve an approximate MRR of 35.8mm³/min and a SR of 2.5 µm in practical laboratory conditions. The Copper Tungsten electrodes showed maximum material removal rate of 42.3 mm³/min and the surface quality achieved implies its suitability at high discharge currents. Superior performance of the cryogenically treated superalloys was achieved with improved tool life, minimized wear, and optimized machining rate of MRR= 2.464 g/min and SR = 5.06 µm. Furthermore, Jatropha oil-based bio dielectric provided the same pull-out force, MRR (235.29×10⁻⁵mm³/min) and TWR (0.39 µm) comparable to kerosene with additional benefits of sustainability factor. Therefore, the confirmation tests from the lab experiments support that these material combinations innovate the EDM processes effectively, making it more efficient and environment friendly. These findings demonstrate new advancements in cutting hard materials and minimizing the adverse effects on invoking ecology among mass industries.

Keywords: Hastelloy C-276, Eco friendly dielectric, Electric Discharge Machining, Material Removal Rate, Electrode, SR

ICMIT/MM66

Microstructural and Mechanical Characterization of Wire-Fed Stainless Steel in Direct Energy Deposition Additive Manufacturing

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ABSTRACT

Direct Energy Deposition (DED) is one of the latest advancements in additive manufacturing of high-performance metallic components that has gained much attention in the fabrication and repair of such components. Besides, this study focuses on the DED process using stainless steel wire as a feedstock material for investigating microstructure evolution, mechanical properties, and process parameter optimization. Being corrosion-resistant and having excellent mechanical strength, stainless steel becomes a very acceptable candidate for the DED application under conditions that are well-controlled. As far as this research is concerned, the focus is on the effect of critical process parameters- laser power, wire feed rate, and scanning speed- on the quality and consistency of the deposited structures. The results of detailed analysis will reveal the possible variations in the said parameters and their overall effects on the melt pool behaviour, grain morphology, porosity characteristics, and mechanical performance thereof. The study, therefore, demonstrates, through systematic experimentation and characterization, that the optimized parameter sets can yield dense, defect-free builds with desirable mechanical and corrosion-resistant properties. In general, the resultant findings highlight the promise of wire-fed DED as a low-cost and thus sustainable manufacturing route, particularly valuable for applications in aerospace, automotive, and biomedical sectors. Furthermore, DED endows manufacturing applicants with near-net-shape solutions coupled with efficient repair of parts, thus promising an obvious alternative towards continued life extension and material waste reduction.

Keywords: Direct Energy Deposition (DED); Additive Manufacturing; Stainless Steel Wire; Microstructural Evolution; Mechanical Properties

ICMIT/MM67

Mechanical performance of 3D-printed AlSi10Mg honeycomb-structured protective gears

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ABSTRACT

The mitigation of impact-related injuries while designing the protective gear is a prime concern. Even though the current solutions are capable of reducing the severity of the impacts, it's difficult to fully eliminate the force transmission. Therefore, to enhance the effectiveness of these protective gears, the incorporation of advanced lattice structures such as honeycombs can be advantageous due to their high energy-absorbing capabilities. These structures are inspired by the natural geometry of beehives and are widely used in transportation, engineering, architecture, and biomedical industries. The present work focuses on analyzing the impact of honeycomb lattice structures on the mechanical properties of the AlSi10Mg alloy. These alloys offer a high strength-to-weight ratio, weldability and strength which makes it suitable for the fabrication of protective gears. The finite element analysis (FEA) was conducted to determine the mechanical behavior of the specimens fabricated at different process parameters. This study will provide insight into the lattice structures' influence on the energy absorption property of honeycomb structures of AlSi10Mg, contributing to the advancement of materials for high- impact resistant applications.

Keywords *Lightweight alloys; AlSi10Mg Alloy; Mechanical properties; Impact resistance; Compressive force.*

ICMIT/MM68

Microstructural and mechanical properties of the Al-Zn-Al sheet lamination through friction stir additive manufacturing**Preeti Gautam, Akash Nag*, Jana Petrů***Faculty of Mechanical Engineering, VSB – Technical University of Ostrava, Ostrava, Czech Republic***Corresponding Author Email ID: akash.nag@vsb.cz***ABSTRACT**

The additive manufacturing method is becoming rapidly the most pursued manufacturing strategy for customised batch production of components. There are different additive manufacturing procedures which are now being utilised in industries to manufacture various components. Friction stir additive manufacturing is one of the additive manufacturing techniques that utilises the solid state friction stir welding technology to join metallic/non-metallic thin sheets layer-by-layer by persuading sufficient plastic deformation and dynamic recrystallisation. In the present study, three different layers of Al5083-Zn-Al5083 of three mm thickness each are joined layer by layer through the friction stir additive manufacturing technique to produce the multifunctional additive component. A Zn plate between the two Al5083 plates is used to improve the bonding strength at both interfaces. The temperature distribution profile at the joining of each layer was investigated by thermal imaging camera to discuss the peak temperature throughout the joining direction and the effect of frictional heat on the first interface, while the joining of the second interface was also discussed in detail. Further, microstructural, morphological and elemental characterisation has been carried out by FESEM images and EDS line diagrams carried out at each interface. Since the proposed method is well known for microstructural modification through refining the grain structure, mechanical properties like tensile strength and hardness variation through cross-section of the additive component are examined to measure the effect of grain refinement at interfaces while joining the layers for three specimens at the same process parameters. The results indicated the improved yield strength of 257 MPa and 336 MPa at 12.5% elongation compared to 221 MPa yield strength and 303 MPa UTS of Al5083 alloy. Microhardness reached 110 HV at interface 1 and 97 HV at interface 2 compared to 70 HV of the base Al5083 alloy.

Keywords: Friction stir additive manufacturing, Al5083, Tensile strength, microhardness variation

ICMIT/MM69

Multi-Modal Characterization of Crack Initiation Zones in Brittle Materials Under Grinding Stresses**Tanmay Tiwari¹, Chandra Sekhar Rakurty²**¹*Center for Precision Manufacturing, Department of Mechanical Engineering, University of Akron, Akron, Ohio, USA 43325*²*The M. K. Morse Company, Canton, Ohio, USA 44707***ABSTRACT**

Early failure of brittle materials such as tungsten carbide remains a significant challenge in precision manufacturing, primarily due to crack initiation during the grinding process. Tungsten carbide is particularly susceptible to crack formation under the combined influence of high mechanical and thermal stresses generated during grinding. This study presents a comprehensive framework to predict and validate crack initiation zones on the surface of tungsten carbide substrates. The approach integrates experimental procedures, real-time sensor data acquisition, and surface characterization techniques. Post-grinding analysis using scanning electron microscopy (SEM) and X-ray diffraction (XRD) enabled detailed examination of surface and subsurface crack development. Simultaneously, real-time acoustic emission (AE) monitoring detected signal spikes indicative of crack initiation events. These AE signals were correlated with SEM observations and XRD-derived residual stress profiles to accurately localize and confirm crack zones. The framework offers new insights into the complex mechanisms of crack initiation and propagation in brittle materials during grinding. By linking real-time monitoring with microstructural validation, the study enhances understanding of how thermal and mechanical stresses contribute to surface damage. The findings provide a foundation for optimizing grinding parameters and improving the reliability and service life of tungsten carbide components in high-precision applications.

Keywords: *Grinding Stress, Multi-Model characterization, Brittle failure*

ICMIT/MM71

The Role of WAAM Process Parameters in Engineering Titanium- Based Functionally Graded Materials**Amit Kumar Mandape, Srinivasu Gangi Setti***

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ABSTRACT

Wire Arc Additive Manufacturing (WAAM) is a Directed Energy Deposition (DED) method employed for the fabrication of large-scale metallic components, offering high deposition rates and material efficiency. Titanium alloys are frequently selected for WAAM due to their high specific strength, corrosion resistance, and stability at elevated temperatures, making them ideal for aerospace, biomedical, and other high- performance applications. Functionally Graded Materials (FGMs), characterized by gradual spatial variation in composition and microstructure, enable the optimization of material properties across different regions of a component to meet specific mechanical and thermal demands. The combination of WAAM with titanium-based FGMs facilitates the production of parts with tailored, location-dependent properties. This review provides an overview of the WAAM process and examines the influence of key processing parameters—such as heat input, wire feedstock composition, and deposition strategies—that enable the fabrication of titanium-based FGMs. It explores the impact of these parameters on the formation of material gradients, microstructure evolution, and mechanical properties of the components. The limitations of various additive manufacturing (AM) technologies in producing FGMs are briefly discussed, and potential strategies to address these challenges are outlined. Furthermore, the review highlights future opportunities for the development of titanium-based FGMs in high-performance applications.

Keywords; Additive Manufacturing (AM); Directed Energy Deposition (DED); Functionally Graded Materials (FGMs); Titanium Alloys; Process Parameters;

ICMIT/MM72

Thermal Analysis of Dissimilar Friction Stir Welding of Al 6061-T6 and Cu 11000 Using Finite Element Method**Kumar Shambhav^{1*}, Bishal Saha Durjoy¹, Rakesh Kumar¹, Somnath Chattopadhyaya¹, Raj Das²**¹*Department of Mechanical Engineering, Indian Institute of Technology (ISM), Dhanbad, India*²*Defense and Aerospace Centre, RMIT University, Melbourne, Australia)***Corresponding Author Email ID: 22dr0107@iitism.ac.in***ABSTRACT**

Friction Stir Welding (FSW) is an advanced solid-state joining process that can be used for welding dissimilar metals, with superior joint quality, and refined microstructure compared to conventional fusion welding methods. In the analysis, a numerical simulation is performed to examine the thermal behaviour during the FSW of Al 6061-T6 and Cu C11000, used in aerospace and electrical applications. The simulation is carried out using the Finite Element Method via ANSYS Coupled Field Transient analysis, focusing on temperature distribution and heat flux. Equivalent time steps are used in the analysis to replicate the process within computational limits and the tool modelled as a rigid body to simplify calculations, and thermal contact resistance at the tool-workpiece interface is assumed to be negligible, ensuring consistent heat transfer throughout the simulation. The results reveal a peak temperature and maximum heat flux in the weld zone with RPM, sufficient to induce intense plastic deformation without reaching the melting point of either base material. This study shows the thermal profile of dissimilar FSW joints with varying RPM and provides a foundation for optimizing process parameters to get defect-free, high-strength welds.

Keywords: Friction Stir Welding, Thermal Simulation, Finite Element Method, ANSYS Coupled Field Transient, Numerical Modelling

ICMIT/MM73

Developing Scheduling Policies to Reduce the Total Weighted Completion Time in Additive Manufacturing

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ABSTRACT

Additive Manufacturing (AM), generally known as 3D printing, has shown greater possibilities for transforming the manufacturing sector by enabling the production of lightweight parts or products of highly varied quality in smaller batches and lower quantities. And as AM advances from basic prototyping to full-scale manufacturing, efficient job scheduling will become a critical requirement to meet tight delivery windows and high-priority demands. However, the intermittent availability of AM machines - due to maintenance, operator shifts, or material changeovers - poses a significant challenge for high-priority production. Regardless of extensive research on AM cost structures and make span optimisation, the scheduling problem targeting Total Weighted Completion Time (TWC) with consideration to realistic downtime constraints remains unexplored. This paper proposes a comprehensive MILP model for both single and identical-parallel AM machines with multiple arbitrary non-availability periods, and jobs that are fully, semi, and non-resumable. Each job carries a weight reflecting its priority, and machines may undergo scheduled maintenance, operator breaks, or shift changes. Our formulations extend location-based scheduling by integrating multiple non-availability intervals per machine, ensuring fully, semi, and non-resumable job processing against interruptions. Implemented in Gusek/GLPK, we employ scheduling policy strategies and a 1,200-second time cap to balance solution quality and tractability. Detailed numerical examples—including a 3-machine benchmark reproducing the known optimal TWC of 56 and a 2-machine, 5-job scenario demonstrating weight-driven priority shifts—validate correctness. Our results demonstrate that prioritising weighted completion time under realistic downtime yields schedules that better align with industrial AM's quality-of-service demands.

Keywords: Additive Manufacturing; Scheduling; Total Weighted Completion Time; Machine Downtime; MILP; Non-resumable Jobs, Gusek; GLPK

ICMIT/MM74

Impact of Process Variables on Roughness of Turned Surfaces using Biodegradable Cutting Fluids

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ABSTRACT

This research focuses on improving the surface finish during mild steel turning by optimizing cutting parameters, with a particular emphasis on the role of cutting fluids. The primary goal is to achieve better surface quality and enhance machining performance by determining the best combination of speed, feed, and depth of cut. An important aspect of the study was to evaluate how biodegradable cutting fluids affect the machining process. To measure performance, surface roughness was used as the key indicator. All experimental trials were performed on a center lathe under varying machining conditions. Various cutting fluids, including sunflower, coconut, and soyabean oils, were tested to determine their effectiveness in enhancing the surface finish. Statistical methods, including Response Surface Methodology (RSM), were utilized to refine the parameters and establish the relationship between machining conditions and surface roughness. The findings indicated that cutting speed and feed had a significant impact on surface quality. Additionally, biodegradable cutting fluids demonstrated a notable influence on the surface roughness of mild steel. The best cutting fluid and ideal cutting parameters have been identified in the study. A surface roughness value of 0.2858 microns was found to be minimum with soybean oil as the cutting fluid. The associated cutting speed was 25m/min. The corresponding feed-rate and cut-depth were 0.10mm/rev and 0.7879mm respectively.

Keywords: Biodegradable Cutting Fluids, Turning Operations, Surface Roughness, Surface Finish Optimization, Response Surface Methodology (RSM)

ICMIT/MM77

Advancements in 3D Printing of Composite Materials: Challenges and Prospects

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ABSTRACT

Three-dimensional (3D) printing of composite materials has revolutionized manufacturing, enabling the creation of lightweight, durable, and complex structures with customizable properties. This paper examines the fundamentals of 3D printing technologies and composite materials, emphasizing the synergy between technological innovations and material science. Key advancements include the development of continuous fiber-reinforced polymers (CFRPs), hybrid manufacturing techniques, and the incorporation of recycled materials, enhancing sustainability and mechanical performance. Despite these opportunities, challenges such as material compatibility, economic feasibility, and process reliability hinder the widespread adoption of this technology. The study also highlights recent breakthroughs in multifunctional composites, advanced computational tools, and hybrid additive manufacturing processes, which have expanded the capabilities of 3D printing. Future directions emphasize the need for research into self-healing materials, sustainable resins, and enhanced process control to unlock the full potential of 3D-printed composites.

Keywords: 3D Printing, Composites, Additive Manufacturing, Industrial Applications

ICMIT/MM78

Development of TiB₂-Al₇Si_{0.3}Mg in-situ Functionally Graded material through Centrifugal Casting Process

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ABSTRACT

Aluminum Al₇Si_{0.3}Mg functionally graded composite material (FGM) reinforced with 5 wt.% submicron TiB₂ particles was developed through in-situ salt reaction process produced via horizontal centrifugal casting technique. The FGM was characterized for its microstructure using optical microscope and scanning electron microscope (SEM). Also, the hardness behaviour was accessed through Vickers microhardness test. The FGM showed aluminium phase, needle structured eutectic Si and TiB₂ clusters. A slight difference in microstructure was noticed between the inner and the outer region of the cast. The addition of TiB₂ strengthened the FGM showing nearly 12% improvement in hardness from inner region (79 HV_{0.1}) to the outer zone (88.5 HV_{0.1}). This shows potential of TiB₂-Al₇Si_{0.3}Mg FGM for manufacturing high performance automotive components.

Keywords: Functionally graded material; Titanium diboride; Aluminium Al₇Si_{0.3}Mg; in-situ process; Hardness.

ICMIT/MM79

Effects of Process Variables and Tool Geometry on Friction Stir Welding of Similar Aluminium Alloy Joints AA5083 and AA6061

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ABSTRACT

Friction Stir Welding (FSW) is a key solid-state welding technique for aluminum alloys. This technology conquers the key problems caused by conventional fusion welding, namely oxidation and cracking. The review focuses on the role of material attributes, design of tool, and process variables on the FSW joint quality. The degree of variables including rotational speed, feed rate, and tool geometry frequently affect the tensile strength, hardness, and ductility. Tilt angle, plunge depth, and pin design were the parameters on which optimal control was made to produce welds free of defects. The review is quite keen on establishing a balance between input heat and material flow to reduce defects while allowing for designs in hybrid tooling to enhance mechanical properties. Future research directions include more efficient designs of the tool along with optimization of parameters for achieving higher strength within the joint along with general performance of the weld.

Keywords: Friction stir welding (FSW), Tool profiles, Process parameters, Mechanical properties, HAZ, Microstructure

ICMIT/MM80

Optimization of EDM Process Parameters for Al-MMC by Composite Desirability Approach

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ABSTRACT

The Aluminum Metal Matrix Composites are gaining significance importance in industrial, automobile and aerospace applications due to superior mechanical properties. Unconventional machining processes are required for a number of complex applications in machining industries where conventional process cannot be used. The aim of this study is to explore the optimal parameters for the Electrical Discharge Machining (EDM) process on Aluminum Alloy 6061 reinforced with SiCp and B4C Hybrid, where copper metal is employed as the tool electrode. We investigated how different process parameters affect machining performance, focusing on current, pulse duration, and pulse pause time. We analyzed their impact on Material Removal Rate and Tool Wear Rate. To structure our experimental setup, we utilized the Taguchi method alongside traditional desirability function analysis to consider multiple responses. By doing so, we pinpointed the optimal machining conditions for the specified tool and work piece combination. Our findings were consistent with existing literature data, affirming the validity of our results.

Keywords: EDM; MMC; Desirability analysis; Taguchi; MRR; TWR.

ICMIT/MM81

Analysis of Calotropis Gigantea Plant Stem Fiber Reinforced Epoxy Composite Material

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ABSTRACT

This study focuses on analyzing the mechanical behavior of Calotropis gigantea plant stem fiber-reinforced epoxy composite materials. The fibers were manually extracted from the plant's stem and subjected to chemical treatment to enhance their compatibility with the epoxy matrix. Composite samples were fabricated using the hand-layup method, with varying fiber-to-resin weight percentages to evaluate their influence on mechanical properties. The primary objective of this research is to assess key mechanical characteristics, including tensile strength, impact resistance, density, and hardness, to determine the material's suitability for industrial applications. The results indicate that the Calotropis gigantea fiber-reinforced composites show good mechanical properties, making them a practical fiber composite. The study highlights that an optimal fiber-resin ratio significantly improves tensile and impact strength while maintaining a favorable strength-to-weight ratio. Additionally, the composites show suitable hardness and density, suggesting durability under mechanical stress. Their lightweight, eco-friendly, and cost-effective nature, these composites have potential applications in various industries.

Keywords: Calotropis Gigantea, Natural Fiber, Mechanical Properties, Fiber weight ratio

ICMIT/MM86

Influence of Process Parameters on Surface Modification Using Electric Discharge Machining: A systematic review

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ABSTRACT

Electric Discharge Machining (EDM) has emerged as a vital non-traditional machining process widely used for shaping hard-to-machine materials while also offering the capability to modify and enhance surface characteristics. Beyond its primary function of material removal, EDM has proven to be an effective surface engineering technique that can alter surface morphology, microstructure, mechanical properties, and functional behaviors of work materials. This systematic review explores the influence of key EDM process parameters on the surface modification outcomes across various material systems. The study focuses on understanding how process variables—such as discharge voltage, peak current, pulse duration, pulse interval, electrode polarity, dielectric medium, and tool material, influence the formation of surface layers, recast zones, micro-cracks, and surface topography. Special attention is given to powder-mixed EDM (PM-EDM), a technique that introduces conductive or semi-conductive powders into the dielectric to alter discharge characteristics and promote favorable surface alloying. The effect of powder type, concentration, and dielectric flow rate is examined in detail to understand their role in enhancing surface quality, reducing surface defects, and introducing desired elemental composition into the workpiece surface. Key findings indicate that increasing discharge energy typically leads to higher surface roughness and thicker recast layers, accompanied by micro-cracks and porosity. However, controlled discharge energy and optimized pulse parameters can minimize thermal damage and produce fine surface textures with uniform material deposition. The polarity of the electrode significantly influences the direction of material transfer and energy distribution, impacting the nature of surface modification and compound formation. Similarly, tool material selection affects thermal conductivity.

Keywords: Electric discharge machining (EDM); Surface modification; Recast layer; Surface

alloying; Surface roughness; Material transfer; Microhardness

ICMIT/MM87

Numerical Simulation and Experimental Analysis on Cold Forging of Commercially Pure Aluminum

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ABSTRACT

Near net shape forging represents a manufacturing philosophy aimed at achieving components that closely approximate their final geometries in the as-forged state. Unlike a specific forging process, it emphasizes minimizing post-forging machining and material wastage while optimizing efficiency and cost-effectiveness. The present work targets the preform design of the commercially pure Aluminum for closed-die forging conditions by analyzing their flow in the die cavity with the help of Deform 3D software. The simulations have been conducted using DEFORM 3D V11.2 software to determine the optimal shape and size of the preform, thereby minimizing costly and inefficient shop floor iterations. After optimizing the preform, experiments were carried out to produce a near-net shape forged sphere with a diameter of 40 mm. This methodology will be extended to more complex-shaped products in the future. The findings highlight the effectiveness of simulation in refining the forging process, offering practical insights for manufacturing complex components with improved quality and efficiency.

Keyword: Near net shape; Forging; commercially Pure Aluminum; Finite Element Analysis; Friction

ICMIT/MM90

A Review Paper on parametric Optimization of TIG Welding for Stainless steel

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ABSTRACT

In today's manufacturing market, quality and productivity are crucial factors. The primary goal of industries is to boost production and provide higher-quality products at the lowest possible cost. TIG welding is a crucial and widely used process that joins two similar or dissimilar parts by heating the material or occasionally adding pressure to increase productivity while reducing time and expense. Tungsten Inert Gas Welding (TIG) is also known as Gas Tungsten Arc Welding (GTAW). In the present research project an attempt is made to understand the effect of TIG welding parameters such as welding current, gas flow rate, welding speed, that are influences on responsive output parameters such as quality of weld, microstructure, hardness and tensile strength of welding, by using optimization philosophy. Increasing welding current, increases the deposition rate and reduces the hardness. The main objective of this review paper is investigate optimal machining parameters and their contribution on producing improve weld quality and increase Productivity.

Keywords: TIG Welding, Welding Parameters, Electrode, Hardness, Tensile Strength

ICMIT/MM91

A Critical Review on Reinforcement Effects in SAC-Based Lead-Free Solders: Base Metal Degradation and IMC Formation

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ABSTRACT

The advancement of lead-free solder materials provides optimal solutions for various industries and, ensuring durability in critical situations. This progress can be achieved through various methods, with reinforcement being the most commonly preferred approach to enhance material properties. This study scrutinizes the existent research on the impact of growing base metal disintegration as well as establishment of intermetallic compound sheet from Sn-Ag-Cu (SAC) set lead independent solder ingredient along Graphene reinforcement along wt% (weight ratio). The analyses are inspected carefully to emphasize ultimate meaningful work launched whereas the SAC succession lead free solder ingredient accompanying various reinforcement like metal particles (Mn, Co, Al), oxide particles (TiO₂, Al₂O₃, SnO₂), graphene nanosheets (GNS), single wall carbon nanotubes (SWCNT), and carbide (SiC, and TiC).

Keywords: Lead-free solder; Sn-Ag-Cu (SAC) alloy; Intermetallic compounds (IMCs); Base metal degradation

ICMIT/MM92

Chemically functionalized MoS₂ Nano additive-enriched bio-lubricants for superior tribological performance**Gaurav Raj*, Hemant Kr. Choudhury***Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur, Bihar, India.***Corresponding Author Email ID: raj7374819085@gmail.com***ABSTRACT**

Mineral oil-based lubricants lack degradability, resulting in environmental hazards and natural loss when disposed of in the environment. Therefore, replacing mineral oil-based lubricants with renewable and biodegradable alternatives has become necessary due to various health concerns and limited availability. This study aims to examine the influence of chemically functionalized molybdenum disulfide (MoS₂) as an additive on the tribological performance of two different vegetable base oils (i.e. sunflower oil and soyabean oil). The functionalized MoS₂ were characterized by X-ray diffractometer (XRD), high-resolution transmission electron microscopy (HR-TEM), and Fourier-transform infrared spectroscopy (FTIR). To formulate the Nano lubricants, MoS₂ with varying concentrations (0.025, 0.05, 0.075, 0.1 and 0.15 wt.%) were dispersed separately in both base oils using a magnetic stirrer followed by ultrasonication. The tribological experimentation was carried out using a ball-on-disc type test rig. All tests were conducted as per ASTM standards. The tribological results showed that the incorporation of MoS₂ nanoadditive in both bio-lubricants exhibited a significant enhancement in the tribological performance. Furthermore, for the better unveiling of the lubrication mechanism of MoS₂, worn surfaces were characterized using various analytical techniques such as scanning electron microscopy (SEM), scanning probe microscope (SPM), and energy-dispersive X-ray spectroscopy (EDS).

Keywords: Bio-lubricants, sunflower oil, soyabean oil, nanoadditive, MoS₂, friction, wear.

ICMIT/MM93

**Comparative Study of Mechanical and Functional Properties of Al6061-SiC-Cenosphere
And Al6061-WC-Cenosphere Metal Matrix Composites with As-cast Al6061**

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ABSTRACT

Al6061 is a strong and versatile aluminum alloy extensively used in various industrial sectors. This research focuses on the fabrication and evaluation of Al6061-based metal matrix composites reinforced with Tungsten Carbide and Cenosphere, as well as Silicon Carbide and Cenosphere, using the stir casting method. The primary goal is to explore their suitability for automotive engineering applications. Mechanical tests revealed notable enhancements in hardness, tensile strength, and compressive strength compared to as-cast Al6061. Additionally, wear resistance improved and density reduced with the incorporation of reinforcements. These improvements demonstrate the composite's potential for high-performance automotive uses, where strength, durability, and lightweight properties are crucial.

Keywords: Cenosphere; Metal matrix composites; Silicon Carbide; Tungsten Carbide

ICMIT/MM99

Colloidal Formulation and Robocasting of Silicon Nitride Scaffolds with Tailored TPMS Structures**Amit Kumar¹, Siddharth Yadav¹, Anuj Kumar Sharma¹, Santosh Kumar², Govind Kumar Verma^{2*}**¹*Centre for Advanced Studies, Dr. A.P.J. Abdul Kalam Technical University, Lucknow, 226031, Uttar Pradesh, India*²*Department of Mechanical Engineering, Indian Institute of Technology (BHU), Varanasi, 221005, Uttar Pradesh, India***Corresponding Author Email ID: govindkumarverma.rs.mec20@iitbhu.ac.in***ABSTRACT**

Silicon nitride (Si₃N₄), combined with alumina and yttria as liquid-phase sintering aids, is a promising bio-ceramic for orthopedic and dental applications due to its biocompatibility, wear resistance, and mechanical reliability. Despite its advantages, the fabrication of complex, porous Si₃N₄ architectures with tailored mechanical and biological functionality remains challenging due to limitations in slurry stability, rheology control, and geometric accuracy during additive manufacturing. This study addresses this gap by developing a non-Newtonian, shear-thinning colloidal suspension containing ~42 vol.% solid loading of Si₃N₄ with sintering aids, explicitly optimized for robocasting. The ink formulation utilizes polyvinyl alcohol (PVA) as a binder, carboxymethyl cellulose (CMC) as a viscosifier, and glycerol as a gelation agent to achieve suitable rheological properties and shape fidelity. Different triply periodic minimal surface (TPMS) scaffolds were successfully fabricated using this optimized ink. Process parameters were fine-tuned for each geometry, resulting in printed parts with relative densities ranging from ~56% to ~92%. After debinding and liquid phase sintering, the structures exhibited a fibrous β -Si₃N₄ microstructure and a fracture toughness of approximately ~5 MPa·m^{1/2}. These findings demonstrate the potential of 3D printed Si₃N₄ scaffolds with bioinspired TPMS architectures for patient-specific, load-bearing biomedical implants.

Keywords: Additive manufacturing, Silicon nitride, Robocasting, Rheology, TPMS.

ICMIT/MM100

Designing a Boron Nitride Polyethylene Composite for Shielding Neutrons**Raviraj Kumar***Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur, Bihar, India.***Corresponding Author Email ID: ravi.118mech@gmail.com***ABSTRACT**

The growing need for lightweight, and non-toxic neutron shielding material in nuclear power, space exploration and health care, motivates the development of advanced polymer-based composites. Neutrons interacting with a biological target produce secondary charge particles that are damaging the human health. This study investigates boron-nitride polyethylene (BN-PE) composite as an alternative to conventional shielding materials. The most effective way to shield neutron is to slow them to thermal energies and then capture the thermalized neutrons. These factors leads us to consider potential material solution for neutron shields that maximize the protection of human while minimizing the shield mass and which adapt well to modern additive manufacturing technology. Using hexagonal boron nitride as a capture medium and high density polyethylene (HDPE) as a thermalization medium. Composite were fabricated using melt blending and compression molding with BN contain of 5, 10, 20 weight % respectively. We aim to design optimal internal structure of BN/HDPE composite by minimizing the effective dose, which is a measure of the estimated radiation damage exposure for human. The 20wt % BN-PE composite achieved a 41.6% reduction in thermal neutron flux compared to pure PE. Mechanical testing showed a moderate decrease in tensile strength (27.3 MPa to 22.1MPa) and thermal analysis indicated a 15 degree Celsius improvement in degradation temperature. Optimal structure reduced the effective dose up to a 72x over aluminum and 4x over HDPE. This is the significant improvement in shielding effectiveness that could dramatically reduce the radiation exposure of occupational workers. These results demonstrate that BN-PE composite offer a promising balance of shielding performance, mechanical strength and thermal stability, making them suitable for practical application in radiation protection. The synergy between neutron moderation and capture in the BN-PE matrix presents a promising pathway for lightweight, non-toxic neutron shielding materials suitable for reactor shielding, spent fuel storage, and radiation protection in compact nuclear systems.

Keywords: Neutron shielding, Boron nitride, Polyethylene composite, Thermal neutron, Radiation protection

ICMIT/MM101

Nano Finishing of Additively Manufactured Parts**Atul Singh Rajput***National Institute of Technology Karnataka, Surathkal, India***Corresponding Author Email ID: atulsingh@nitk.edu.in***ABSTRACT**

Additive Manufacturing (AM), particularly Laser Powder Bed Fusion (LPBF) of Ti-6Al-4V alloys, enables the customization of biomedical implants to match patient-specific needs. However, the inherent poor surface quality of AM components leads to elevated wear rates, generating debris that can cause systemic toxicity and clinical complications such as Aseptic Loosening, ARMD, ALVAL, and Pseudotumours. To address this challenge, this study introduces a novel Hybrid Electrochemical Assisted Magnetorheological (H-ECMR) finishing method to improve surface quality without compromising surface integrity. The process leverages the combined effects of mechanical abrasion and electrochemical action, resulting in a significant increase in oxide layer thickness from 8 nm to 78 nm, enhancing corrosion resistance. Surface features like holes and pockets hinder uniform polishing due to the entrapment of Carbonyl Iron Particles (CIPs). To overcome this, a Feature-based Hybrid MRF Planning System (FHMRP-PS) was developed to automate paraffin wax deposition into these features before H-ECMR finishing. Additionally, toolpath strategies—Zig-Zag, Hilbert, Peano, Spiral, and Trochoidal—were compared for uniformity and effectiveness in reducing surface roughness. The Trochoidal path exhibited the most consistent results across the work surface. The process is applied to femoral head and bone plate implants fabricated by LPBF. Initial roughness ($R_a \sim 14.67 \mu\text{m}$) is reduced to 33.14 nm after H-ECMR finishing. SEM, AFM, and optical profilometry confirm the improvement in surface morphology and dimensional integrity. Post-processing also significantly improves corrosion resistance (rate reduced from 0.081 mm/year to 0.0103 mm/year) and wear resistance (wear rate reduced from 18.86×10^{-5} to $0.96 \times 10^{-5} \text{ mm}^3/\text{min}$). Negative skewness and low kurtosis values further affirm enhanced tribological properties. This integrated finishing approach advances the functional performance and biocompatibility of AM-fabricated biomedical implants.

Keywords: Additive Manufacturing, Nano-additive, implants, LPBF

ICMIT/MM102

**Performance Enhancement of μ -ECDM for Exotic Non-Ferrous Materials Using On-Process
Sonication of Electrolyte Solution**

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ABSTRACT

The classical micro electrochemical discharge machining (μ -ECDM) process has shown potential for machining exotic materials commonly used in the defense and aerospace industries. However, its effectiveness is limited when fabricating high aspect ratio micro-components due to inadequate electrolyte availability at deeper machining zones. On-process sonication of the electrolyte has emerged as an effective solution to enhance electrolyte circulation and spark stability during machining. In this study, a micro-hole was fabricated in a titanium alloy, an exotic non-ferrous material, using the classical μ -ECDM process, and its performance was compared with that of the ultrasonic-assisted ECDM (UECDM) process, which incorporates on-process sonication. The investigation involved varying key electrical parameters such as applied voltage, duty factor, and electrolyte concentration. Results revealed that UECDM significantly improved dimensional accuracy, material removal rate (MRR), and surface quality compared to the classical μ -ECDM process. Specifically, μ -ECDM resulted in a larger hole diameter, a 1.32 times thicker recast layer (RCL), and a 1.42 times wider heat-affected zone (HAZ). In contrast, the UECDM process achieved a 47% increase in MRR and a 26.05% improvement in surface quality.

Keywords: Ultrasonic vibration, μ -ECDM, Titanium alloy, Heat affected zone, Recast layer thickness, Surface roughness

ICMIT/MM103

Effect of Alkali Treatment and Fiber Orientation on the Tensile Properties of Jute Fiber Reinforced Composites

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ABSTRACT

The increasing environmental concern and demand for eco-friendly materials have created interest in natural fiber-reinforced polymer composites for engineering purposes. Among the different natural fibers, jute has been identified as a potential reinforcement because it is biodegradable, economical, and possesses satisfactory mechanical properties. In this paper, the influence of fiber orientation angles (45°, 60°, and 90°) and alkali treatment (5% NaOH for 1, 3, 5 and 7 hours duration) on tensile characteristics of jute fibre-reinforced epoxy composite is explored. The jute fibers were treated with sodium hydroxide solution to improve fiber matrix adhesion prior to being embedded in the epoxy matrix. The results show that maximum tensile strength was observed in composites with 90° fiber orientation. In addition to this, alkali-treated (5 hours duration) jute fibers showed good interfacial bonding, and therefore the composites showed better tensile performance as compared to untreated ones.

Keywords: Jute fibre, orientation angle, tensile strength

Track 2

Fluid Flow & Thermal Engineering

ICMIT/TT04

Experimental Investigations of Eigen Frequencies on Bellow Pipes used in thermal plants with free-free conditions**Md Asjad Mokhtar*, Ankit Kumar***Darbhanga College of Engineering, Mabbi, Darbhanga, Bihar***Corresponding Author Email ID: ankitr606@gmail.com***ABSTRACT**

Industries such as power generation, HVAC System, chemical processing sectors, etc. encounter very adverse/tough operating conditions to achieve the efficiencies good enough to sustain in today's competitive environment. Of the many important components in such industries, The Bellow Pipes, also known as expansion joints, undergoes through sudden expansion and contraction of the working fluid, experiences random vibrations in the system. In order to have them function robustly, it is important to understand and analyze the fundamental eigen-frequencies, critical speeds and its harmonics. The present work investigates extensively about the fundamental natural frequencies of a bellow pipe used in thermal power plant connected flexibly in the system. With the help of suitable fixtures, free-free conditions were obtained for experimental measurements using accelerometers, and data acquisition system. A thorough investigations were carried out on the fundamental eigen-frequencies in axial as well as lateral directions using sweep tests and rap tests with multiple measurements. Three separate samples of Bellow Pipes were tested with free-free boundary conditions and the estimated eigen-frequencies were averaged out. Vibration waveforms picked up from manual and auto sweep tests and rap tests are processed using Fourier Transforms to identify natural frequencies associated with different modes. The results of different test are found to be in close agreement with each other and the final eigen-frequencies associated with different modes are reported. The sensors, uniaxial and tri-axial accelerometers and the data acquisition systems are of higher range, better resolutions and good sensitivity are used.

Keywords: Bellow Pipe, Eigen Frequency, Data acquisition, Sweep test, Rap Test

ICMIT/TT05

Combustion Modeling of CH₄-Air Mixture in an Axisymmetric Channel with an Obstacle**Vinayak Jha^{1*}, Kumar Amit², Sandeep Tiwari¹**¹*Department of Mechanical Engineering, Darbhanga College of Engineering, Bihar-846005*²*Department of Mechanical Engineering, IIT Patna, Patna-801106, India***Corresponding Author Email ID: jhavinayak8@gmail.com***ABSTRACT**

This study investigates the impact of an obstacle on the combustion characteristics of a methane-air (CH₄-air) mixture in an axisymmetric channel. The presence of obstacles in combustion systems significantly influences flow dynamics, heat transfer, and flame stability, making it essential to understand their effects for optimizing combustion performance. The research focuses on analyzing how varying the distance of the obstacle from the inlet affects the combustion process under different inlet velocities of air and methane. The simulation employs a computational fluid dynamics (CFD) approach to model the combustion process, incorporating detailed chemical kinetics and turbulence-combustion interactions. The input parameters include the inlet velocities of air and methane and the obstacle's position relative to the channel inlet. The output parameters analyzed are the temperature distribution within the channel, temperature at the obstacle surface, and inlet pressure variations. Preliminary results indicate that the obstacle induces complex flow patterns, such as recirculation zones and turbulence intensification, which significantly affect flame propagation and thermal characteristics. The study aims to identify optimal configurations that minimize thermal losses, enhance flame stability, and improve overall combustion efficiency. These findings have potential applications in designing efficient combustion systems for industrial burners, gas turbines, and other energy systems.

Keywords: Combustion Modeling, Computational fluid dynamics, Optimization

ICMIT/TT06

Design, Analysis and Fabrication of an IoT-based refrigerator monitoring system
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ABSTRACT

A mechanical refrigerator with internet of things monitoring increases functionality and safety by providing information on the state of the machine such as gas leakage, food quality and compressor health as well as level of refrigerant and age of parts. Refrigerant leakage is monitored by attaching a gas sensor like MQ-135, whereas temperature and humidity necessary to preserve foods are detected by DHT11 & DS18B20 and finally to detect the performance of the compressor both acceleration and temperature sensors have been used. Refrigerant control is maintained by pressure sensors; thermal and current sensors manage other components including fans and motors, indicating when to be serviced. The system has a linked dashboard in cloud form and is controlled through a mobile application that also provides real-time info and warning. In Wen's refrigerator, the sensor data are analysed through the AI/ML algorithms which can identify the probable failure in the near future, and thus maintain which can increase the lifetime of the refrigerator. Further, the energy intake and usage is monitored with an energy management system and an alarm notification system can even send the alarm notification via SMS, E-mail, or Push Notification in an emergency. It uses communication interfaces like Wi-Fi (ESP8266, ESP32) and microcontrollers (Arduino or Raspberry) that allows it to report data to cloud services like AWS IoT, or Microsoft Azure. It can also run within home systems such as Google Home or Amazon Alexa to make controlling easier. Through the proposed IoT based solution, the refrigeration systems are made smarter, safer, and efficient, and arable interventions are performed continuously to enhance quality, minimize expenses for repair and energy efficiency. This paper presents a method for determining refrigerant charge using low-cost, noninvasive measurements (i.e., surface mounted temperature measurements). The method could be used as part of a protocol for verified service providers (VSPs) in California AC diagnostic tune up or refrigerant charge, air flow (RCA) verification programs. Ultimately, the method could be embedded within a portable virtual refrigerant charge gauge for a technician's use or permanently installed on the AC unit. The accuracy of the virtual refrigerant charge sensor method is evaluated in this paper using laboratory data for a number of different systems and over a wide range of operating conditions with and without the presence of other faults. Seven different systems are considered, including a window unit, residential split systems, and light commercial packaged systems employing either fixed orifice expansion devices or thermostatic expansion valves and R-22 or R-410a as the refrigerant. The virtual refrigerant charge sensor is shown to have good performance in terms of accuracy and robustness and has the potential to be easily implemented and installed.

Keywords: IOT, DHT11 & DS18B20, MQ-135, AI/ML algorithms

ICMIT/TT07

Performance Assessment of Eco-Friendly Refrigerant R513A Versus R134a: Heat Transfer Coefficient and Pressure Drop Analysis

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ABSTRACT

This paper presents an experimental investigation into heat transfer coefficient (HTC) and frictional pressure drop (FPD) of R134a and its low-GWP alternative, R513A, under flow boiling conditions in micro-fin tubes. experiments were conducted using 1000 mm long, 9.52 mm outer diameter micro-fin tube across range of operating parameters: mass fluxes from 50 to 300 $\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, vapor qualities between 0.02 and 0.96, saturation temperatures of 17°C and 22°C, and heat fluxes of 12, 24, and 36 $\text{kW}\cdot\text{m}^{-2}$. study aimed to evaluate impact of micro-fin enhancement on flow boiling performance and compare flow boiling characteristics of two refrigerants. Results demonstrate that R513A delivers improved heat transfer performance over R134a, with reasonable increase in FPD. Experimental data were validated using established empirical correlations, and sensitivity analyses were performed to assess influence of key operating parameters. findings provide valuable insights for application of environmentally friendly refrigerants in high-performance heat exchanger design and thermal system optimization.

Keywords: Flow boiling; micro-fin tube; Heat transfer coefficient; Frictional pressure drop; R134a; R513A

ICMIT/TT08

A Comparative Study and Analysis of Shell and Tube Heat Exchanger Using Nanofluid

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ABSTRACT

This paper explores the effects of nanofluid, namely Aluminium Oxide (Al_2O_3), Titanium Oxide (TiO_2), and Magnesium Oxide (MgO) as working fluids in a compact shell and tube heat exchanger. The study focuses on assessing heat transfer performance by analyzing the heat transfer coefficient, and Nusselt number, under different parameters using CFD simulation. The nanofluid flows through the tube side as cold fluid, while water acts as the hot fluid in the shell side. Parameters such as nanofluid concentration (0.8%, 1.5%, and 2% by volume) and inlet velocity of the cold fluid (0.4 m/s, 0.7 m/s, and 1 m/s) are varied. The result indicates that Al_2O_3 exhibits higher heat transfer performance followed by MgO , while TiO_2 exhibits the lowest performance among the three nanofluids. Moreover, the heat transfer performance of all nanofluids is enhanced with increasing velocity and nanofluid concentration. This research also explores the potential of compact shell and tube heat exchangers, which are increasingly favored for their efficiency and space-saving design.

Keywords: CFD, Heat Transfer Coefficient, Nusselt Number, Inlet Velocity, Nanofluid, Heat Exchanger

ICMIT/TT09

A Sustainable Comparative Investigation on Diesel Pumps and Solar Irrigation Pumps

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ABSTRACT

The economy of Bihar is significantly influenced by agriculture, and it is the backbone of the state's economy. Despite its strength, the sector faces challenges in the carbon emissions due to the use of diesel-operated pumps. Keeping this in mind, the state government of Bihar implemented several schemes advocating the use of solar irrigation pumps as a sustainable option. The authors investigated the assessment of solar irrigation pumps, necessities, benefits, government schemes, challenges, and the future road ahead for Bihar. This investigation also supports the notion that with the transition to solar irrigation, dependence on fossil fuels will diminish, thereby increasing crop productivity and the farmer's income. Comparative investigations of a diesel pump and a solar pump for irrigation purposes are performed. In addition to diesel consumption, carbon emissions for different crops are also investigated.

Keywords: Solar Pump, Diesel pump

ICMIT/TT12

REDESIGNING AN AIRCRAFT ENGINE BRACKET FOR AN ENHANCED PERFORMANCE**Atul, Prabhanjan P Bharadwaj*, R Sahana, Roopa R, S Pavish Kumar***Alliance School of Applied Engineering Alliance University Bengaluru, Karnataka, India***Corresponding Author Email ID:prabhanjanpbharadwaj@gmail.com***ABSTRACT**

The engine bracket of an aircraft is a structural part that holds the engine and distributes its load to the airframe. In advanced aerospace applications, it is very important to optimize such parts to minimize weight, increase strength, and decrease vibrations. The goal of this research is to design the engine bracket anew with the help of advanced engineering methods like topology optimization, finite element analysis (FEA), and principles of additive manufacturing. The initial goal is to minimize the weight of the bracket without sacrificing structural integrity, as even minor decreases in mass will result in notable fuel savings and increased aircraft efficiency. Vibration damping is also seen as a major consideration since engine-induced vibration can impact both performance and passenger comfort. The redesign will study the application of high-performance light materials, e.g., aluminium alloys, titanium alloys or carbon fibre reinforced polymers, and examine integrating damping material into high-frequency regions of the bracket. By utilizing simulation-based design practices, such as stress analysis and modal analysis, the research picks out points of high stress and where to remove unnecessary material. The final structure should be both light in weight and long lasting, and enhanced manufacturability through the application of 3D printing technology. The present work has prospects to enhance brackets performance in the next-generation planes and electric motors where control overweight and vibrations is more necessary.

Keyword: - Brackets, modal analysis, Finite element, weight, stress analysis.

ICMIT/TT13

Assessing the Socio-Economic Impact of Cow Dung Briquette Production: A Sustainable Approach to Biowaste Conversion and Energy Utilization

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ABSTRACT

The growing energy demand and the need for sustainable waste management practices have catalyzed innovative solutions that merge environmental stewardship with socio-economic development. This study primarily discusses the transformation of cow dung—an abundant agricultural byproduct—into efficient, eco-friendly fuel briquettes. Central to this research is the design, implementation, and evaluation of a novel semi-automated cow dung briquette-making machine. This research investigates the socioeconomic implications of cow dung briquette manufacturing as a decentralized bioenergy solution. By transforming locally available cow dung and agricultural byproducts into energy-dense briquettes, rural communities may reduce their dependence on traditional fuels like firewood, charcoal, and kerosene, which are often uneconomical and environmentally hazardous. The study combines field data, stakeholder interviews, and performance metrics from pilot communities across rural and peri-urban regions to assess the impact of this biowaste-to-energy solution. The study evaluates the socioeconomic benefits of cow dung briquetting across several aspects, including employment generation, income diversification, women's empowerment, improved health outcomes, and energy security. The findings show that small-scale briquette production units not only provide employment chances but also encourage sustainable waste management practices and alternative fuels for domestic usage. Moreover, the involvement of women and self-help groups in the briquette production process fosters social inclusion and community-oriented entrepreneurship. This study concludes that cow dung briquette production serves not only as an energy solution but also as a catalyst for overall rural development, facilitating energy access, environmental conservation, and socioeconomic advancement. Environmentally, the shift to briquettes reduces CO₂ emissions w.r.t. unmanaged dung and contributes to cleaner combustion practices. The findings underscore the transformative potential of integrating automation into rural energy solutions. This study advocates for broader investment in semi-automated briquette production as a pathway toward circular economy practices, rural energy resilience, and inclusive green development.

Keywords: Cow Dung Briquettes, Decentralized Bioenergy, Sustainable Waste Management, Rural Development, Circular Economy.

ICMIT/TT14

Smart Grid Simulation with AI, Renewable Energy, and Battery Integration

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ABSTRACT

This research informs us on how to design and simulate a smart grid using sophisticated renewable energy sources like solar and wind energy, energy storage systems and AI- based control. The aim is to explore how smart systems can make the processes of power generation, transmission, and storage more efficient. The system was designed using MATLAB/Simulink in and Simscape Power Systems such that solar panels, a wind turbine, a battery storage, and various electrical loads were modeled. The model comprises two variable renewable energy power sources (solar and wind power), four loads representing various energy needs, and a battery storage that takes up excess energy. A simple artificial intelligence controller was also included in order to manage the flow of energy between the grid, the storage system, and the loads based on power availability. To enhance the realism of the system, we employed moderate configurations that balance idealized with realistic conditions, thereby making the variables of power levels and their variations feasible but not overly complicated. The system also has a simple dashboard to control and observe the grid in real time, complemented by data logging features to track important parameters such as power generation, battery condition, and energy consumption. This study demonstrates how the integration of AI control with renewable energy and battery storage can make smart power grids more efficient and reliable. The simulation is a valuable tool for testing and optimizing smart grid technologies and could be extended to incorporate more advanced features, such as machine learning to make improved decisions in the future.

Keywords: Smart grid, AI-based control, renewable energy, solar, wind power, battery storage, MATLAB/Simulink, data logging

ICMIT/TT15

Sustainable Briquetting of Cow Dung: A Low-Cost Biofuel Production Technique for Rural and Industrial Energy Needs

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ABSTRACT

The growing demand for sustainable and affordable energy solutions has inspired renewed interest in biomass-based fuels, particularly in the rural and industrial sectors. This paper offers an explanation of cow dung briquette production and its cost analysis. The briquettes so produced are low-cost, environment-friendly, meet energy shortage, and reduce reliance on traditional fossil fuels. Cow dung, when combined with agro-residues like sawdust, straw, or rice husk, can easily be converted into dense, energy-rich briquettes employing simple procedures. This paper proposes an approach for producing cow dung briquettes, which includes raw material collection, binder integration, compaction, and shape. High calorific value, lower smoke emissions, simplicity of storage, and effective combustion are just a few advantages of these briquettes. Additionally, the method offers a sustainable way to manage waste and eliminates burning of stubble in the field. Briquettes made from cow dung are highlighted for their potential as a dual-use energy source, meeting small-scale industrial settings' energy needs while also providing home fuel for rural communities. In rural areas, this briquetting method can also lead to job creation and business prospects. By creating a scalable and easily accessible briquetting paradigm, the technology seeks to advance energy self-sufficiency, rural entrepreneurship, and sustainable waste management. In conclusion, briquetting cow dung exhibits potential as a scalable solution to meet energy demands while promoting environmental and economic sustainability.

Keywords: - Briquette production, Agro-residue, Sustainable waste management, economical, rural entrepreneurship.

ICMIT/TT18

Optima Design of Wavy Channel for Heat Transfer with Nano-Fluid Flow

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ABSTRACT

Compared to straight channels, the wavy shape improves heat transmission efficiency and fluid mixing. The wavy design encourages fluid element mixing by creating secondary flows and vortices. This is particularly useful in applications where thorough mixing is essential. The increased surface area and turbulence caused by the wavy path improve heat transfer rates. This makes the wavy microchannel suitable for cooling systems and other heat management applications. The wavy geometry generally results in a higher pressure drop compared to straight channels. This needs to be considered in the design to ensure that the pumping system can handle the increased resistance.

Keywords: Nano-fluid, Design of wavy channel, Thermal behavior

ICMIT/TT20

Design and Fabrication of a Peltier-Based Air-Cooling Device

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ABSTRACT

This project explores the design and development of a compact and energy-efficient air cooler utilizing a Peltier module (thermoelectric cooler). Traditional air conditioning systems are often bulky, consume significant power, and rely on refrigerants that may have environmental impacts. The Peltier-based air cooler presents an eco-friendly alternative by leveraging the thermoelectric effect, where the application of an electric current across dissimilar semiconductor materials creates a temperature difference. In this system, the Peltier module is used to absorb heat from the surrounding air on its cold side and dissipate it through a heat sink and fan arrangement on the hot side. The setup includes aluminium heat sinks, DC fans, and proper insulation to enhance cooling performance. A DC power supply powers the device and offers benefits such as silent operation, portability, and low maintenance. Performance tests indicate that the cooler can effectively reduce the ambient temperature in a small, enclosed space. It demonstrates potential for use in personal cooling devices, electronics cooling, and other niche applications where conventional systems are impractical. The study highlights the viability and limitations of thermoelectric cooling and suggests improvements for future iterations, such as enhanced heat dissipation mechanisms and improved power efficiency.

Keywords: Peltier Module, Thermoelectric Cooling, Compact Cooling System, Energy Efficiency, Environmentally Friendly Cooling

ICMIT/TT22

Advances in Bubble Dynamics and Heat Transfer in Confined Two-Phase Flow Systems: A Comprehensive Review

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ABSTRACT

Two-phase flow systems, especially bubble dynamics in confined geometries, are central to most industrial applications such as heat exchangers, microchannel cooling, and bubble column reactors. This review integrates recent developments in experimental, theoretical, and computational studies of two-phase bubble flow in small-scale systems. We start by describing the mechanisms of bubble formation and detachment in crossflow liquid environments, with emphasis on innovative nozzle designs that improve bubble detachment frequency and minimize bubble size. The experimental findings show that nozzle geometry and orientation are critical to bubble properties, providing opportunities for improved mass and heat transfer performance. Next, we discuss the intricate phenomena controlling flow boiling in small channels, where boiling regimes, heat transfer coefficients, and instabilities are extremely sensitive to channel dimensions and thermofluidic properties. The channel size classification and corresponding dimensionless parameters, such as the confinement, are discussed in detail. Finally, we discuss the application of Computational Fluid Dynamics (CFD) in simulating two-phase bubble columns, with emphasis on multiphase flow regimes, interfacial force modeling, and the application of population balance models for predicting bubble size distribution. The review points out existing challenges, such as regime transition modeling and interfacial force closure limitations, and opportunities for further improvement of experimental methodologies and CFD formulations. This synthesis provides a comprehensive overview of optimizing bubble dynamics for efficient thermal and mass transfer in small-scale two-phase systems.

Keywords: Two-Phase Flow, Bubble Dynamics, Heat Transfer Enhancement, Computational Fluid Dynamics (CFD)

ICMIT/TT23

A comparative review of shape of splitter plate for enhancement of heat transfer

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ABSTRACT

Due to increasing drive for small-scale systems, especially microelectronic devices, substantial attention has been devoted in recent years to their thermal management system to maintain optimal operating temperature to enhance their durability and reliability. Applying splitters in a microchannel improves fluid distribution, as well as nanofluids provide a promising approach for enhancing heat transfer efficiency. This paper presents a depth review of viscous flow around different bluff body shapes, circular, rectangular and square cross-section. It emphasizes how the Reynolds number (Re) and shape of geometry affect the wake characteristics. This study includes the Strouhal number, vortex shedding, blockage ratio, irreversibility rate inside the fluid flow. In this study we analyzed and summarized detailed characteristics of wakes for various bluff body shapes at different Reynolds numbers.

Keywords: Splitter plates, Microchannel, Bluff body, Vortex shedding, Nanofluids

ICMIT/TT25

Implication of Ethanol in Diesel-Moringa blend: An experimental and numerical investigation

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ABSTRACT

This paper investigates the performance and emission traits of a diesel engine running on mixtures of diesel, moringa biodiesel, and ethanol. Key performance metrics such as brake-specific fuel consumption (BSFC), brake thermal efficiency (BTE), and exhaust gas temperature (EGT) were assessed across loads from 25% to 100%. Results showed that BSFC dropped significantly from low to medium loads; D100 had the lowest BSFC (about 0.28 kg/kWh), while DM20 had the highest (approximately 0.38 kg/kWh). BTE increased with load to roughly 34% at maximum load with D100 reaching greater efficiency. EGT rose from around 450 K at 25% load to over 700 K at full load. Ignition delay dropped with increasing load because DM20 had the least delay. Emission experiments revealed DME20 generated the highest CO₂ (~1250 g/kWh), least NO_x (~300 ppm), and smoke across all loads, hence underlining its improved combustion and environmental advantages over pure diesel and other blends.

ICMIT/TT26

Experimental analysis of V-shaped with dimple and protrusion rib on absorber plate in solar air heater

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ABSTRACT

Solar air heater is an equipment that converts solar energy into thermal energy and transfer heat through air for low and moderate temperature in various applications, such as space heating, seasoning of timber and drying of agricultural products etc. However, solar air heaters have low thermal efficiency due to the low heat transfer coefficient between the absorber plate and the flowing air. Applying artificial roughness element is a common technique to enhance thermal performance of a solar air heater. Artificial roughness is provided in the form of protrusions and dimples of various shapes, sizes, and orientations on the underside of the absorber plate. Experimental analysis has been carried out to study the thermal performance of solar air heater having V-shaped with dimple and protrusion rib on absorber plate. The performance of the V-shaped with dimple and protrusion rib model with different roughness pitch such as 10 mm, 20 mm and 30 mm has been compared with smooth V-shaped absorber plate on the basis of heat transfer results. The height of the roughness has been set to 1 mm. This height is equal to the thickness of the viscous sublayer. Experiment has been performed in the range of Reynolds number 4000-16000. It has been observed that all roughened models are obtained higher performance than the smooth V-shaped absorber plate solar air heater. The optimum results is obtained for roughness pitch (P) at Reynolds number.

Keywords: Solar air heater, Dimple and protrusion rib, Reynolds number

ICMIT/TT27

Performance Evaluation of Microchannels with Integrated Microinserts for Enhanced Heat Transfer

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ABSTRACT

This study investigates the enhancement of thermal performance in circular microchannels through the integration of rectangular micro-inserts. Utilizing distilled water as the working fluid under single-phase flow conditions, experiments were conducted on microchannels of 1 mm and 2 mm sizes across a wide range of Reynolds numbers. The impact of micro-inserts on fluid flow resistance and heat transfer efficiency was analysed by evaluating parameters such as pressure drop, Nusselt number, and thermal performance factor (TPF). Results demonstrate that the incorporation of micro-inserts significantly intensifies heat transfer performance, with a concurrent increase in flow resistance. For both channel sizes, the presence of micro-inserts enhanced the Nusselt number and overall heat transfer coefficient, with particularly notable improvements observed at moderate Reynolds numbers. The study concludes that the synergistic effect of reduced channel size and micro-insert addition yields a substantial increase in overall thermal performance, providing valuable insights for the design of high-efficiency microchannel cooling systems.

Keywords: *Microchannel Heat Sink, Micro-inserts, Thermal Performance Factor, Single-Phase Flow, Nusselt Number, Heat Transfer Enhancement, Reynolds Number*

ICMIT/TT30

Heat Distribution in Proton Exchange Membrane (PEM) Fuel Cells using Monte Carlo Simulation

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ABSTRACT

The quest for efficient and sustainable energy systems has placed fuel cells at the forefront of renewable energy technologies. Fuel cells represent a promising and sustainable source of electricity, offering a clean and efficient alternative to traditional power generation methods. Harnessing electrochemical reactions, fuel cells convert the chemical energy stored in fuels such as hydrogen into electricity with minimal environmental impact. However, one of the critical challenges in fuel cell operation is uneven heat distribution, which can adversely impact their efficiency, reliability, and lifespan. This research paper delves into the application of the Monte Carlo technique—a stochastic simulation method—to analyse and model heat distribution within fuel cells. By employing this computational approach, we aim to enhance the understanding of thermal behaviour and develop strategies to optimize fuel cell design for improved performance. The study begins by exploring the fundamental principles of the Monte Carlo method, highlighting its suitability for simulating complex systems with random or probabilistic elements. Fuel cells, characterized by their intricate heat transfer mechanisms involving conduction, convection, and radiation, are particularly well-suited to analysis using this technique. The research outlines the development of a Monte Carlo-based simulation framework, which models the particle interactions and heat transfer processes within the fuel cell. A key focus of the study is the investigation of temperature gradients and their influence on electrochemical reactions in the fuel cell. Uneven heat distribution can lead to localized hot spots, which may cause degradation of the electrolyte membrane and reduce overall efficiency. Through detailed simulations, this research identifies critical zones within the fuel cell where temperature variations are most pronounced. It also examines the interplay between thermal management and fuel cell performance, offering insights into how design modifications can mitigate these issues. In conclusion, this research highlights the significance of heat distribution analysis in fuel cells and showcases the potential of the Monte Carlo technique as a powerful tool for thermal modelling. The outcomes not only pave the way for improved fuel cell designs but also demonstrate the synergy between computational methods and sustainable energy systems, offering promising directions for future research and innovation.

Keywords: PEM, Monte Carlo simulation, PEM

ICMIT/TT31

Experimental Investigation and Performance Analysis of Solar-Powered Electric Vehicle System

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ABSTRACT

Energy is the important factor for improving civilization. Due to increasing the industries and comfort life of human being uses more and more fossil fuel. Uses of fossils fuel disturbed the environment. The transportation industries consume more fossils fuel and causes of global warming. The government focusing and trying to shift transportation industries to renewable source of energy. Solar energy is one of the abundant and free of cost renewable source of energy available in the earth. Application of solar operated vehicles will be a revolution toward a sustainable transportation system. This paper presents the experimental performance analysis of the Solar Powered Electric Vehicle (SPEV) system. The various parameters influencing the vehicle's performance have been studied and analysed. The maximum efficiency was found 40% at 12 kg brake load and system demonstrates stable operation up to a maximum load.

Keywords: Solar powered vehicle; Electric Vehicle; Brake load; Solar energy.

ICMIT/TT32

Experimental cooling performance analysis of thermoelectric module

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ABSTRACT

Energy crisis and environmental deterioration are the major concerns in the 21st century. The refrigerant used in available cooling cycle are not ecofriendly. The researcher and scientist are looking ecofriendly cooling system. The Thermoelectric cooling is achieved by P and N Type semiconductor and it has advantage over commercial refrigeration in reliability, size, weight and temperature control. The present work reported about cooling performance, coefficient of performance and thermal output on the basis of high flow (80l/h) and low flow (20l/h) rate of water circulation through hot side heat exchanger. At a fixed voltage (13V), different properties were obtained and analyzed on hot and cold side of Peltier module.

Keywords: Peltier module, Seebeck effect, thermoelectric cooling, finned tube heat exchanger

ICMIT/TT33

Prismatic Solar Still Using Forced Convection and Steel Balls: An Experimental Approach

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ABSTRACT

Solar energy is one of the energetic elements for life on earth surface and its only source is sun. It is found an abundant quantity and renewable in nature. solar still driven by renewable energy is an advantageous device for household and industry to extract distilled water from salinized water. In this paper experimental investigation have been conducted on prismatic solar still for improvement in productivity. For the investigation Normal Solar Still (NSS), Solar Still with Coated Still Balls (SSCSB) Solar Still with Forced Convection at Low Speed (SSFC1) and Solar Still with Forced Convection at High Speed (SSFC2) have been considered. The comparative improvement in performance have been presented in paper for Prismatic Solar Still (PSS). All the experiment has been conducted in BIT Sindri, Dhanbad, Jharkhand, India at Latitude 23.6595 N and Altitude 86.4741E in actual climatic conditions. A significant improvement was found as SSCSB, SSFC1 and SSFC2 of 12.72%, 34.54% and 58.18% respectively.

Keywords: Prismatic solar still, Coated Still Balls, Forced Convection, productivity

ICMIT/TT36

Experimental investigation of diesel pool fires inside a naturally ventilated compartment

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ABSTRACT

An experimental approach has been shown to investigate the burning of diesel fuel using fuel pans with diameters ranging from 0.3 to 0.6 meters, within a 64 m³ cubical compartment. During the pool fire test, combustion efficiency, centerline temperature, incident heat flux, and heat release rate (HRR) were recorded. The heat flux was measured using SBG01 water-cooled heat flux sensors, manufactured by Hukseflux, which were installed in the walls, floor, and ceiling of the compartment. Pool centerline temperatures were measured using a K-type thermocouple and compared with various correlations available in the literature from similar studies. The peak heat release rates calculated using the Large-Scale Heat Release Calorimeter (LSHRCal) were 34.28, 89.87, 153, and 224.92 kW for pool diameters of 0.3 m, 0.4 m, 0.5 m, and 0.6 m, respectively. The combustion efficiency of these pool fires ranged from 0.92 to 0.96, indicating that the experiments were carried out under well-ventilated conditions. The findings of the present study may serve as a valuable reference for validating design fire numerical models applicable to Nuclear Power Plants (NPPs).

Keywords: Diesel pool fire, compartment fire, large-scale heat release calorimeter, heat flux sensors.

ICMIT/TT38

Design, Development, Testing and Performance Evaluation of an Earth-Subsurface Air-Cooling System Using Buried Copper Pipes

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ABSTRACT

This study presents the design, fabrication, and experimental evaluation of a passive air-cooling system that utilizes the earth's subsurface thermal stability for energy-efficient cooling. The system consists of copper pipes buried at a depth of 12 feet, forming a closed-loop system where ambient air is circulated and cooled by virtue of conductive heat exchange between air, copper pipe and the surrounding soil. The reason that copper pipes were chosen for maximum efficiency of heat transfer is due to its superior thermal conductivity. This system was installed and tested under real climatic conditions to evaluate its cooling performance. The parameters considered are inlet/outlet air temperature and air velocity. Results shows a significant temperature drop in the circulating air, exhibiting the potential of such systems for reducing dependence on conventional air conditioning, particularly in hot and humid regions. The study further analyses the influence of pipe length and flow rate on system performance. The findings support the potential of earth-subsurface air-cooling systems as a sustainable alternative for building thermal comfort.

Keywords: Thermal Conductivity, Passive Cooling, Subsurface Thermal Energy, Copper Pipe Heat Transfer, Sustainable Building Systems, Geothermal Air Cooling, Energy-Efficient Ventilation

ICMIT/TT39

Some Studies on the straight parallel flow field design in proton exchange membrane fuel cells

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ABSTRACT

In an era of where environmental concerns have reached a critical point, the quest for clean efficient energy sources have become paramount. Proton Exchange Membrane fuel cells (PEMFCs) offer a glimpse of hope, with their ability to directly convert the chemical energy of fuels into useful work, leaving negligible environmental impact. However, maximizing the performance of PEMFCs requires innovative design approaches, particularly in the realm of flow field designs (FFDs) of the bipolar plates (BPs). The study will present a comprehensive analysis of the performance and local species transport phenomenon of PEMFCs, focusing on the impact of geometric designs and FFDs. An enhanced FFD provides more homogenous reactant diffusion in the cell, lowers the pressure drop and provides higher electrical performance. The FFDs can be improved by adding baffles in the flow channel, developing completely new and innovative FFDs or adopting nature/bio-inspired designs. However, a detailed analysis of various aspects of FFDs like cell performance, cost and manufacturing difficulties has to be done. The aim of the project is to analyse and optimise the design of flow in PEMFCs using computational fluid dynamics (CFD) simulations and Design of Experiment (DoE) methodologies. The objective is to identify channel geometries and operating conditions that maximise power density, minimize pressure drop and ensure uniform reactant distribution. Different channel geometries, including serpentine, interdigitated and parallel channels will be investigated. The effect of parameters such as channel width, depth and spacing will be explored through CFD simulations. DoE Techniques will be employed to efficiently explore the design space and identify optimal channel configurations. The results of this study will provide valuable insights into design and development of high-performance PEMFCs for various applications including automotive, stationery power generation and portable electronics.

Keywords: Computational fluid dynamics (CFD), Proton Exchange Membrane fuel cells (PEMFCs), Design of Experiment

ICMIT/TT41

Performance Evaluation of a 1.14 kWp Solar Panel for Lucknow Region Using “PVsyst”

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ABSTRACT

In the modern era, the energy demand is increasing due to a high rate of consumption. In 2023, over 60% of global electricity was generated from fossil fuels, which have various adverse effects on the environment, such as global warming, air pollution, and water pollution, etc. To address this issue, the focus is shifting towards renewable energy sources such as solar and wind energy. Solar energy is considered the most prominent due to factors like India's geographical location and the availability of maximum sunlight hours. This research focuses on evaluating the performance of a 1.14 kWp grid-connected solar PV system in the Lucknow region using PVsyst simulation software. This study was conducted to evaluate the performance in advance by making it a practical tool for pre-stage planning instead of a physical setup. Weather data from meteonorm8.2 was used to evaluate the system's accuracy, focusing on important parameters such as annual energy generation, performance ratio (PR), and system losses. Monthly weather parameters such as global horizontal irradiation (GHI), diffuse irradiation, ambient temperature, and wind speed were factored in for the emulation of realistic operating conditions. These parameters help in determining how much energy can be generated by the system and its efficiency. It also represents how weather patterns affect performance annually. Losses caused by temperature, inverter, inefficiencies, and other components were examined and visualized in loss diagrams. It is estimated that 1528.8 kWh of electricity could be generated annually, with a high PR of 83.61%. This signifies that the system would operate efficiently under the normal weather conditions found in Lucknow and show favorable results in electricity generation. Overall, this research analysis supports smarter system design and more efficient execution, for better planning, resource use, and sustainability for solar PV projects.

Keywords: Solar PV; PVsyst; Performance Ratio; Renewable Energy; Pre-installation Analysis; Weather Data; Climate Impact; Loss Analysis

ICMIT/TT43

A Numerical Study on Heat Transfer Enhancement in Solar Air Heater with Circular Perforated Finned Absorber

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Abstract

The current study presents a numerical analysis of heat transfer enhancement in a solar air heater using circular perforated fins attached to the absorber plate along the fluid flow path. A mathematical model has been developed to evaluate the thermal performance of this modified solar air heater design. The effects of fin pitch, hole diameter, number of perforations, and duct depth on the Nusselt number and thermal efficiency were analyzed at varying mass flow rates. Compared to solid fins, the use of perforated fins with an 8 mm hole diameter at a mass flow rate of 0.04 kg/s resulted in a 16.9% increase in the Nusselt number. Both the Nusselt number and thermal efficiency increased with a higher number of perforations and larger hole diameters.

Keywords: Perforated fins, Thermal performance, Nusselt Number

ICMIT/TT45

A Comprehensive Review of Ocean Energy Sources and Their Challenges in India

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ABSTRACT

This paper presents a summary of ocean energy technologies as a promising form of renewable energy. It explores various ocean energy resources and the innovative techniques developed to harness them. Ocean energy also called marine energy comes from the movement of waves and tides, as well as from temperature and salinity differences in the sea. With the potential to significantly contribute to future energy systems, ocean energy can also help reduce carbon emissions and stimulate economic growth, particularly in coastal and remote areas where electricity access is limited and transmission is expensive. Interest in ocean energy is rising globally, including in India, which has a long coastline. Many nations are now working to generate electricity from marine resources. However, widespread adoption is currently limited due to technological, economic, and environmental challenges, as most marine energy systems are still in the developmental phase. To overcome these barriers, increased investment in research and development from both governments and the private sector is essential. This study aims to support scientific and industrial efforts to utilize diverse ocean energy sources, paving the way for a cleaner and more sustainable energy future.

Keywords: *Ocean; Renewable Energy; Tidal; Wave; Ocean thermal Energy*

ICMIT/TT46

Adaptive Design, Fabrication, and Performance Analysis of a Thermosyphon Solar Water Heater using a Flat Plate Collector

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ABSTRACT

This study investigates the adaptive design and performance analysis of a thermosyphon solar water heater [TSWH] using a flat plate collector [FPC], focusing on optimizing thermal efficiency and system reliability. This design incorporates a copper absorber plate with selective coating and riser tubes bonded to maximize heat transfer to water. By studying various previous thesis and research work we optimized the collector parameters [50-80°C] under varying solar irradiance [400-1000W/M²]. Under real-world conditions, experimental validation demonstrated efficient collector performance, achieved through thermosyphon- driven circulation that removes pump dependency. Results show that a 0.62×0.62 m FPC paired with a storage tank meets daily hot water demands for households of Indian subcontinent.

Keywords: Thermosyphon solar water heater [TSWH], Flat plate collector [FPC], Copper absorber plate

ICMIT/TT47

Computational Analysis of Soaking of Steel Bloom in Reheating Furnace

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ABSTRACT

Steel blooms and billets are reheated in the reheating furnaces before hot working. This soaking of the steel blooms are to be precisely controlled for proper rolling afterwards. Many a times it is found that during rolling defective products are generated. That happens due to either overheating or incomplete or insufficient heating. Therefore, detecting the proper heating is very important for reducing the defective products and reducing the productivity hour losses. This work aims to numerically predict the condition of soaking from the pyrometer data of the exit of the reheating furnace, using heat conduction equations to calculate temperature distribution over time. Based on the predictions, the soaking time needed to reach the target temperature can be determined.

Keywords: Soaking pit, Reheating furnace, Computational fluid dynamics, Bloom, Bilet

ICMIT/TT48

Analysis of thermo-viscoelastic waves inside a Polymer micro-rod caused by instantaneous heat under Atangana-Baleanu fractional derivatives framework

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ABSTRACT

Classical thermo-viscoelastic model is observed to be unsuccessful to forecast the thermo-mechanical characteristics inside the thermo-viscoelastic materials. The current study establishes an efficient modified thermo-viscoelastic model including fractional-order derivatives at the micro-scale to record the memory-dependent effect and the size-dependent effect during thermal-mechanical interactions in a polymer rod type material influenced from a thermal shock. Governing equations are formulated and solved by the method of Laplace Transform. Graphical results are illustrated to express the effects of the fractional- order parameter, the fractional- order strain parameter, the mechanical relaxation parameter and the non-local parameter on the variations of the physical fields- temperature, displacement and thermal stress. Author believes that this work may be advantageous for the academicians to understand the effects of key parameters on a polymer type material.

Keywords: Thermo-viscoelastic model, Fractional-order derivatives, Polymer rod

ICMIT/TT49

**Combustion Characteristics of Fuel Briquettes Produced from Coal Fines and Mangifera Indica
Sawdust with Jaggery as a Binder**

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ABSTRACT

Fuel briquettes made from Mangifera Indica sawdust and fine coal with jaggery powder used as a binder. Sawdust and fine coal were mixed in different ratios such as, 100:0, 90:10, 80:20, 70:30, 60:40, 50:50, 40:60, 30:70, 20:80, 10:90 and 0:100 (wt/wt). In each briquette 10% jaggery powder was added for binding purpose. Manual screw press machine was used to fabricate the cylindrical briquette (radius 1 cm and height 5 cm) with loading time 15 min. Proximate and ultimate analysis were done for each briquette according to ASTM methods. Comparison for analysis of results of each fabricated briquette were compared. Maximum heat content and minimum ash value was found in 100:0 (wt/wt) fabricate briquette. Fixed carbon content and heating value were decreased when saw dust in different ratio was added in coal fines. Elemental composition of carbon, hydrogen, and oxygen were found in ranged of 50 to 44 %, 5.48 to 5.55%, and 38.43 to 41.43% for different fabricated briquette. Optimum results were found with 70:30 (wt/wt) fabricate briquette.

Keywords: Fuel Briquette; Binding; Proximate Analysis; ASTM Methods; Fixed Carbon

ICMIT/TT50

Multi-Objective Optimisation of Shell and Tube Heat Exchanger Using Nsga-Ii Model

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ABSTRACT

This study examines the use of NSGA-II for multi-objective optimization of shell-and-tube heat exchanger (STHE) design, balancing thermal efficiency and cost. The algorithm optimises nine vital design variables, achieving a solution with 0.50 effectiveness and an annual cost of \$1337, outperforming the theoretical calculations (0.326 effectiveness, \$2211.16 cost). Important factors like tube length and outer diameter play a critical role in shaping the performance of the heat exchanger. Visualization tools like parallel coordinate plots make spotting the optimal design combinations easier. This method highlights how effective is NSGA-II in achieving cost-efficient, high-performance shell and-tube heat exchanger designs tailored for industrial use.

Keywords: Shell and tube heat exchanger, genetic algorithm, multi-objective optimization, NSGA-II, optimization.

ICMIT/TT51

Experimental Investigation of Nanoparticle Integration in Biodiesel Blends for Enhanced Performance and Reduced Emissions

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ABSTRACT

The growing energy demand, exhaustion of fossil fuel resources, and the rising speed of fuel utilization along with the generation of hazardous release have driven extensive research into eco-friendly fuels. Biodiesel was generated from Wheat germ oil using a two-step process: acid esterification followed by transesterification. A 20% Wheat germ oil biodiesel mix (by volume) was equipped by mixing biodiesel with conventional diesel. To enhance the fuel properties, nano-additives such as iron oxide (Fe_2O_3) were incorporated. These additives were blended with the biodiesel blend at concentrations of 25, 50, 75, and 100 ppm, respectively. This study aims to evaluate the performance and exhaust emissions of a diesel engine operating on a wheat germ oil biodiesel blend enhanced with nano-additives. An experimental test rig was set up on a single-cylinder diesel engine to evaluate engine performance and emissions under varying load conditions. The results indicated that the biodiesel blend with nano (Fe_2O_3) (WGOA175) achieved the highest improvement in thermal efficiency, with a gain of 7.2% correlated to all other tested fuels. Additionally, the Wheat germ oil biodiesel blend enclosing (Fe_2O_3) (WGOA150) bring about the greatest reduction in emissions, lowering carbon monoxide (CO) and nitrogen oxides (NO_x) by approximately 31% and 41.6%, respectively, when compared to all other fuel blends.

Keywords: Nanoparticle, Biodiesel, iron oxide (Fe_2O_3)

ICMIT/TT52

Flow Past an Oscillating Cylinder: Influence of Reynolds Number and Oscillation Frequency on Wake Development

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ABSTRACT

This manuscript delineates a computational investigation into the two-dimensional fluid dynamics surrounding a circular cylinder subjected to transverse oscillations across a spectrum of Reynolds numbers ($Re=100,500,1000,2500$) and non-dimensional oscillation frequencies (F). The simulations were executed using ANSYS Fluent 2025 R2, employing uniform inlet velocity and pressure outlet boundary conditions, by the methodology outlined in. The analysis encompassed wake characteristics, vortex dynamics, and flow asymmetries, evaluated based on the velocity contours generated for each scenario. The findings elucidate the significant impact of oscillation frequency and Reynolds number on vortex shedding, wake elongation, and the transition to turbulence. The outcomes provide valuable insights into flow-induced vibrations relevant to engineering applications.

Keywords: Oscillating cylinder Two-dimensional flow, Reynolds number, Wake Dynamics, Vortex shedding

ICMIT/TT53

Emerging Innovations in Thermal Insulation materials to promote energy conversation
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ABSTRACT

The increasing demand for energy efficiency and sustainable practices across diverse industries has intensified the need for innovative thermal insulation materials. This paper explore emerging trends in thermal insulation materials and techniques and highlights the involvement in the field of energy conservation and also addressing the limitations of traditional approaches. The trends indicate a move towards more efficient, sustainable and technologically advanced insulation solutions to meet the growing demand for energy conservation in industrial facilities, buildings and even transportation to prevent heat loss or gain. Some of key trends which participate in insulation are aerogels, vacuum insulation panels (VIPs), bio based and sustainable materials, phase change materials (PCMs), smart and adaptive insulation, nanotechnology enhanced insulation, reflective and radiant barrier systems. All these trends are having different proportions, such as some of them are highly porous materials, some of them have low thermal conductivity, good insulation, and eco-friendly and renewable options, and so on. Additionally, smart insulation systems capable of adopting to environmental change are gaining attention. These advancements contribute to energy conservation by minimizing heat loss, lowering heating and cooling demands, and supporting global efforts toward carbon neutrality. Considering these characteristics, the solution could substantially improve thermal and also acoustic comfort for energy-poor households, while also delivering notable environmental advantages and meaningful social benefits. This paper explores these emerging trends, evaluating their potential and challenges for widespread adaptation.

Keywords: Thermal Insulation, Innovative Materials, Sustainable construction, PCMs, VIPs, Aerogels.

Track 3

Design and Simulation

ICMIT/DS03

SIMULATION OF PASSIVE AIR-COOLING SYSTEM FOR AN ARC WELDING MACHINE

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ABSTRACT

Welding is a critical process in various industries, where metal components are joined together to create strong and durable structures. However, one of the significant challenges faced during welding is managing the heat generated in the welding area. Excessive heat can lead to problems such as warping, distortion, and reduced weld quality. Passive air-cooling system uses natural circulation of air around the welding zone to reduce temperature, improving weld quality, reducing overheating risks, and extending equipment lifespan. This demand requires critical examination of Passive air-cooling during arc-welding. A mathematical model has been developed and simulated and observed the dissipation of heat in the weld-bead as well as the work-pieces. The result obtained can be extended to develop alternative cooling systems.

Keywords: Passive cooling system, Arc welding, Simulation, Natural circulation, Overheat

ICMIT/DS05

Theoretical Simulations and Experimental Studies on the Crack Growth of Cr-Mo alloy steel using Finite Element Method**Md Asjad Mokhtar, Ankit Kumar*, Dr. Sandeep Tiwari***Darbhangha College of Engineering, Mabbi, Darbhanga, Bihar**Corresponding Author Email ID: ankitr606@gmail.com***ABSTRACT**

In the modern developing world structural steel, alloys and other metals are subjected to extreme and difficult loading conditions which leads to its failure in various modes. The ductile properties of such structural members must exhibit significant amounts of crack propagation prior to failure to avoid any catastrophe. Therefore, for estimation of structural reliability, computational simulation and experimental justifications on the stable crack growth turn out to be an important aspect. Recent studies confirmed Crack Opening Displacement (COD) as a parameter for predicting the onset of crack growth in thin sheet materials. Similarly, the crack initiation and propagation in ductile materials are better described on the basis of critical CTOD (crack tip opening displacement) and J-Integral concept. The present work investigates crack propagation using finite element approach in the software package ABAQUS. Two specimens, namely Three Point Bend Specimen and Compact Tension (CT) Specimen, subjected to Mode I loading were analyzed for fracture with COD criteria. Contour integral (J-Integral) method is also implemented to find J values and related curves such as δ -Resistance and J-Resistance curves. The material used are Cr-Mo alloy steel and mild steel and the obtained results are compared with the experimental work. Experiments were performed on the Compact Tension (CT) specimen. Couple of samples have been tested to find out the fracture toughness in terms of critical CTOD and Critical J-integral. Load Vs COD plot was obtained directly from experimental results. Finite Element Analysis of CT specimen using cohesive element to simulate crack is carried out and the results are compared with another FE simulation.

Keywords: Crack Growth, Crack Tip Opening Displacement (CTOD), Resistance curves, J-Integral, Fracture criteria, Cohesive element.

ICMIT/DS06

Role of artificial intelligence and machine learning in metal casting- A Review

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ABSTRACT

Metal casting plays very important role in manufacturing components with desired features which involves pouring liquid metal into a mould to create 3D objects. In present time artificial intelligence and machine learning is used to control process and quality in metal casting. In addition to that development of optimization methods using artificial intelligence and machine learning helps to minimize defects and increases reliability of the system. Different researchers did wide research work in applications of these methods/tools in aspects of casting process optimisation, casting quality checking along with control of defects, intelligent design and materials selection. Artificial intelligence (AI) can help monitor and control heat treatment processes in real time, which can lead to better quality and fewer defects. In present time to achieve the target of sustainable manufacturing in metal casting, *artificial intelligence (AI)* is applied to reduce scrap and energy Usage in the *casting process in addition to greenhouse gas (GHG)* emission reduction. The review article helps foundry engineers and researchers to provide the data for improving different steps in the metal casting process for enhancing net-shape casting with high reliability and integrity. It is challenging task to make metal casting process more efficient and full-scale control, with the defects, microstructure, performance, and service life of the fabricated castings can be accurately predicted and tailored.

Keywords-intelligent casting; 3D printing; artificial intelligence , machine learning; process control; quality casting system; heat treatment

ICMIT/DS08

Fault Diagnosis and Condition Monitoring in Robotic arm joint using DCNN

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ABSTRACT

This paper aims to predict and classify sensor faults in a robotic arm joint using Deep Convolutional Neural Network (DCNN). The study aims to work solely on encoders sensor to classify drift and noise occurrence so that a course of action can be taken to prevent a catastrophe. The methodology is proposed using a Universal Robot 3e (UR3e) robotic arm's dataset. The open-source dataset of UR3e robotic arm simulation gave various insights to parameters involved in a robot's task initiation, task processing and task completion. Feature engineering was done to combine parameters, study the relation between the combined parameters, find out the patterns that lead to the fault so that the root cause can be treated. Earlier studies on other versions of UR robotic arms or any other robotic arms have worked using Wavelet transformation, Fuzzy Logic and Artificial Neural Networks (ANN); it has been realized that the modern approach of ANN is more efficient. We adopted the similar approach of using deep learning technique like DCNN to classify encoder faults for UR3e robotic arm which find its application in screwdriving, painting, and pick-and-place tasks.

Keywords: Fault Diagnosis, Encoder, Deep Convolutional Neural Network, Robotic Joint

ICMIT/DS10

Effect of molding sand constituents on casting quality: an RSM-based study

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ABSTRACT

The casting process is a fundamental manufacturing method used for producing complex metal components with varying geometries and applications. However, the quality and efficiency of cast products are significantly influenced by multiple process parameters such as moisture percentage, clay percentage, silica oxide (SiO₂) percentage etc. This study focuses on the experimental investigation and optimization of key casting process parameters using the Response Surface Methodology (RSM) & Statistical Analysis (ANOVA), a powerful statistical tool for modeling and analyzing problems where several variables influence the output responses. A series of experiments were conducted to understand the influence of selected parameters on critical output responses such as surface finish, dimensional accuracy and mechanical properties of the cast product. The collected data was used to develop regression models and generate response surface and contour plots. The interaction effects between process parameters were analyzed to identify optimal conditions that yield the best performance. The results indicate that the most significant factor affecting surface finish of the cast product is clay percentage and next important factor in this respect is moisture percentage followed by SiO₂.

Keywords: Casting, RSM, ANOVA, Surface Finish, Mechanical Properties

ICMIT/DS11

Design and Performance Analysis of Sustainable Automotive Bumpers Using Glass Reinforced Plastic and Recycled Rubber Composites

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ABSTRACT

With growing emphasis on environmentally friendly and lightweight materials in the automotive industry, this research aims to evaluate the mechanical viability of sustainable bumper alternatives. The bumper model was designed using CAD software and subjected to static structural analysis using Finite Element Analysis (FEA) in ANSYS simulation software to assess stress distribution, deformation, and energy absorption capacity under impact loading conditions. To validate the simulation results, physical prototypes were fabricated and tested using a Universal Testing Machine (UTM) for static load-bearing capability and mechanical strength. The comparative analysis between simulated and experimental results confirmed good agreement, demonstrating that the Glass Reinforced plastics (GRP)-recycled rubber composite bumper offers sufficient structural integrity, excellent energy dissipation, and a reduced environmental footprint. The study concludes that such sustainable composite materials can effectively replace traditional materials in automotive bumper systems without compromising safety or performance.

Keywords: Sustainable bumper, CAD software, ANSYS simulation, Universal Testing Machine (UTM), Glass Reinforced plastics (GRP), Traditional materials

ICMIT/DS12

Optimizing Front Axle Design for Commercial Vehicles: Enhancing Strength-to-Weight Ratio through FEA and Experimental Validation

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ABSTRACT

In the competitive manufacturing landscape, accelerating new product design cycles while minimizing costs is critical. The front axle, occupying up to 40% of a vehicle's cargo capacity, requires optimization to enhance its strength-to-weight ratio while maintaining a specified safety factor and assembly constraints. The front axle serves as a central shaft facilitating wheel rotation, supporting the vehicle's frontal structure, enabling steering, and absorbing road-induced shocks. Failure under diverse loading conditions can lead to severe consequences, necessitating robust analysis. This study investigates the front axle of a commercial vehicle under varying road surface-induced loading conditions. The axle was modeled using CAD software and analyzed via finite element analysis (FEA) in ANSYS 2021, employing shape and material optimization tools. Experimental validation was conducted using a universal testing machine (UTM), with results correlated against FEA outcomes to ensure accuracy.

Keywords: Front axle, Strength-to-weight ratio, Finite element analysis, Shape optimization, Material optimization.

ICMIT/DS17

Design and Simulation Analysis of an In-Plane Comb Drive MEMS Accelerometer
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In this study, we detailed the design and simulation of a MEMS-based in-plane comb drive accelerometer. Our research focuses on understanding and optimizing its performance through differential capacitive sensing. The accelerometer consists of interdigitated comb fingers attached to a suspended proof mass, which is supported by flexible beams. This configuration enables precise in-plane motion in response to acceleration within a $\pm 10g$ range. Analytical methods were used to determine critical design parameters such as number of fingers, gap spacing, proof mass dimensions, and suspension beam stiffness to ensure optimal sensitivity, stability, and mechanical reliability. Simulation studies were performed using COMSOL Multiphysics to evaluate the accelerometer's displacement behavior, natural frequency, capacitance variation, and dynamic response. The results demonstrate high linearity, efficient sensitivity tuning, and strong shock tolerance under dynamic loading conditions. The simulated performance closely aligns with theoretical predictions, validating the design and highlighting its effectiveness for capturing rapid motion with accuracy. This work lays a strong foundation for the future fabrication of the device and suggests potential enhancements in beam geometry and damping mechanisms to further improve robustness for integration in automotive, navigation, and portable consumer electronic systems.

Keywords: MEMS, Comb Drive, In-Plane Accelerometer, Capacitive Sensing, COMSOL Multiphysics, Sensitivity, Shock Response

ICMIT/DS18

Experimental Study and Vibration-Based Condition Monitoring of Spur Gearbox Under Variable Load Conditions for Enhanced Reliability and Performance Optimization**Saicharan Suryavanshi*, Patil Rupesh J, Jitendra Atmaram Hole, Nilesh S. Navagale***Rajashri Shahu College of Engineering Tathwade, Pune-411033 saicharansuryavanshi@gmail.com***Corresponding Author Email ID: suryavanshisaicharan@gmail.com***ABSTRACT**

The reliability of gearboxes, particularly those with spur gear arrangements, is critical in various industrial applications, as they are subjected to varying load conditions that can lead to wear, failure, and costly downtime. This paper presents an experimental investigation and vibration-based condition analysis of a spur gear gearbox under varying load conditions. The primary objective is to assess the impact of different loading scenarios on the gearbox's vibration characteristics and develop strategies for early fault detection. Using advanced vibration analysis techniques such as Fast Fourier Transform (FFT), wavelet transform, and machine learning models, we analyze vibration signals collected from the gearbox under light, medium, and heavy load conditions. The results show that vibration patterns can be effectively used to identify gear faults and predict potential failures. By integrating machine learning-based predictive maintenance models, the study demonstrates how vibration data can be utilized to enhance the reliability and longevity of gearboxes. The findings provide valuable insights into vibration-based condition monitoring, offering a proactive approach to maintaining gearbox performance and reducing unplanned downtime in industrial environments.

Keywords: vibration analysis, gearbox, spur gear arrangement, varying load conditions, condition monitoring, fault detection, Fast Fourier Transform (FFT), wavelet transform, machine learning, predictive maintenance, reliability, early failure prediction, industrial applications

ICMIT/DS19

Design and Implementation of an IoT-Based Smart Water Quality Monitoring System Using Turbidity Sensors and Node-RED

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ABSTRACT

Access to clean water is a critical global challenge, and traditional water quality monitoring methods often require manual sampling and laboratory analysis, leading to delays and limited real-time data availability. To address this issue, this work focuses on the development of an IoT-based water quality monitoring system capable of providing real-time feedback. The proposed system was simulated using Node-RED software, offering a robust platform for modeling and analysis. For hardware implementation, components such as a turbidity sensor, NodeMCU ESP8266 microcontroller, and Arduino IDE were employed. In the monitoring setup, the turbidity sensor effectively detected the turbidity levels of different water samples. The project successfully demonstrated a basic yet functional IoT water quality monitoring system using readily available and cost-effective components. A step-by-step methodology was followed, covering hardware connections, code implementation, data acquisition, and online data visualization. The Arduino code processed the turbidity sensor readings and categorized the water quality into "clean," "cloudy," and "dirty" based on predefined value ranges. The serial monitor provided immediate feedback for each sample, validating the sensor's capability to distinguish between varying turbidity levels. Additionally, the integration of an MQTT broker and Node-RED enabled real-time visualization of the water quality data on an interactive dashboard. This work highlights the potential of IoT technologies in remote environmental monitoring and provides a foundation for the development of more comprehensive water quality monitoring systems in the future.

Keywords: Water Quality Monitoring, Turbidity Sensor, NodeMCU ESP8266, Arduino

ICMIT/DS20

Parametric Analysis and optimization of Sound absorption coefficient in anechoic chamber

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ABSTRACT

This study presents a comprehensive parametric analysis and optimization of the Sound Absorption Coefficient (SAC) within an anechoic chamber using both theoretical modeling and experimental validation. In response to increasing demands for quieter environments across consumer and industrial applications, three commonly used acoustic materials—polyurethane foam, felt cloth, and glass wool—were evaluated over a frequency range of 125–2500 Hz. A wedge-shaped absorber was designed based on Leo Beranek's criteria, with dimensions optimized for low-frequency performance. The Transfer Function Method was employed within an impedance tube setup to measure SAC experimentally. Simultaneously, theoretical predictions were obtained using the Delany and Bazley model, incorporating key material properties such as density, flow resistivity, and porosity. The results show that felt cloth performs best at lower frequencies (125–500 Hz), while PU foam and glass wool demonstrate superior absorption at higher frequencies (1000–2500 Hz). A clear trend of increasing SAC with rising frequency was observed for all materials. Comparative analysis highlights a good correlation between experimental and theoretical results, with felt cloth exhibiting the lowest percentage error, indicating its reliability in modeling. These findings offer valuable insights for the optimized selection of acoustic materials in anechoic chambers and similar noise-control environments.

Keywords: Anechoic chamber, Frequency, Absorbing Material, polyurethane foam, felt cloth, Glass-wool, Impedance Tube Methods, transfer matrix method

ICMIT/DS21
Finite Element Analysis and Structural Improvement in Industrial Helmet using Composite Material

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ABSTRACT

This study investigates the development of composite materials for improving the safety and performance of protective helmets used in hazardous environments. By integrating coir and carbon fibers through advanced manufacturing techniques, the design leverages their synergistic properties to enhance structural integrity and impact resistance. Finite Element Analysis (FEA) is utilized to evaluate the mechanical performance of helmets made from hemp and Kevlar blends. Hemp, a sustainable and lightweight natural fiber, is combined with high-strength carbon fibers to create an eco-friendly yet robust helmet. The research involves formulating composites with varying hemp-to-Kevlar ratios, followed by prototyping helmets. A parametric model is created using SolidWorks 2022's 3D modeling module, and FEA is conducted via ANSYS 2021 to analyze the composite helmets' behavior.

Keywords: Composite materials, Helmet safety, Hemp fiber, Carbon fiber, Finite Element Analysis

ICMIT/DS22

Modelling and Analysis of Self-Balancing and Anti- Collision System in Two-Wheeler

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ABSTRACT

Two-wheeled vehicles constitute a primary mode of transportation globally, despite their advantages, a significant drawback remains their inherent instability, especially during maneuvering and sharp turns, which increases the likelihood of accidents. This project aims to address this critical safety concern by developing a self-balancing two-wheeler model incorporating a gyroscopic stabilization mechanism and an Arduino-based anti-collision system. A gyroscope is strategically integrated at the midpoint between the two wheels to enhance dynamic stability by leveraging the principle of gyroscopic precession. The common issue in conventional two-wheeler designs is the fixed position of the engine and power transmission components, which cannot feasibly be relocated to the centre due to structural and functional constraints. Furthermore, the model incorporates a basic anti-collision system, implemented through Arduino programming, which employs ultrasonic sensors to detect and respond to nearby obstacles, thereby further reducing the risk of accidents. The proposed system is modelled to determine its effectiveness in maintaining balance during turns and preventing collisions. Preliminary results indicate a significant improvement in the maximum permissible turning angle and overall vehicular stability.

Keywords: Two-Wheeler, Balancing, Precession, Gyroscope, Anti-Collision System, Mass Distribution, Stability, Safety

ICMIT/DS23

EVALUATION OF JOHNSON-COOK MATERIAL MODEL PARAMETERS OF AA5083

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ABSTRACT

The Johnson-Cook (J-C) material model is widely used to predict the behaviour of materials under extreme conditions, such as high strain rates and temperatures. This study aims to develop and validate the J-C material model parameters for Aluminium 5083, a non-heat-treatable alloy known for its excellent corrosion resistance and high strength-to-weight ratio. The parameters include strain hardening, strain rate sensitivity, and thermal softening components, which are crucial for accurately simulating the material's response under dynamic loading conditions. The proposed model will be calibrated using experimental data from quasi-static and dynamic tests, including tensile, compression, and impact tests at various temperatures. The results of this study will provide valuable insights into the mechanical properties of Aluminium 5083 and enhance the predictive capabilities of the J-C model for this specific alloy. This research will contribute to the design and optimization of lightweight, high-performance structures in various engineering fields.

Keywords: Johnson-Cook Model, Aluminium 5083, Strain Hardening, Strain Rate Sensitivity, Thermal Softening, Finite Element Simulation

ICMIT/DS24

A Review on Analysis of Automobile lower control arm

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ABSTRACT

The lower control arm (LCA) is a fundamental component of an automobile suspension system, playing a crucial role in vehicle stability, handling and ride comfort. The lower control arm is subjected to many loads due to variation in gross weight and impacts due to fluctuation of road surface and additional forces. The lower control arm is a unique type of independent suspension used in machine vehicles. The experimental findings are validated using Finite Element Analysis (FEA) performed in ANSYS Workbench to analyse stress distribution and deformation behaviour. New design of automobile lower control arm is manufactured. Future research directions are also outlined to enhance the reliability of lower control arm. In this paper, a review of recent literature has been conducted, and conclusions have been drawn based on the findings.

Keywords: Cast Steel, Aluminum, Stress Concentration, Finite Element Analysis (FEA), Control Arm etc.

ICMIT/DS25

Stress Optimization and Shape Optimization of Automobile lower control arm Using FEA and UTM Testing

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ABSTRACT

In automotive suspension, a control arm is a hinged suspension link between the chassis and the suspension or hub that carries the Wheel. A detailed design of the existing LCA is conducted, followed by the application of FEA to simulate stress distribution and identify critical failure points under varying loading conditions. Then static analysis will be performed to find out stress deformation of existing sheet metal lower control arm. Stress is critical position while considering design of lower control arm to fatigue failure of lower control arm therefore stress optimization will be performed. The results obtained from FEA highlight areas of excessive stress concentration, guiding the optimization process. Subsequently, shape optimization techniques, including topology optimization, are employed to develop a more efficient design with reduced material usage while maintaining mechanical integrity. Then new optimized design of lower control arm manufactured in pipe structure. SOLIDWORKS 2022 software will utilized to design the suspension arm. ANSYS 21 software will also use for analysing the structural strength and optimize the parts weight. Experimental testing will be performed on universal testing machine. (UTM)

Keywords: Cast iron, Aluminium, Stress Concentration, UTM testing, Finite Element Analysis (FEA), Ansys, SOLID WORKS etc.

ICMIT/DS26

A Review on Optimization of Engine Mounting Bracket Using Finite Element Analysis

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ABSTRACT

Engine mounting brackets play a crucial role in securing components such as engine and transmission system to the vehicle's chassis. They also help in reducing noise, vibration and harshness caused due to engine and thus improving vehicle's comfort. These brackets are subjected to a variety of static and dynamic loads, including vibrations, impacts and thermal stresses during vehicle operation. The design and optimization of engine mounting brackets are important to ensure structural integrity, reduction in vibration and improved performance in automotive applications. This paper includes the study of analysis and optimization of different engine mounting brackets. Proper design and analysis of the brackets helps in reducing vibrations and keeping the stresses within a safety limit. The analysis of the bracket includes static and dynamic analysis, which can be carried out through Finite Element Analysis (FEA). Optimization of design can be achieved by topology optimization, by which stress distribution and implicit points of failure can be determined. It is a technique by which the surface of a component is optimized to get the required shape and hence reducing the weight. Thus, the weight reduction can be achieved without compromising the strength.

Keywords: Engine Mounting Bracket, Finite Element Analysis (FEA), Static Analysis, Optimization, etc.

ICMIT/DS29

Classification of Eye Refractive Error Using Machine Learning Techniques**Dharmendra Kumar^{1*}, Narayan Kumar¹, Suyash Vikram¹, Vivek Kumar²,
Kumar Abhinav²**¹*Dept. of Mechanical Engineering, Government Engineering College, Vaishali, India*²*Dept. of Electrical Engineering, Government Engineering College Vaishali, India***Corresponding Author Email ID: dharmendra.kr.gecv@gmail.com***ABSTRACT**

Refractive error is a major public health concern that hurts people's socioeconomic development, particularly those who do not have access to effective treatment. Over 1 billion people worldwide have issues as a result of uncorrected refractive defects, which have an influence on their education, productivity, and quality of life. Although government and private institutions organise eye camps in underprivileged communities, they frequently struggle to maintain an effective supply chain for spectacles, as well as access to eye care doctors and other important services. This study focuses on detecting and categorising refractive error issues in order to aid with improved planning and resource allocation in community eye camps. The study examines four basic visual impairments—myopia, hypermetropia, presbyopia, and astigmatism—and divides them into twelve distinct groups, which include combinations such as My_As_presbyopia and Hy_astigmatism, as well as the Normal eye. The suggested method uses machine learning techniques to analyse historical eye prescription data obtained from Arvind Eye Care Hospital's camps in Theni, Tamil Nadu. The workflow consists of four phases: data collection, preprocessing, model selection, and model tweaking. SVM, Random Forest, Decision Tree, K-Nearest Neighbours, Naive Bayes, and Logistic Regression are among the classification algorithms evaluated in terms of accuracy, precision, recall, and F1-score. The Decision Tree classifier outperformed all other models, attaining 97.6% accuracy while maintaining good sensitivity (0.95), precision (0.97), recall (0.98), and F1-score (0.97). A case study demonstrates the model's robustness, even with twelve complex classes. This research can help to guide effective service delivery and reduce preventable blindness, which aligns with Vision 2020's goals.

Keywords: Refractive error, Eye camps, Machine learning, Myopia, Hypermetropia, Decision Tree.

ICMIT/DS30

**Design and Finite Element Analysis of Banana Coir Fiber-Reinforced Composites for
Automobile Door Intrusion Beams**

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ABSTRACT

The pursuit of lightweight and environmentally sustainable materials has driven significant research into the utilization of natural fibers as reinforcing agents in composite materials for automotive applications. This study focuses on the design and Finite Element Analysis (FEA) of banana coir fiber particulate composites intended for use in automotive door intrusion beams. The mechanical characteristics of banana coir fibers are evaluated to determine their effectiveness as reinforcement material in enhancing impact resistance and structural performance. A systematic modelling approach was adopted to simulate various composite configurations and fiber orientations. The simulations were carried out using ANSYS Workbench 21 to assess the response of the proposed materials under impact loading conditions. Additionally, physical testing of the banana coir fiber specimens was performed using a universal testing machine to validate the simulation results.

Keywords: Banana Coir Fiber particulate, biodegradability, Finite Element Analysis (FEA), structural performance, physical testing, Crashworthiness, Federal Motor Vehicle Safety Standard (FMVSS) 214, automotive industry, operational stresses.

ICMIT/DS31

Design and Shape Optimization of Engine Mounting Bracket of a Truck Using Hybrid Material

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ABSTRACT

Engine mounting brackets play a significant role in securing components such as the engine and transmission system to the vehicle's chassis and thus reducing noise and vibration. Traditional mounting brackets are typically made from single material, such as steel or aluminum. However, recent advancements in materials science have led to the development of hybrid materials, which combine the properties of different materials to create novel and improved structures. Hybrid materials, which combine two or more different materials, e.g. metals and composites, can offer unique advantages in terms of strength, stiffness, weight reduction and improved resistance to environmental factors such as corrosion and wear, etc. This study investigates the use of hybrid materials to improve the mechanical properties of the engine mounting bracket of a truck, while enabling effective shape optimization. Initially, a comprehensive analysis of the existing design of bracket was conducted to identify areas for improvement in terms of weight reduction, strength, and durability. Using computer-aided design (CAD) software i.e. Solidworks 2022 and for Finite Element Analysis (FEA) i.e. ANSYS workbench 21, engine mounting bracket was modelled, and the mechanical performance was evaluated under static loading scenarios. The results for static analysis were analyzed for stress, strain and deformation. Topology optimization was employed to minimize weight without compromising strength. The bracket was then modelled for hybrid materials and then analysis was performed. The results indicate significant improvements in performance metrics, including stress distribution and deformation, compared to conventional single-material designs.

Keywords: Engine Mounting Bracket, Hybrid Material, Finite Element Analysis (FEA), Topology Optimization, etc.

ICMIT/DS34

Real-Time Simulation-Based Performance Assessment of Photovoltaic Systems

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ABSTRACT

The growing demand for clean and sustainable energy has accelerated the deployment of photovoltaic (PV) systems worldwide. However, their performance is significantly influenced by real-time environmental conditions such as solar irradiance and ambient temperature. This study presents a simulation-based approach to assess the performance of a photovoltaic system under real-time operating conditions using MATLAB/Simulink. The model incorporates actual weather data inputs to simulate the dynamic behavior of a grid-connected PV system. Key performance indicators such as energy output, performance ratio (PR), and system efficiency are analyzed over a selected period. The results demonstrate that real-time fluctuations in irradiance and temperature notably impact the energy yield and overall efficiency. This work emphasizes the importance of integrating real-world data into simulation environments to achieve accurate and reliable performance evaluations, providing valuable insights for system optimization and energy forecasting.

Keywords: Photovoltaic system, MATLAB, Performance ratio, System efficiency

ICMIT/DS36

Urban Airflow Modelling for Pollution Dispersion in Cities**Amit Kumar¹, Jigesh Yadav^{2*}**¹*Dept. of Mechanical Engineering, Motihari College of Engineering, Motihari*²*Dept. of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur***Corresponding Author Email ID: jigesh@mitmuzaffarpur.org***ABSTRACT**

The rapid urbanization and industrial growth of modern cities have led to increased air pollution, posing significant risks to public health and the environment. Understanding urban airflow dynamics is crucial for predicting pollutant dispersion and developing strategies to mitigate its effects. This study employs Computational Fluid Dynamics (CFD) to model urban airflow patterns and analyse pollutant dispersion in a densely populated cityscape. A high-resolution 3D representation of urban geometry, including buildings, roads, and vegetation, is developed to capture complex flow phenomena, such as turbulence, vortex formation, and street-canyon effects. The simulations utilize Large Eddy Simulation (LES) to accurately resolve turbulent flows and capture pollutant behavior under varying meteorological conditions. Different pollutant sources, such as vehicular emissions and industrial stacks, are modeled, and their dispersion patterns concerning wind speed, direction, and urban layout are analyzed. Additionally, the study investigates the impact of green infrastructure, such as urban trees and parks, on mitigating pollution levels. The results highlight critical zones of high pollutant concentration, revealing the influence of urban design and atmospheric conditions on air quality. This research provides valuable insights into the effectiveness of sustainable urban planning measures, such as optimized building placement and increased vegetation, to improve air quality. The findings serve as a tool for policymakers and urban planners to design healthier and more sustainable cities.

Keywords: Computational Fluid Dynamics, Large Eddy Simulation

ICMIT/DS37

**Residual Stress Evaluation in LPBF-Printed Stainless Steel 316L Lattice Structures:
Experimental and Simulation-Based Approach**

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ABSTRACT

Understanding and predicting residual stress is crucial for enhancing the structural reliability of components manufactured by Laser Powder Bed Fusion (LPBF). This study investigates the residual stress behaviour in six stainless steel 316L lattice structures— Dodecahedron, Diamond, Gyroid, Honeycomb, Octet, and Split_P using a combined experimental and simulation-based approach. Residual stress was experimentally determined using X-ray diffraction (XRD) via the Williamson-Hall (WH) plot method, and numerically simulated using Simufact Additive. The WH method revealed varying levels of residual stress across lattice geometries: Dodecahedron (492.89 MPa), Diamond (498.83 MPa), Gyroid (1454.92 MPa), Honeycomb (932.34 MPa), Octet (344.43 MPa), and Split_P (617.60 MPa). Corresponding simulation results were 510 MPa, 475 MPa, 1380 MPa, 960 MPa, 370 MPa, and 600 MPa, respectively. The percentage deviations between experimental and simulated values ranged from 2.85% (Split_P) to 7.42% (Octet), demonstrating strong agreement and validating the accuracy of the numerical model. These findings highlight the influence of lattice geometry on stress accumulation and underscore the effectiveness of integrating XRD-based WH analysis with process simulation for residual stress evaluation in LPBF-manufactured lattice structures.

Keywords: Residual Stress, LPBF, Lattice Structures, Williamson-Hall Method, Simulation, Simufact Additive,

ICMIT/DS39

Design, Finite Element Analysis, and Experimental Validation of Banana Coir Fiber-Reinforced Composites for Automotive Door Intrusion Beams

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ABSTRACT

The growing demand for lightweight, sustainable, and high-performance materials in the automotive sector has driven research into natural fiber-reinforced composites as alternatives to conventional materials. This study presents the design, finite element analysis (FEA), and experimental validation of banana coir fiber-reinforced composites for use in automobile door intrusion beams. Banana coir fibers, known for their low density, biodegradability, and moderate mechanical strength, were processed and incorporated as particulate reinforcements in a polymer matrix. Composite specimens were modeled with various fiber orientations and volume fractions, and structural simulations were performed using ANSYS Workbench 21 to evaluate their behavior under impact loading conditions. The results from the FEA were compared with experimental data obtained through impact and tensile tests conducted on a universal testing machine. The findings indicate that banana coir fiber composites demonstrate promising mechanical performance, making them a viable eco-friendly alternative for structural applications in automotive safety components.

Keywords: Banana Coir Fiber particulate, Finite Element Analysis (FEA), structural performance, physical testing, Crashworthiness, Federal Motor Vehicle Safety Standard (FMVSS) 214, automotive applications, door Intrusion beam, experimental validation, ANSYS Workbench, operational stresses

ICMIT/DS41

Design and Fabrication of a Multi-Directional Air Cooling System for Enhanced Thermal Comfort in Living Spaces

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ABSTRACT

India faces increasing energy demands, with a significant portion of electricity wasted in inefficient building designs, particularly through conventional HVAC systems that account for nearly 50% of building energy consumption. In response to this growing concern, this study presents the design and fabrication of a multi-directional air cooling system aimed at improving thermal comfort while significantly reducing energy usage in residential environments. The proposed device utilizes the principle of evaporative cooling, offering a sustainable and energy-efficient alternative to traditional vapor compression systems. By incorporating a 360-degree airflow mechanism, the system ensures uniform air distribution across the living space, enhancing indoor comfort without the high operational costs of conventional air conditioners. The design prioritizes portability, low power consumption, and adaptability for use in dry and semi-dry regions, aligning with the urgent need for eco-friendly and cost-effective cooling solutions. The successful implementation of this system highlights its potential to reduce greenhouse gas emissions and contribute to energy conservation efforts on a national scale.

Keywords: Evaporative cooling, multi-directional airflow, energy-efficient cooling, thermal comfort, 360- degree air circulation, sustainable design, HVAC alternative, low-power air cooler, indoor climate control, eco-friendly cooling system

ICMIT/DS42

Design and Fabrication of Portable Pulse Jet Bag Filter House

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ABSTRACT

The project, titled "Design and Fabrication of Portable Pulse Jet Bag Filter House," aims to create a flexible and efficient solution for controlling air pollution across various industries. The focus is on developing a portable system that can capture dust and particles from exhaust gases, keeping the air cleaner. This involves carefully selecting the best filter materials, fine-tuning the air-to-cloth ratio for optimal performance, and adding a pulse jet cleaning feature to ensure the system runs smoothly. The portable nature of the design makes it perfect for temporary or adaptable installations, especially in industries like cement, steel, and construction. Ultimately, this project helps promote a healthier environment by providing a reliable way to reduce industrial emissions and meet air quality standards.

Keywords: Portable Baghouse, Pulse Jet Cleaning, Air Pollution Control, Filter Media, Emission Standards

ICMIT/DS44

DESIGN AND FABRICATION OF NOZZLE FOR IMPROVED LAYER ADHESION IN CONCRETE 3D PRINTING -USING SOLID WORKS

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ABSTRACT

Concrete 3D printing has emerged as a promising technology for construction, offering rapid construction, reduced labor costs, and design flexibility. However, achieving optimal layer adhesion remains a significant challenge, impacting the structural integrity of printed concrete structures. This research focuses on designing and fabricating a specialized nozzle to enhance layer adhesion in concrete 3D printing processes. Using SolidWorks, a 3D CAD software, a novel nozzle design was conceptualized, incorporating features such as optimized flow channels, pressure distribution mechanisms, and innovative layer deposition techniques. The nozzle was meticulously designed to ensure precise material extrusion, minimizing voids and ensuring uniform layer thickness. The designed nozzle was fabricated using advanced manufacturing techniques, such as 3D printing or CNC machining, to achieve the desired precision and durability. The fabricated nozzle was integrated into a 3D concrete printing system and subjected to rigorous testing to evaluate its performance in terms of layer adhesion, print quality, and overall build speed. The results of the experimental investigation demonstrated that the novel nozzle significantly improved layer adhesion compared to conventional nozzles. The optimized flow channels and pressure distribution mechanisms facilitated the formation of strong interlayer bonds, leading to enhanced structural integrity of the printed concrete components. Additionally, the innovative layer deposition techniques enabled the creation of complex geometries with minimal defects. This research contributes to the advancement of concrete 3D printing technology by addressing the critical issue of layer adhesion. The developed nozzle design and fabrication methodology offer a promising solution for improving the quality and efficiency of 3D concrete printing processes.

Keywords: Nozzle, 3D Concrete Printing, Solid works, Layer Deposition

ICMIT/DS45

Investigation of Stresses and Fatigue Life of Crankshafts with Oilways using Finite Element Fatigue Analysis

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ABSTRACT

The crankshaft is a critical engine component that operates under highly variable and complex loading conditions, primarily torsional loads arising from the inertia of rotating parts and bending loads caused by gas pressure within internal combustion (IC) engines. These alternating stresses generate fully reversible cyclic loading, making the crankshaft subjected to catastrophic fatigue failure. Finite Element Analysis has emerged as an indispensable tool for assessing complex geometries under real-world loading conditions, enabling precise estimation of fatigue life, reducing development time, and lowering manufacturing costs. In the present study, a Finite Element Fatigue Analysis (FEFA) is conducted on a crankshaft model developed in CATIA V5, with subsequent analysis performed in ANSYS Workbench. The study investigated the effect of design features, particularly the crankpin oilway hole and fillet radius, on stress distribution and fatigue performance. The analysis is carried out using three engineering materials widely employed in crankshaft manufacturing: QT 7000, AISI 4130, and Ti-6Al-4V. Each crankshaft model is subjected to maximum bending and torsional loads to simulate realistic engine operating conditions. The presence of the oilway hole is found to significantly affect local stress concentration and fatigue life. Among the materials analyzed, Ti-6Al-4V demonstrates superior fatigue resistance due to its excellent strength-to-weight ratio, followed by AISI 4130 and QT 7000.

Keywords: Crankshaft, FEM, Stress Analysis, Fatigue Analysis, QT 7000, AISI 4130, Ti-6Al-4V, Bending Load, Twisting Load

ICMIT/DS46

**Design Optimization and Fatigue Behavior of Crankshafts with Hollow Crankpins
Subjected to Torsional and Bending Loads using FEM**

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ABSTRACT

The crankshaft is a critical component in internal combustion engines, responsible for transforming the reciprocating motion of pistons into rotational energy, while enduring continuous bending and torsional stresses. Traditionally, crankpins are designed as solid structures; however, the pursuit of weight reduction and enhanced efficiency has led to the exploration of hollow crankpin designs. This study examines the structural and fatigue behaviour of crankshafts with hollow crankpins, analysing varying hollow percentages from 5% to 50% of the crankpin diameter. A solid crankpin configuration is included as a baseline reference. Finite Element Analysis (FEA) is conducted using ANSYS Workbench to assess the impact of hollow geometry on stress distribution, deformation, and fatigue life under peak bending and torsional loading conditions. Crankshaft models are developed in CATIA V5, and two high-performance materials AISI 4130 and Ti-6Al-4V are selected for comparative analysis due to their prevalent use in automotive and aerospace industries. The results indicate that hollow crankpins offer considerable weight reduction potential, though higher hollow ratios lead to increased stress concentrations, particularly near fillet regions. Ti-6Al-4V exhibits superior fatigue life attributed to its high strength-to-weight ratio, while AISI 4130 provides a cost-effective solution with reasonable fatigue resistance. An optimal hollow percentage range is identified for both materials, achieving a balance between lightweight design and mechanical durability.

Keywords: Crankshaft, Crankpin, Finite Element Analysis, Fatigue Analysis, Torsional and Bending Load, FEM

ICMIT/DS47

Selection of Flywheel Materials Using Mechanical Properties and Weighted Decision-Making Techniques

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ABSTRACT

The flywheel plays a critical role in mechanical systems by storing rotational energy and ensuring smooth power delivery, vibration reduction, engine balancing, and clutch operation. Given its high-stress environment, selecting suitable materials for flywheel fabrication is essential to optimize performance and durability. This study presents a systematic approach to material selection based on key mechanical characteristics: Young's modulus, bulk modulus, hardness, density, and Poisson's ratio. A comparative evaluation of materials such as cast iron, steel, and aluminium is conducted, guided by insights from a panel of three decision-makers. Their inputs are integrated using a weighted decision-making technique to determine the most appropriate material for flywheel construction. The analysis indicates that cast iron, steel, and aluminium are the most suitable materials with the least usage by volume due to their advantageous property profiles. The methodology demonstrated in this study provides a scalable framework for future material selection in similar applications and can be extended by incorporating additional materials and property metrics for enhanced accuracy.

Keywords: Flywheel, Mechanical systems, Material Selection, Mechanical Properties, Weighted Decision-Making Technique

ICMIT/DS48

THE SCIENCE BEHIND PRINCE RUPERT'S DROP: A REVIEW OF FRACTURE MECHANICS

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ABSTRACT

Prince Rupert's Drops (PRDs) exhibit a paradoxical combination of extreme mechanical strength and explosive fragmentation upon tail failure. This study reviews experimental analyses of stress distribution and thermal effects in PRDs, emphasizing their internal residual stresses and fracture mechanisms. The high compressive stress on the outer shell (400–700 MPa) and tensile stress within the core contribute to their unique properties. Using high-speed photography and microcomputed tomography, previous research has revealed systematic fragmentation patterns influenced by environmental conditions such as air, water, and syrup. Thermal contraction and quench granulation play a critical role in their formation, with bubbles in molten glass preventing PRD formation and leading to granulation instead. Additionally, bifurcation of cracks and self-sustained fracture waves contribute to their rapid disintegration. This review consolidates findings on the stress distribution, thermal effects, and fractographic patterns of PRDs to enhance understanding of glass fragmentation mechanics. The insights gained apply to fields ranging from materials science to volcanology and impact physics.

Keywords: Prince Rupert's Drops, Stresses, Thermal contraction, Quench granulation, Glass fragmentation, Crack bifurcation, Photo-Elasticity

ICMIT/DS49

Optimized Design Approaches for Enhanced Performance of Brake Pedals: A Review

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ABSTRACT

This paper reviews different design methods and optimization techniques used to improve brake pedal performance, durability, and weight efficiency. As a key part of a vehicle's braking system, the brake pedal plays an important role in safety, driver comfort, and overall effectiveness. To enhance its lifespan and functionality, ongoing advancements in materials and design are essential. The study examines both traditional and modern materials, incorporating tools like Finite Element Analysis (FEA), structural analysis, and topology optimization. It also looks at how ergonomic design and electronic components contribute to better performance. Additionally, new technologies, including lightweight composites and additive manufacturing, are explored for their potential in brake pedal development. By analysing these advancements, this research highlights effective strategies to maintain safety while improving cost efficiency and overall braking system performance.

Keywords: Finite Element Analysis (FEA), Cost efficiency, Braking system performance

ICMIT/DS52

Design and Prototyping of Advanced Driver Assistance System

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ABSTRACT

This article introduces a practical approach to road lane and object detection for an autonomous car prototype, leveraging OpenCV and Python. The lane detection system applies pre-processing techniques like edge detection and Hough Transform to accurately identify lane boundaries in real-time. For object detection, the model integrates YOLO via OpenCV's DNN module to detect vehicles, pedestrians, and other obstacles. The system is powered by an Arduino-based motor control setup with a laptop handling image processing instead of a Raspberry Pi. Experimental tests demonstrate its ability to function under varying road and lighting conditions, proving its potential for real-world driver assistance and autonomous navigation. Future enhancements include integrating LiDAR/Radar for improved obstacle detection and transitioning to deep learning-based lane tracking.

Keywords: Lane Detection; Object Detection; OpenCV & YOLO.

ICMIT/DS53

Finite Element-Based Comparative Analysis of Core Loss in PMSMs Using Various Magnetic Materials

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ABSTRACT

This study proposes a systematic approach for selecting optimal magnetic core materials in Permanent Magnet Synchronous Motors (PMSMs) by focusing on core loss minimization using Finite Element Method (FEM) analysis. The PMSM models are developed and simulated using ANSYS software with different commercially available core materials. Material loss characteristics across varying frequencies are incorporated from manufacturer datasheets to ensure realistic and accurate simulations. The comparative analysis of three magnetic materials reveals significant differences in core loss behavior, validating the effectiveness of the proposed selection strategy. Reducing core loss is particularly critical in electric vehicle (EV) applications and other high-efficiency systems, as it directly contributes to lower thermal stress, improved power density, and enhanced motor longevity. The findings highlight the vital role of core loss analysis in the optimal design of PMSM drives, where efficiency, compactness, and thermal performance are key priorities in modern energy systems.

Keywords : PMSM, Electrical Steel, material data, BH curve, Core loss, Finite Element Method (FEM)

ICMIT/DS54

Biomedical Scaffolds from 3D-Printed PLA/MgO Composites: A Novel Approach

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ABSTRACT

Creating bone scaffolds are an important part of research in tissue engineering. This work focuses on fixing problems with bones and helping heal injuries. This research shows how using Polylactic Acid-MgO materials can improve scaffold designs. Previous studies reported that the strength and breakdown speed of pure PLA often do not meet the needs for real-world use. In this study, MgO was intended in PLA to make the scaffold stronger, improve its mechanical properties. We fabricate and tested a specimen made of polylactic acid (PLA) and magnesium oxide (MgO) mixed together in a ratio of 90% PLA and 10% MgO. The PLA-MgO composite was made using very cold grinding, a special type of extrusion with one screw, and a 3D printing was done using method called fused deposition modeling (FDM). Tensile and compressive tests were conducted both virtually and physically using CATIA software and a universal testing machine (UTM) respectively. Mechanical testing revealed that the addition of MgO improved the compressive strength by 10% and tensile strength by 22% compared to pure PLA scaffolds.

Index Terms— Bone scaffold, PLA, MgO, 3D printing, FDM

ICMIT/DS55

Assessing the Crashworthiness of 4th Generation Pressure Vessels in Hydrogen Cars Using Finite Element Analysis

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ABSTRACT

The escalating utilization of hydrogen as a sustainable energy carrier underscores the imperative for secure and effective storage systems, particularly in automotive applications. This study evaluates the crashworthiness of Type IV composite overwrapped pressure vessels (COPVs) for hydrogen-fuelled vehicles, combining computational modelling and structural integration analysis. Leveraging ANSYS 2022 R2 for finite element simulations, the research investigates stress distribution, deformation, and energy absorption under impact loading, aligning with methodologies from prior studies on composite pressure vessels. The vessel design employs carbon-fibre-reinforced polymer (CFRP) materials, optimized for gravimetric efficiency and safety compliance, and simplifies the vehicle's rear-end structure to assess performance in frontal and rear-impact crash scenarios. By addressing turbulence these issues, this work advances lightweight, high-strength pressure vessels capable of enduring real-world impact conditions, contributing to the safe adoption of hydrogen mobility solutions.

Keywords: - Hydrogen storage, Type IV pressure vessel, Crashworthiness, Composite materials, Finite element analysis (FEA), LS-DYNA simulation, Carbon fibre reinforced polymer (CFRP), Impact simulation, Energy absorption, Hydrogen fuel cell vehicles

ICMIT/DS56

Effect of Insoles on Kinetics Responses during Walking**Rizwan Ahmad¹, Shailendra Singh Chauhan¹, Madhusudan Pal³, Neha Gupta¹**¹*Babu Banarasi Das Institute of Technology and management, Lucknow, India.*²*Centre of Excellence (CoE), Footwear Design and Development Institute (FDDI), A- 10/A, Sector 24, Gautam Budh Nagar, Noida-201301, Uttar Pradesh, India.*^{*}*Corresponding Author Email ID: sschauhan3456@gmail.com***ABSTRACT**

Ground reaction forces (GRF) play a central role in gait biomechanics and inform footwear design to promote comfort and minimize risk of injury. With advanced insole technology, there are as yet few comparative studies assessing bespoke insole designs, on GRF profiles while level-ground walking, and hence limited evidence-based optimization of footwear. The purpose of this study was to assess and compare the Ground Reaction Forces (GRF) of level-ground walking with one shoe model (Own Sample) having two other types of insoles (D17A and D17B), by gaining kinetic data with force plate analysis. Four subjects walked at their own natural and comfortable pace on a Kistler 3D force plate incorporated in a walkway, wearing standardized shoes with D17A or D17B insoles. Vertical (Fz), mediolateral (Fx), and anteroposterior (Fy) GRF parameters, such as peak forces, midstance forces, and timing measures, were recorded for each foot. Statistical analysis compared differences between insoles and bilateral asymmetries. Result of present study showed no significant differences in any GRF parameters (Fz, Fx, Fy) between D17A and D17B insoles, confirming biomechanical equivalence. However, significant left-right asymmetries were noted in several parameters. For Fz, differences were observed for peak forces bilaterally. For Fx, Heel Strike Peak, Midstance Phase, Time for Midstance, and Push-off Peak were found to have significant asymmetries, whereas Time for First and Second Peaks did not. For Fy, Braking Force, Time for Braking Force, Propulsive Force, and Time for Propulsive Force, there were asymmetries present with Braking Force only significant in D17A. There were asymmetries that persisted throughout both types of insoles. The functional equivalence of D17A and D17B insoles indicates they are interchangeable for general walking use, making footwear design easier. Yet, recurring gait asymmetries indicate the necessity of individualized footwear to correct bilateral differences, possibly lowering injury risk and improving performance. Future studies should investigate insole effects across various populations and activities, including material analyses and wearable technologies to connect laboratory and real-world environments.

Keywords: Ground reaction forces, Kinetics, Insole type, Kistler 3D force plate, Walking.

ICMIT/DS57
**Deep Learning-Based Weld Defect Detection in Underwater Welding on AH36 Using
Convolutional Neural Networks**
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ABSTRACT

This study investigates the advantage of Convolutional Neural Networks (CNNs) in an automated underwater welding defect detection system for AH36 steel. The study intends to improve the consistency of underwater weld inspections, reduce dependency on humans, and expedite fault identification procedures by utilizing deep learning (DL). Only the binary categorization of weld images as defective, which is lack of fusion (LOF) and non-defective based on visual indicators in underwater conditions, is the subject of this work. Two CNN architectures, ResNet (Residual neural network)-18 and VGG (Visual geometry group)-16, were trained and evaluated for this study. The comparative advantages of these designs in managing the difficulties of underwater visual data are demonstrated by empirical findings.

Keywords: Underwater welding, weld defect, deep learning, defect detection, CNN, AH36.

Track 4

Industrial Technologies & Miscellaneous

ICMIT/IT01

Performance investigation of highly sensitive Si_3N_4 material-based MRR sensor for biosensor applications

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ABSTRACT

In this paper, the modeling and simulation of a mirroring resonator (MRR) and its application as a biosensor for detecting cancer cells are carried out. The mathematical modeling to find the transfer function is done in the z-domain using the concept of delay-line signal processing. The MRR-based biosensor is fabricated on the Si_3N_4 platform, which offers low losses and a small footprint. It is susceptible to changes in the refractive index of the cladding environment of MRR due to the interaction of cancer cells or chemicals. This paper primarily focuses on the shift in the resonance wavelength to identify different types of cancer cells. The simulation results indicate shifts in the resonance wavelength or frequency compared to healthy cells have been obtained using MATLAB and OptiFDTD software. The sensitivity of the proposed MRR-based biosensor is in the range of 734.5-1528.77 nm/RIU.

Keywords: Micro Ring resonator, Biosensor, Sensitivity, Q-factor, refractive index sensor

ICMIT/IT02

Piezoelectric Energy Harvesting: A Review**Atul***Alliance School of Applied Engineering**Alliance University***Corresponding Author Email ID: mindmakeupatul@gmail.com***ABSTRACT**

Alternative energy is quite a lucrative option to harvest from waste energy which can overcome the challenges pertaining to the environmental and energy generation using conventional methods. Increasing utilization of conventional energy sources will lead towards environmental problems, which attracted the researchers to identify the alternative sources of energy which can be trapped from the available waste energy like vibration. Vibration energy is available in ample amounts which can be captured and converted into useful energy. Therefore, interest in piezoelectric materials plays a crucial role in harvesting energy using smart structures. Development of MEMS and advanced Production technology helped to fabricate orthotropic and isentropic materials together with improved characteristics like integration, stiffness, flexibility, coupling factors and coefficient pertaining to the piezoelectric for multiple applications. Apart from that minimizing the size of the devices with ambient power is already recognized by some of the industries like marine, aircraft and electronics. It is also predicted that many devices are powered by smart structures using PZT materials in the form of piezo generators. This research is going to present the overall development of piezoelectric energy harvesting process with the available waste energy like vibration energy. This paper describes the functioning modes of various piezo generators and the mechanics behind it. There are various piezoelectric materials which are reviewed based on their derivatives. The claim of piezoelectric energy harvesting at micro and nano scale are also appraised in this paper. In this paper, the development of piezoelectric materials and structures are also discussed briefly which showcased the sustainability in developing the autonomous power generation where battery replacement is a challenge and supply to the multiple devices with respect to wireless technology.

Keywords: Piezoelectric, energy, harvesting, smart structures, vibration

ICMIT/IT04

Robotic Bidirectional Path Planning in Static and Dynamic Environment**Narayan Kumar^{1*}, Amit Kumar², Ankit Kumar³**¹Assistant Professor, Department of Mechanical Engineering, Government Engineering college, Vaishali²Professor, Department of Mechanical Engineering, National Institute of Technology, Patna;³Assistant Professor, Department of Mechanical Engineering, Darbhanga College of Engineering, Darbhanga.*Corresponding Author Email ID: narayan.gecvaishali@gmail.com**ABSTRACT**

The use of robots in daily life is growing. Determining a robot's course is one of the key components of using it. Path planning explains how the robot will go from its starting location to its destination. In robotics, sampling-based methods are frequently preferred for high-dimensional motion planning due to their quick search space exploration. However, because of their intrinsic stochasticity, they can only provide asymptotic guarantees on the quality of the solutions. When planning, most of the work is often concentrated in the areas closest to the start and target configurations, leaving a lot of empty space in between. This work offers a bidirectional search strategy for path planning finding in difficult maps by a mobile robot. This method builds the grid surface model first, and then uses a global minimal number of turns approach to further analyse the various least-distance paths that the algorithm produced. This approach can accelerate convergence and smoothen the global path. The path with the least amount of robot diversion from the initial to the final position is employed in the combinatorial selection of the optimum path from the set of the shortest paths. With it, the robot may cover the maximum distance in a single spin. The robot can jump and cover the most significant distance in a single leap while maintaining a constant speed while it follows a straight path.

Keywords: Path planning, Bidirectional search strategy, Grid surface model, Combinatorial selection, shortest path algorithms, Path smoothening

ICMIT/IT05

Impact of Membrane Housing Height on Oil Retention and Filtration Efficiency in Quartz-Based Separation Systems

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ABSTRACT

The increasing global demand for cost-effective and high-performance oil–water separation systems has intensified interest in optimizing membrane-based technologies. This study explores the influence of membrane housing height on the filtration efficiency of quartz-based membranes modified with hydrophilic nanoparticles. Using synthetic oily wastewater with an initial oil concentration of 39,567.4 mg/L, the separation performance was evaluated at membrane housing heights of 1 mm, 2 mm, 3 mm, and 4 mm. A pressure-driven quartz membrane system was employed, and separation efficacy was quantified via gravimetric oil and grease analysis. The results revealed a substantial inverse relationship between membrane height and residual oil concentration, decreasing from 3,761.5 mg/L at 1 mm to 1.2 mg/L at 4 mm. Enhanced performance at increased heights is attributed to improved fluid dynamics and more effective membrane–oil interactions. Supporting characterizations using Scanning Electron Microscopy (SEM) and X-ray Diffraction (XRD) confirmed structural stability and more uniform oil distribution at higher membrane positions. These findings underscore the importance of membrane housing design in maximizing oil rejection and optimizing water recovery in industrial oily wastewater treatment. This research provides valuable insights for the development of next-generation quartz-based filtration systems, promoting energy efficiency, operational stability, and sustainable water reuse.

Keywords: Quartz Membranes, Oil–Water Separation, Wastewater Treatment, Membrane Housing Height, Filtration Efficiency, hydrophilic nanoparticles

ICMIT/IT06

EFFECT OF GRAPHENE NANOPARTICLES ON Sn-Zn-Bi SOLDER ALLOY

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ABSTRACT

With growing demand for environmentally friendly and high-performance alternatives to traditional lead-based solders, this study explores the development and characterization of a Sn-Zn-Bi lead-free solder alloy reinforced with graphene nanoparticles. A base alloy composition of Sn-90%, Zn-9%, and Bi-1% was prepared using the stir casting method, with graphene nanoparticle additions of 0.1%, 0.3%, and 0.5% by weight. The addition of graphene aimed to enhance the mechanical strength, thermal stability, and microstructural uniformity of the alloy. The casting process parameters were optimized to overcome challenges related to the high viscosity of the molten alloy, especially at elevated temperatures (around 450°C). Preliminary results suggest that graphene reinforcement may significantly influence grain refinement and improve the overall performance of the solder. Further investigation through mechanical, microstructural, and thermal characterization will provide deeper insight into its applicability in electronic packaging.

Keywords: Lead free solder; Sn-Zn-Bi alloy; Graphene nanoparticles

ICMIT/IT08

Multi-Point Motion Planning of Service Robot for Cleaning and Sanitization in Dynamic Obstacle Conditions Using A-Star Algorithm**Anup Kumar^{1*}, Narayan Kumar², Tarun Dev³**¹*Department of Mechanical Engineering, Indian Institute of Technology, Patna, Bihar.*²*Department of Mechanical Engineering, Government Engineering College, Vaishali, Bihar.*³*Department of Mechanical Engineering, Muzaffarpur Institute of Technology, Muzaffarpur, Bihar***Corresponding Author Email ID: mcekumaranup@gmail.com***ABSTRACT**

Mobile robots have been widely used in challenging real-world situations in recent years. Reduced interpersonal contact and autonomous routine cleaning, sanitization, and support in crowded environments are the main uses of service robots. Service robots are desperately needed on congested platforms such as train platforms, hospital patient areas, and employee and public contact places in companies, institutions, schools, and so forth. And numerous additional locations, including offices, hospitals, colleges, and schools, can use the same set of systems. Autonomously cleaned and sanitised public roads, highways, and toll areas can be achieved. A different industry began experimenting with service robots during the COVID-19 pandemic. Both stationary and movable barriers can be found on railway platforms. The only source of obstacle position coordinates is the installed cameras. A service robot uses the A star algorithm to avoid or alert itself to impediments in order to carry out regular cleaning and sanitization. The programme covers all spots as quickly as possible, saving time and effort. The trials were conducted on a grid area featuring moving barriers. MATLAB is used for simulations and analysis. The considerable reduction in calculations and memory requirements is the key advantage of utilising A-Star.

Keywords: Motion Planning; A-star algorithm; Dynamic obstacles environment; Obstacles avoidance.

ICMIT/IT10

Credit Card Fraud Detection through Machine Learning Techniques: A Comparative Study

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ABSTRACT

In the modern era, peoples are using latest technology for betterment of life, in this one of is online transactions. Day to day increasing online transactions through internet, challenges to handle cybersecurity and fraudulent cases during online transaction. Major issues with this are imbalance dataset, fraudulent cases are less than 0.2% of the legitimate transactions. In this work focus on fraud detections in credit card transaction using various machine learning techniques. We have reviewed latest research article and performed comparative analysis of different ML (Machine Learning) techniques for fraud detection these are: Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and K-nearest neighbour (KNN). Second issues is highly imbalanced dataset, for the solution of this examine different techniques to overcome the dataset problem, these are: imbalanced data, Under Sampled (US) data, Over Sampled (OS) data, sampled data using Synthetic Minority Over Sampling Technique (SMOTE). we compare and discuss the performance of various machine learning algorithms and sampling Techniques. Our study provides a useful source to guide academic and industrial researchers in evaluating financial fraud detection systems and designing robust solutions.

Keywords: Credit Card Fraud detection, Imbalance dataset, Machine Learning, Support Vector Machine (SVM), SMOTE.

ICMIT/IT11

Multi-objective optimization of fused deposition modeling process parameters to enhance quality of graphene-reinforced PC-ABS composite part using Taguchi -Grey Relational Grade

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ABSTRACT

The Fused deposition modeling is an additive manufacturing technique used at large scale to produce complex and intricate shape and prototypes. This paper aims to find the optimal setting of process parameters of fused deposition modelling and identify the effect of various process parameters of fused deposition modelling using Analysis of variance. Experiments are designed using Taguchi “L27” orthogonal array. This research considered five factors (layer height, infill density, infill style, printing temperature, and printing speed) to analysis the effect on printing time, material consumption and dimensional accuracy. The Mathematical Model (MM) was developed by using Taguchi methodology. The developed Mathematical Model is used as fitness function in GA in order to find out the optimal settings of process parameters and to predict the corresponding printing time, material consumption, and dimensional accuracy.

Keywords: Fused deposition modelling, FDM, Taguchi, optimization, Genetic algorithm

ICMIT/IT12

Towards Sustainable Development: Prioritizing Lean, Green, Circular, and Sustainable (LGCS) Practices to Mitigate Fruit and Vegetable Wastage Using AHP

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ABSTRACT

The global challenge of food loss and wastage, particularly in the fruit and vegetable (F&V) supply chains, poses significant risks to food security, environmental sustainability, and economic development. This study aims to contribute to sustainable development by exploring and prioritizing Lean, Green, Circular, and Sustainable (LGCS) practices that can effectively mitigate F&V loss and wastage. A comprehensive framework was developed by integrating the principles of LGCS, supported by literature review and expert opinion. The Analytic Hierarchy Process (AHP) was employed as a multi-criteria decision-making tool to evaluate and rank the identified practices based on their relative importance and potential impact. The findings reveal that Circular practices, particularly those promoting resource recovery and closed-loop systems, are highly prioritized, followed closely by Green initiatives aimed at reducing environmental footprints. Lean practices that enhance process efficiency and Sustainable practices ensuring long-term socio-economic balance also play a critical role. The research provides a strategic road-map for stakeholders, including policymakers, supply chain managers, and sustainability advocates, to focus on high-impact practices that align with the broader goals of sustainable food systems. This study bridges a critical gap by offering a structured prioritization of LGCS practices, enabling informed decision-making in tackling food loss in perishable agricultural supply chains.

Keywords: Sustainable production, Green Production, Supply Chain Management, Multi Criteria Decision Making, Food loss and wastage.

ICMIT/IT19

Development of Electrochemically Deposited and Annealed Surfaces for the Study of Wettability through evaluation of Contact Angle, Contact Angle Hysteresis, and Roll- Off Angle**Vishwajeet Kumar¹, Abhik Majumder^{1*}, Jyoti Kumar¹, Alok Ranjan²**¹*Department of Mechanical Engineering, National Institute of Technology Agartala, Tripura India- 799046.*²*Department of Mechanical Engineering, Muzaffarpur Institute of Technology Muzaffarpur, Bihar India-842003.***Corresponding Author Email ID: onlyabhik@gmail.com***ABSTRACT**

Over the past two decades, the biomimicry of the lotus leaf effect for industrial applications has become a significant research focus. The lotus leaf, known for its high contact angle and low contact angle hysteresis, holds great promise for self-cleaning, anti-corrosion, anti-fouling, and anti-icing applications, provided its surface characteristics can be replicated on a large scale. In this study, a superhydrophobic coating was developed on a copper substrate through electrochemical deposition, followed by annealing and surface energy modification. The electrochemical deposition of zinc onto the copper substrate, followed by annealing, imparts the necessary surface roughness, while myristic acid was used for surface energy modification. The resulting coating achieved a high-water contact angle of 156°, a roll-off angle of 3°, and a contact angle hysteresis of 5°. To examine the coating's morphology, scanning electron microscopy (SEM) was conducted on the coated samples at various magnifications. Additionally, Fourier-transform infrared spectroscopy (FTIR) and X-ray diffraction (XRD) analyses were performed to assess the effects of surface energy modification. The coated samples were also tested for mechanical robustness and maintained a high contact angle despite repeated mechanical abrasion. The results indicate promising practical applications.

Keywords: Contact angle, XRD, SEM, Roll-Off angle

ICMIT/IT21

Electromechanical characterization of VHB-like dielectric elastomer-based out-of-plane actuator for soft robotics

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ABSTRACT

Out-of-plane actuation provides significant advantages for soft robotic systems by enabling multidimensional deformations essential for operations such as lifting, grasping, and adapting to various surface contours. This study details the design and evaluation of a cost-effective dielectric elastomer actuator, with a focus on how mechanical boundary conditions affect the generation of electrostatic forces. An extensive investigation examines how pre-strain and weight impact actuator performance. The results support established findings regarding pre-stretch optimization, which promotes uniform electric fields, while also revealing new insights into the role of weight in modifying energy output. Furthermore, the study analyzes electromechanical behavior under different electric fields, highlighting the influence of applied voltage on both deformation and force generation. By evaluating the actuator's energy conversion efficiency, the research clarifies how electrical input is translated into mechanical work. These findings are particularly critical for soft robotics, where precise control, adaptability, and efficiency are necessary. In conclusion, the optimized design parameters that take into account pre-strain and applied load significantly enhance displacement, actuation force, and energy efficiency, positioning this actuator as a promising candidate for integration into low-cost, lightweight, and adaptive soft robotic systems.

Keywords: Out-of-plane actuation; Dielectric elastomer; Pre-strain; Electrostatic forces; Energy conversion efficiency.

ICMIT/IT22

Mechanical and Dielectric Properties of Porous Elastomer for Flexible Sensors**Tejasvi Anant, Raj Kumar Sahu***

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ABSTRACT

Porous dielectric elastomers are ideal for flexible pressure and strain sensors due to their tunable elastic modulus, which enhances sensitivity by amplifying deformation-induced changes in dielectric properties. In this work, porous and solid silicone elastomers are compared to understand how porosity influences key performance metrics—mechanical compliance, dielectric permittivity, and sensitivity—for optimizing high-performance flexible strain sensors. Porous elastomers were fabricated using a sugar-templating method to introduce controlled porosity. Mechanical tests, including stress–strain analysis, evaluated their elasticity and compliance, which determine how effectively the material conforms and responds to deformation. Dielectric measurements under varying strains assessed changes in permittivity, directly linked to sensor sensitivity. Cyclic loading tests were conducted to examine signal stability, repeatability, and hysteresis, essential for reliable operation in wearable devices. Results show that porous elastomers exhibit significantly lower modulus and higher dielectric responsiveness compared to solid samples, making them more effective in detecting subtle mechanical changes. This study offers valuable insights into the structure–property relationship of silicone-based elastomers and underscores the potential of porous architectures in developing next-generation wearable and soft electronic sensors.

Keywords: Porous dielectric elastomers; Flexible strain sensors; Silicone elastomers; Mechanical compliance; Dielectric sensitivity

ICMIT/IT23

A Review of Photostrictive Actuators

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ABSTRACT

Smart materials such as piezoelectric & magnetostrictive materials require electrical wiring for their operation. These metallic wires make systems heavy and complex. Recently, researchers proposed a photostrictive innovative material, which does not need any wiring. The principle of photostrictive materials is based on the photostrictive effect, which is a combination of photovoltaic & converse piezoelectric effects. The photostrictive phenomenon was observed in four main groups of materials: (1) Ferroelectrics, (2) Polar, (3) non-polar semiconductors, and (4) Organic-based materials. This paper provides an overview of present photostrictive materials and their actuation applications.

Keywords: Smart Material, Piezoelectric Material, Magnetostrictive Material, Photostrictive Material, Photovoltaic & Converse Piezoelectric Effects.

ICMIT/IT25

Infrared Suppression Strategies for Warships: An Analytical Efficacy Evaluation**Khagesh Kumar Choudhary^{1*}, Dr. P K Sharma², Capt. (Dr.) AVSN Murty³**¹IR Division, Warship Technology²Naval Science and Technological Laboratory³Defence R & D Organisation, Ministry of Defence, Vigyan Nagar, Visakhapatnam – 530 027**ABSTRACT**

Stealth technology has emerged as a cornerstone of contemporary warfare, delivering considerable tactical advantages across both terrestrial and maritime theaters. In the marine domain, the incorporation of stealth features into the design and architecture of warships plays a vital role in enhancing their survivability by decreasing the probability of detection by adversary systems. Warships are inherently susceptible to identification and tracking due to the emission of multiple detectable signatures, including but not limited to Radar Cross Section (RCS), Infrared (IR), Electromagnetic, Magnetic, Acoustic, Wake, and Laser signatures. Among these, infrared (IR) radiation holds particular significance due to its passive detection nature, allowing hostile systems to monitor and engage targets without actively revealing their own presence. As such, the suppression of IR signatures has become a central consideration in naval stealth engineering. This research paper focuses on the strategic importance of reducing IR emissions from naval vessels and provides a comprehensive overview of the various infrared suppression technologies currently implemented in modern warship systems. Emphasis is placed on understanding the mechanisms of IR signature generation and exploring mitigation strategies designed to reduce thermal visibility. The study also includes a theoretical evaluation of the effectiveness of these suppression techniques, highlighting their relevance to real-world operational scenarios. In addition, the concept of "lock-on range"—defined as the distance at which an infrared-seeking sensor or weapon system can acquire and maintain a target lock based on the ship's thermal signature—is examined. This analysis is crucial for assessing the vulnerability of warships to infrared-guided threats and underscores the role of IR suppression in improving overall stealth and survivability in hostile environments.

Keywords: Infrared, Warship, MWIR, LWIR, Duct, Hull, LOR.

ICMIT/IT26

Revolutionizing Agri-Food Supply Chain Efficiency with Blockchain Technology

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ABSTRACT

The Agri-food industry is an essential part of any economy. Nowadays, it has become dynamic and demanding. The effectiveness of the agri-food industry depends on Agri-Food Supply Chain (AFSC) management. To adapt today's dynamic environment, AFSC needs to be integrated, collaborative, and flexible. Moreover, today AFSC is constantly under increasing pressure to meet consumers' expectations regarding food quality. As a result, applications of newer technologies have increased in the field of AFSC. One of these technologies is Blockchain Technology (BT). BT has been identified as a potential solution to improve traceability, transparency, and efficacy of the food supply chain. This paper provides an overview of the BT and its implementation in the AFSC. The findings indicate that the adoption of the BT has been slow due to several challenges. But if the potential of BT is harnessed properly it could be a great solution to several issues of AFSC.

Keywords: Blockchain Technology; Distributed Ledger Technology; Agri-food sector; Food Supply Chain Management

ICMIT/IT27

Haptic Feedback based Machine Learning models for improving accuracy in surgical Robots

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ABSTRACT

The integration of machine learning (ML) into haptic feedback systems has the potential to revolutionize surgical robotics by enhancing precision, reducing latency, and improving sensory accuracy. Traditional haptic feedback mechanisms often suffer from limitations, such as inadequate real-time responsiveness and suboptimal sensitivity, which impact the effectiveness of robotic-assisted surgery. This paper highlights recent advancements in ML-driven haptic feedback, focusing on techniques such as neural network-based force estimation, reinforcement learning for adaptive control, and deep learning for tactile perception. Current challenges are examined, and future research directions aimed at optimizing data-driven haptic systems for enhanced surgical performance are proposed. By leveraging ML algorithms, it is possible to refine force-feedback mechanisms, ensure seamless interaction between robotic systems and human operators, and ultimately improve surgical outcomes.

Keyword: Machine Learning, Haptic Feedback, Surgical Robotics, Force Prediction, Minimally Invasive Surgery

ICMIT/IT28

Design and Development of a Low-Cost Bioelectrical Impedance Analysis System Using Arduino

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ABSTRACT

The design and development of a low-cost Bioelectrical Impedance Analysis (BIA) system using an Arduino microcontroller is presented in this study. Bioelectrical Impedance Analysis is a non-invasive method commonly used for estimating body composition parameters such as body fat percentage, lean body mass, and total body water. Traditional BIA systems can be expensive, limiting their accessibility for personal or widespread use. In this paper, we propose a cost-effective solution that leverages readily available components, including an Arduino platform, to measure bioimpedance and estimate body composition. The system uses a small AC current passed through the body via electrodes, with impedance measurements obtained by analyzing the voltage drop across the body. The data is processed by the Arduino microcontroller and displayed on a digital interface. Calibration and signal processing techniques ensure accurate and reliable measurements. The proposed system offers a viable alternative to commercial devices, demonstrating the potential of low-cost, portable bioimpedance analyzers for personal health monitoring and wellness applications.

Keywords: Bioelectrical Impedance Analysis, Arduino

ICMIT/IT29

Simulation of a LiDAR-Based Obstacle Avoidance Robot in a Hospital Environment using Webots: Enhancing Efficiency and Safety

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ABSTRACT

Hospitals, characterized by complex and dynamic environments, present significant challenges for autonomous robots. This paper explores the simulation of a LiDAR- based obstacle avoidance robot within a simulated hospital environment using Webots, a powerful open-source robot simulator. The primary goal is to investigate the feasibility and effectiveness of employing LiDAR sensors for navigation and safe obstacle avoidance in a real-world hospital setting. The research focuses on developing a robust control system that integrates LiDAR data for real-time mapping, path planning, and execution. We analyze the robot's performance under different static and dynamic obstacles, Furthermore, the paper explores the potential benefits of deploying such robots in hospitals, such as improved logistics, medicine delivery, and reduced human contact, while considering the crucial aspects of safety, ethical considerations, and integration with existing hospital infrastructure. The simulation results provide valuable insights into the design and implementation of autonomous robots for hospital environments and highlight the potential for enhancing efficiency and safety in healthcare settings.

Keywords: LiDAR, Webots, obstacle, navigation

ICMIT/IT30

Performance Evaluation of Swarm Robotics for 3-D Mapping

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ABSTRACT

This study evaluates the performance of a decentralised swarm robotic system comprising multiple autonomous ground robots equipped with Lidar, GPS, Inertial Measurement Unit (IMU), Compass, Camera, and wireless communication modules, as well as other sensors. The robots collaboratively perform real-time 3-D mapping of unknown environments under a leader–follower architecture, where one robot coordinates the swarm's behaviour while the others contribute to distributed data collection. The framework supports the integration of application-specific sensors, such as those for mineral detection, gas sensing, or temperature measurement, which enables intelligent and purpose-driven exploration beyond spatial mapping. Sensor data is aggregated to construct a comprehensive 3d map, which is visualised in real-time and archived for offline analysis. The system's scalability allows for an increased robot count to accelerate mapping performance and information richness. This approach demonstrates potential applications in autonomous exploration, resource surveying, search and rescue, and environmental monitoring.

Keywords: Swarm Robotics, Webot, Swarm behaviour, 3-D mapping, Sensors, Mechatronics

ICMIT/IT31

**A CASE STUDY ON PERFORMANCE IMPROVEMENT USING DMAIC
METHODOLOGY OF SIX SIGMA**

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ABSTRACT

In the fast- changing world of manufacturing, boosting productivity is pivotal for staying competitive and satisfying request need's manufacturing sectors encounter colourful challenges, similar as global competition, a different range of products, shorter supereminent times, and strict client demands for both quality and volume. To attack these issues, Six Sigma has come a thorough quality operation strategy designed to reduce process variation, enhance both product and process quality, and lower functional costs. This document presents a comprehensive analysis of the growth, prosecution, and results of Six Sigma practices in transnational manufacturing sectors. It outlines the essential tools and styles used at each phase of a Six Sigma design, ranging from Define to Control, and emphasizes how these instruments aid in effective design delivery. also, the paper summations important perceptivity and assignments drawn from successful Six Sigma systems, showcasing their real- world connection across colourful manufacturing diligence. The review indicates that while numerous manufacturing areas presently operate at quality situations between two and four Sigma, enforcing Six Sigma strategies can greatly ameliorate process capability and functional effectiveness. The results punctuate the significance of organizational fidelity, data- informed decision- timber, and ongoing enhancement for achieving sustained success through Six Sigma relinquishment.

Keywords: Six Sigma, Lean Six-Sigma, DMAIC Methodology, Uses Tools & Techniques

ICMIT/IT32

Artificial Intelligence in Composite Materials: A Comprehensive Review

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ABSTRACT

The integration of Artificial Intelligence (AI) into the field of composite materials has revolutionized the way materials are designed, analyzed, and optimized. This comprehensive review explores the multifaceted applications of AI, particularly machine learning (ML), deep learning (DL), and data-driven algorithms in the development, characterization, and failure prediction of composite materials. Emphasis is placed on how AI facilitates rapid material discovery, enhances process control in manufacturing, and improves the accuracy of mechanical property predictions. The paper also discusses the use of AI in structural health monitoring (SHM), damage detection, and lifecycle assessment of composites. Key challenges such as data scarcity, model interpretability, and the integration of AI with physics-based modelling are critically analysed. By bridging computational intelligence with materials science, this review highlights the transformative potential of AI in enabling smarter, faster, and more sustainable innovations in composite materials.

Keywords: Artificial Intelligence (AI), Composite materials, Structural health monitoring (SHM), Materials science

ICMIT/IT33

Prediction of Surface Roughness for AlSi10Mg Fabricated via Laser Powder Bed Fusion using Machine Learning Models

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ABSTRACT

This study investigates the predictive accuracy of various machine learning (ML) models for surface roughness in AlSi10Mg alloys manufactured using the laser powder bed fusion (LPBF) process. Data on critical process parameters—laser power, scan speed, layer thickness, and hatch spacing—were meticulously extracted from peer-reviewed research papers. The dataset was split into an 80-20 train-test ratio to evaluate the performance of six ML models: Linear Regression, K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, and Artificial Neural Networks (ANN). The R-squared (R^2), mean absolute error, mean square error, root mean square error was used to assess model efficacy, revealing significant variations in predictive performance. Notably, Gradient Boosting and ANN models exhibited the highest R^2 values (0.934 and 0.910 for train, 0.934 and 0.810 for test, respectively), indicating robust generalization. Validation was conducted using specific LPBF parameters: laser power of 370 W, scan speed of 1300 mm/s, layer thickness of 70 μm , and hatch distance of 0.13 mm. The findings underscore the potential of advanced ML techniques, particularly Gradient Boosting and ANN, in optimizing surface roughness predictions, offering valuable insights for enhancing additive manufacturing processes.

Keywords: Machine learning, Laser powder bed fusion process, Laser power, Scan speed, layer thickness, Hatch spacing.

ICMIT/IT34

Design of VR system to study the internal object in 360° direction

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ABSTRACT

A new technology called virtual Reality, which is recently launched in the market, spreading and exploring its application like a fire. Though this tech is still in under research for future work so that it cannot only be use for entertainment purpose but for as well as in military, medical and manufacturing industry's purpose. The purpose of this paper is to see the live 360° footage of an internal machine with the help of VR headset. Using Google cardboard, having the size of display 5.7inch and the resolution of 720x1440 pixels, gyroscope, and accelerometer sensor to see in every direction, snake camera, mini servo motor for the rotation of camera in 360 direction. This system is generally made for keeping the close eye on those types of things which is very harmful and hazardous for human health. Those extreme cautious things are nuclear plant or that type of system which is worked under in extreme high or low temperature like boiler. With the help of this system the close study of object can be easily done without getting effected by high or low temperature or hazardous gases.

Keywords- 360° live footage, VR headset, snake camera, Arduino

ICMIT/IT36

Experimental Investigation on Delamination of Waste PCBs Using a DMF-Based Solvent**Shayam Anwer¹, Akansha Raj¹, Arunabh Meshram², Anuj Kumar Sharma¹, Rabesh Kumar Singh^{3*}**¹*Centre for Advanced Studies, AKTU Lucknow, Uttar Pradesh 226031, India*²*Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, 208016, India*³*Mechanical Engineering Department, Madan Mohan Malaviya University of Technology, Gorakhpur Uttar Pradesh, 273010, India***Corresponding Author Email ID: rksme@mmmut.ac.in***ABSTRACT**

In recent years, worldwide production of electronic waste (e-waste) increases and stands at 62 billion metric tons. In coming years this quantity is projected to hit 82 million tons by 2030. Waste Printed Circuit Boards (WPCB) offer a considerable amount of e-waste, estimating approximately 2.2 % of the total e-waste and having 30–40% by weight of total e-waste. Even though these devices contain significant amounts of metals such as copper, gold, silver, and many other precious metals. As per the reports, only about 15% of WPCBs are recycled due to the challenges posed by their underlying thermoset epoxy matrix and multilayered structure. The present study incorporates the new and easily scalable technique of delamination via the combination of ultrasound energy above a solvent system-based N,N-dimethylformamide (DMF) plus other reactive agents. This method attempts to dissolve polymeric bonds within WPCBs to expose the metals that are covered. This hybrid solvent-ultrasound technique is a fast and environmentally friendly WPCB pretreatment method. It improves the efficiency of subsequent metal recovery, aligning with sustainable electronic waste (e-waste) disposal strategies. The waste WPCBs are collected from open market and down sized into the specimens of 25 mm², 50 mm², and 100 mm². WPCB were placed in a mixture containing a solid to liquid ratio of 1:3. The ultrasonic bath was at a temperature of 50 °C for a range of 1– 2 hours then proceeded to recirculating the solution to promote the active stirring and the matrix conundrum at later stages of super-radiant collapse. This two-stage process resulted in successful splitting of WPCBs by eliminating the epoxy resins and the binding interfacial adhesion forces. Thereafter, macroscopic analysis is conducted via optical microscopy to see the enhanced layer separation, while scanning electron microscopy (SEM) to analyzed the surface morphology of delaminated WPCBs. The findings indicated successful resin detachment, copper pathway exposure, and maintained structural integrity.

Keywords: Waste Printed Circuit Boards; WPCB Delamination; Ultrasonic Treatment; N, Ndimethylformamide (DMF); SEM-EDX; Optical Microscopy; Metal Recovery

ICMIT/IT38

Dual Stream Cross-Attentional Convolutional neural networks for adaptive anomaly detection in the face of Zero- day attacks**Pragya Bajpayi^{1*}, Sandeep Mathur², and Madhu Sharma Gaur³**¹Amity University, Greater Noida, Uttar Pradesh
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In many domains, such as cybersecurity, fault diagnosis, and industrial monitoring, where accurate identification of rare and anomalous patterns is essential, anomaly detection is essential. A Dual Stream-Cross Attentional Convolutional Neural Network (DSCA-CNN) is presented in this work to enhance anomaly detection efficiency by skillfully capturing spatial, categorical, and channel-specific feature dependencies in high-dimensional datasets. Using a real-world dataset, the suggested DSCA-CNN is evaluated using Python against traditional machine learning classifiers like Support Vector Machines (SVM) and Logistic Regression (LR). Results from experiments show that DSCA-CNN performs better in classification, with the highest accuracy of 93.69%, perfect precision of 1.0, and top F1-score of 66.34%, demonstrating a strong trade-off between accuracy and anomaly sensitivity. While DSCA CNN outperforms baseline models in the majority of performance metrics, at 49.63%, its recall is still comparatively low, suggesting room for improvement in terms of detecting all anomalous cases. These findings support the potential of deep attention-based architectures for anomaly detection tasks and motivate further research into strategies to improve generalization and recall in complex, unbalanced datasets.

Keywords: DSCA-CNN, IoMT, Healthcare, Security, SVM, Anomaly Detection, Python

ICMIT/IT39

Green-Waste Composting: Sustainable Approach

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ABSTRACT

Regular disposal of garden-wastes e.g. ground grass, plant leaves etc needs a lot of waste land for disposal and it also produces fouling smell that is unhygienic to the community. A comparative study of aerobic composting using Acetobacter and anaerobic composting using Bokashi process bacteria (Lactobacillus) has been tried here. The possibilities for reuse of the compost as a seed material for further composting was also studied. All the parameters relevant for a fertilizer are measured and the quality of manure produced is analyzed and found satisfactory. A reactor design was also made in CAD for making a laboratory model for the same.

Keywords: Garden-waste, Compost, micro-organisms, Re-use, Aerobic digestion, Anaerobic digestion

ICMIT/IT40

Design Analysis & Fabrication of Autonomous mobile robot for material Handling in assembly line

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ABSTRACT

In response to the increasing demand for flexible and intelligent intra-logistics solutions in modern manufacturing, this paper presents the design, analysis, and fabrication of an Autonomous Mobile Robot (AMR) optimized for material handling in assembly lines. The proposed system addresses the limitations of traditional AGVs by implementing real-time localization using SLAM (Simultaneous Localization and Mapping), adaptive path planning, and dynamic obstacle avoidance. A centralized task management framework orchestrates multiple AMRs for just-in-time delivery, ensuring synchronized interaction with human operators and workstations. Both simulation and experimental deployment validate the system's effectiveness, demonstrating improvements in travel time, delivery precision, energy efficiency, and adaptability to workflow variations. The solution supports lean manufacturing and aligns with Industry 4.0 objectives, offering a scalable and intelligent automation framework for future smart factories.

Keywords: Autonomous Mobile Robot (AMR), SLAM, Material handling, path planning, obstacle avoidance, Industry 4.0.

ICMIT/IT42

Context-Aware Object Detection in Hospital Environments Using YOLOv11**Piyush Awasthi*, Anuj Kumar Sharma***Centre for Advanced Studies, Dr. A. P. J. Abdul Kalam Technical University, Lucknow, India***Email: piyushawasthi901@gmail.com**#Corresponding Author Email ID: anujksharma@cas.res.in***ABSTRACT**

Correct identification of objects in a hospital setting is essential in improving productivity, safety, and stock control. However, ID application becomes more complicated in settings such as lobbing areas with high patient traffic, patient rooms with medical devices, and restaurants serving food trays. This paper describes a context-aware approach towards YOLOv11 object detection which, after detecting the context of the scene, adjusts its parameters accordingly to increase precision. The model has a training dataset of 2685 images and a validation dataset of 125 images across the nine categories such as ‘Guest’, ‘Food Tray’, ‘Wheelchair’, and others; the model has a mean Average Precision (mAP50-95) of 58.3% and a detection accuracy of 93.23%. Confidence thresholds for each environment are set in the range of 0.3–0.4, IoU thresholds – in the range of 0.4–0.5, and the maximum amount of detections is set from 30 to 300, based on the spatial attention maps. A quantitative analysis of two test images demonstrates flexibility through an ability to correctly identify objects in different situations. Compared to the 39.5% mAP50-95 on 80 classes on the COCO set achieved by YOLOv11n, our system performs better on fewer classes as context awareness improves accuracy by ~2% according to the literature. Given our 80.1% mAP50 as compared to YOLOv6-L’s 51.7% mAP50 in the 56 class HIOD dataset, we have a better performance. The study should be extended to large-scale datasets, and more evidence about the utility of this work in a real-world context is needed further; this work has the potential for usage in robotic support and patient care.

Keywords: Object Detection, YOLOv11, Context-Aware, Hospital Environments, Computer Vision, Healthcare AI

ICMIT/IT43

AI-Driven Object Detection for Sustainable Surveillance Using YOLOv12

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ABSTRACT

The rapid development of artificial intelligence (AI) has changed surveillance systems by letting automated, real-time monitoring indispensable for social media analytics, commercial, and healthcare applications. This work introduces a new object identification framework based on the YOLOv12 model, built using a customized dataset, therefore enabling high accuracy recognition of drones, people, cars, and weapons. With a mean average precision (mAP50) of 0.907 AND mAP 50-95 OF 0.633. the model exhibits remarkable accuracy and robustness in numerous environments. This work uses the light-weight and streamlined architecture of YOLOv12 to provide a scalable and energy-efficient solution for sustainable surveillance systems. Moreover, the method combines advanced data augmentation methods with transfer learning to improve generalization, so solving issues including occlusion and changing environmental conditions. Apart from improving real-time item detection capability, our work addresses the growing demand for environmentally friendly artificial intelligence solutions in public safety and critical infrastructure applications.

Keywords: YOLOv12, Object Detection, AI, Surveillance, Sustainability, Data Visualization

ICMIT/IT/44

Transition from localization to a deep localization strategy: A pathway towards sustainability

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ABSTRACT

For development through decarbonized pathways, many countries have been using solar energy in a big way. China meets over 85% of global solar PV modules and equipment demand and has emerged as the global PV manufacturing hub. The high concentration of manufacturing capabilities and overreliance on China introduces vulnerabilities in the supply chain (SC), affecting economic, environmental, and social sustainability. In response, various countries have adopted localization strategies to enhance self-reliance. Yet, the Indian solar sector still faces significant limitations and challenges under the conventional localization approach. To address this, the study proposes a novel, deep localization (DL) strategy that goes beyond traditional localization by integrating local infrastructure, stakeholder coordination, and value chain development. DL helps bring the stakeholders in close vicinity, enhancing control over the SC, increasing flexibility and agility and reducing risks. This research utilizes an analytic hierarchy process to prioritize factors enabling sustainability in the SC. This quantitative approach helps systematically identify and rank factors, providing a clear decision-making framework for stakeholders. The seven critical enablers are determined through a systematic literature review and discussion with industry experts. Policy support, supplier development, and value chain enhancement emerged as top priorities among these enablers, signifying their role in building a resilient ecosystem. The DL also yields a positive rippling effect, reducing carbonization, enhancing economic robustness, and fostering social development. This study contributes by providing a conceptual framework and actional insight for stakeholders to create a pathway for a sustainable and self-reliant ecosystem.

Keywords: Solar Energy, Solar PV modules, Supply chain

ICMIT/IT46

Depression Detection from Twitter Data Using Machine Learning

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ABSTRACT

In recent decades depression is spreading very fast and affecting lots of people throughout the world. The traditional methods of detecting depression are very time consuming and subjective which could produce biased results. Social media platforms like twitter, are now working as a prominent database for real time information which can be useful for mental health analysis using machine learning (ML) and natural language processing (NLP). This study explores how deep learning, specifically Long Short-Term Memory (LSTM) networks, can help detect depression in tweets. It uses a dataset of 1.6 million tweets from Kaggle, which goes through several preprocessing steps like removing stop words and punctuation, tokenizing, stemming, lemmatizing, and extracting features with GloVe embeddings. To ensure fair training, a balanced set of 40,000 tweets is used, and the model is fine-tuned with the Adafactor optimizer over 20 training cycles. The model shows strong classification performance as it achieves an accuracy of 89% and an AUC score of 0.92. Although there are some challenges in this model as it still depends upon textual data only and shows misclassification of ambiguous tweets. In future more improvements can be done in this project by working on models like BERT and GPT, along with multimodal data integration, to enhance depression detection accuracy.

Keywords: Depression Detection; Machine Learning; Natural Language Processing (NLP); Long Short-Term Memory (LSTM); Sentiment Analysis; Twitter Data; Text Classification

ICMIT/IT50

Optimal PMU Placement Using Binary PSO-Based Optimization Algorithm for Multi-Objective Constrained Functions

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ABSTRACT

This study presents a robust approach for solving the Optimal Phasor Measurement Unit (PMU) Placement (OPP) problem using the Binary PSO-Based (BPSO) Optimization algorithm. The proposed method aims to minimize the total number of PMUs required while simultaneously maximizing network observability and measurement redundancy. Zero Injection Buses (ZIBs) are effectively incorporated into the model by strategically merging them with neighboring buses, which significantly reduces hardware requirements without compromising observability. The BPSO algorithm is evaluated on IEEE 14-, 30-, 57-, and 118-bus systems, where it consistently achieves optimal or near-optimal placement solutions. The influence of weighting factors α (alpha) and β (beta) is also examined to assess the trade-off between PMU count and observability. The results confirm that the BPSO-based approach offers high accuracy, scalability, and computational efficiency, making it a reliable solution for practical power system monitoring and planning applications.

Keywords: Optimal PMU placement (OPP), Binary PSO Based Optimization (BPSO), Complete Observability, system observability redundancy index(SORI)

ICMIT/IT51

**A Deep learning enabled intrusion detection system for *Internet of things*
“IoT” environment**

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ABSTRACT

The rapid integration of IoT devices into critical infrastructures has significantly amplified cybersecurity threats, particularly sophisticated zero-day and botnet attacks. Traditional intrusion detection systems (IDS) often struggle against these advanced threats, prompting the exploration of adaptive deep learning methods. This study comparatively evaluates three distinct approaches an IoT Bayesian model, Federated Learning, and an XGBoost model for IoT intrusion detection using TensorFlow and SHAP plots for enhanced interpretability. The XGBoost model showed strong performance with a validation accuracy of 96.38% and ROC AUC values near 1, while the Bayesian model achieved an accuracy of 98.27%. This research highlights comparative strengths and limitations, informing effective IDS deployment in IoT cybersecurity.

Keywords: IoT, IoT Bayesian model, Federated Learning, XGBoost model

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